



(12) **United States Patent**
Kim et al.

(10) **Patent No.:** **US 12,404,625 B2**
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(54) **HANGER DEVICE**

(56) **References Cited**

(71) Applicant: **LG ELECTRONICS INC.**, Seoul
(KR)

U.S. PATENT DOCUMENTS

(72) Inventors: **Joogyeon Kim**, Seoul (KR); **Hyunsun Yoo**, Seoul (KR); **Hyunjung Kim**, Seoul (KR)

4,625,432 A 12/1986 Baltes
4,989,624 A * 2/1991 Darling D06F 17/04
D6/318

(Continued)

(73) Assignee: **LG ELECTRONICS INC.**, Seoul
(KR)

FOREIGN PATENT DOCUMENTS

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 212 days.

CN 202440683 9/2012
CN 203252394 10/2013

(Continued)

(21) Appl. No.: **18/090,003**

Machine translation of CN-109853202-A, dated Jun. 7, 2019. (Year: 2019).*

(22) Filed: **Dec. 28, 2022**

(Continued)

(65) **Prior Publication Data**

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Primary Examiner — Michael E Barr

Assistant Examiner — Kevin G Lee

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(30) **Foreign Application Priority Data**

Jan. 4, 2022 (KR) 10-2022-0000952

Jan. 4, 2022 (KR) 10-2022-0000954

(Continued)

(51) **Int. Cl.**

A47G 25/14 (2006.01)

D06F 58/20 (2006.01)

(Continued)

(52) **U.S. Cl.**

CPC **A47G 25/14** (2013.01); **D06F 58/203** (2013.01); **D06F 58/22** (2013.01);

(Continued)

(58) **Field of Classification Search**

CPC D06F 58/203; D06F 58/22; A47G 25/14

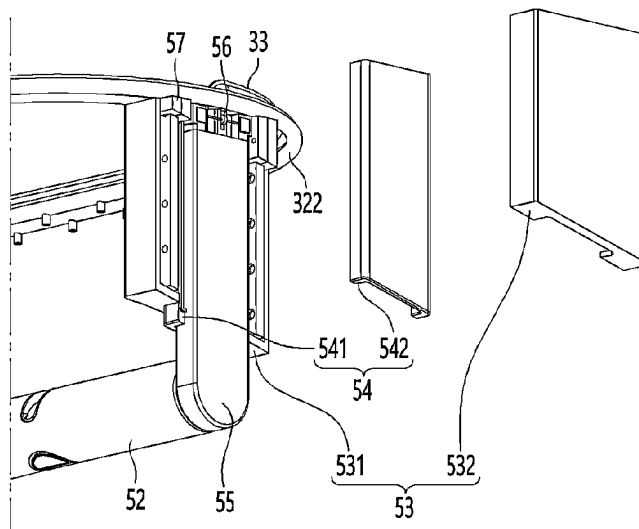
See application file for complete search history.

(57)

ABSTRACT

A hanger device of the present disclosure may include a body having at least opening to allow air to be suctioned into the body and at least one discharge port to allow suctioned air to be forced out of the body. A fan module is provided inside the body and is configured to suction air through the opening and force the suction air to the at least one discharge port. A filtering module removes foreign substances contained in the suctioned air. A hanging unit is coupled to the main body, wherein the hanging unit includes: a pair of supports extending from a bottom of the main body and having at least one lift unit to change a length of the supports in a vertical direction; a hanging bar connecting the ends of the pair of supports; and an air flow path provided in the pair of supports to allow portion of the suctioned air to pass through the pair of supports.

16 Claims, 58 Drawing Sheets



(30) Foreign Application Priority Data

Jan. 4, 2022	(KR)	10-2022-0000961	CN	113737486	12/2021
Jan. 4, 2022	(KR)	10-2022-0000965	CN	113774637	12/2021
Jan. 4, 2022	(KR)	10-2022-0000976	CN	113832691	12/2021
Jan. 4, 2022	(KR)	10-2022-0000981	DE	10-2018-211668	1/2020
Jan. 4, 2022	(KR)	10-2022-0000985	DE	10-2018-008404	4/2020
Jan. 4, 2022	(KR)	10-2022-0000988	EP	2267213	12/2010
Jan. 4, 2022	(KR)	10-2022-0000993	EP	3705618	9/2020
Jan. 4, 2022	(KR)	10-2022-0001153	JP	H07-290944	11/1995
Jan. 4, 2022	(KR)	10-2022-0001155	JP	H11-89700	4/1999
			JP	2003-176930	6/2003
			JP	2003-230476	8/2003
			JP	2011-112251	6/2011
			JP	2012-065864	4/2012
			JP	6098696	3/2017
			JP	2018-057413	4/2018
			JP	2021-101786	7/2021
			KR	200184385	6/2000
			KR	20-0316690	6/2003
			KR	20040053731	6/2004
			KR	10-0497325	6/2005
			KR	2008-0073088	8/2008
			KR	2011-0035030	4/2011
			KR	10-1312024	9/2013
			KR	10-2018-0128746	12/2018
			KR	10-2039829	11/2019
			KR	10-2046259	12/2019
			KR	2020-0080962	7/2020
			KR	2021-0037496	4/2021
			KR	10-2261579	6/2021
			WO	WO 2006/059225	6/2006
			WO	WO 2018/0159313	9/2018
			WO	WO 2020/138931	8/2020
			WO	WO 2020/175959	9/2020
			WO	WO 2021080153	4/2021

(51) Int. Cl.

D06F 58/22 (2006.01)*D06F 58/26* (2006.01)*D06F 59/02* (2006.01)

(52) U.S. Cl.

CPC *A47G 2200/16* (2013.01); *D06F 58/26*
(2013.01); *D06F 59/02* (2013.01)

(56) References Cited

U.S. PATENT DOCUMENTS

5,361,516	A	11/1994	Dahman	
5,815,961	A	10/1998	Estes	
2005/0120757	A1	6/2005	Jackson	
2006/0226177	A1 *	10/2006	McJunkin	<i>D06F 59/02</i>
				223/85
2011/0203130	A1 *	8/2011	Date	<i>D06F 58/203</i> 34/621
2017/0233942	A1	8/2017	Eun et al.	
2018/0064296	A1	3/2018	Jun et al.	
2019/0169789	A1	6/2019	Cho et al.	
2020/0071875	A1	3/2020	Woo	
2021/0156071	A1	5/2021	Cho	
2021/0198836	A1	7/2021	Kitayama	
2023/0210286	A1	7/2023	Kim	
2023/0210287	A1	7/2023	Kim	
2023/0210290	A1	7/2023	Kim	
2023/0212811	A1	7/2023	Kim	

FOREIGN PATENT DOCUMENTS

CN	203440684	2/2014
CN	105133268	12/2015
CN	106337268	1/2017
CN	106676851	5/2017
CN	206873146	1/2018
CN	107724006	2/2018
CN	107815832	3/2018
CN	109355879	2/2019
CN	109853202	A * 6/2019
CN	110373874	10/2019
CN	110438769	11/2019
CN	210529314	5/2020
CN	109853202	4/2021
CN	113047017	6/2021
CN	113123092	7/2021
CN	113718498	11/2021
CN	214855276	11/2021

OTHER PUBLICATIONS

European Search Report dated May 9, 2023 issued in Application No. 22215757.0.

European Search Report dated May 16, 2023 issued in Application No. 22215784.4.

European Search Report dated May 9, 2023 issued in Application No. 22214995.7.

U.S. Office Action dated Mar. 22, 2024 issued in U.S. Appl. No. 18/090,036.

European Search Report dated Mar. 28, 2023 issued in Application No. 22214987.4.

European Search Report dated Apr. 5, 2023 issued in Application No. 22215824.8.

Extended European Search Report dated May 22, 2023 issued in Application No. 22215823.

Extended European Search Report dated Jun. 1, 2023 issued in Application No. 22215783.

European Search Report dated Apr. 6, 2023 issued in Application No. 22216846.0.

European Search Report dated Mar. 30, 2023 issued in Application No. 22216847.8.

European Search Report dated Mar. 30, 2023 issued in Application No. 22216845.2.

Chinese Office Action dated Jul. 22, 2025, issued in Application No. 202210440761.3.

* cited by examiner

FIG. 1

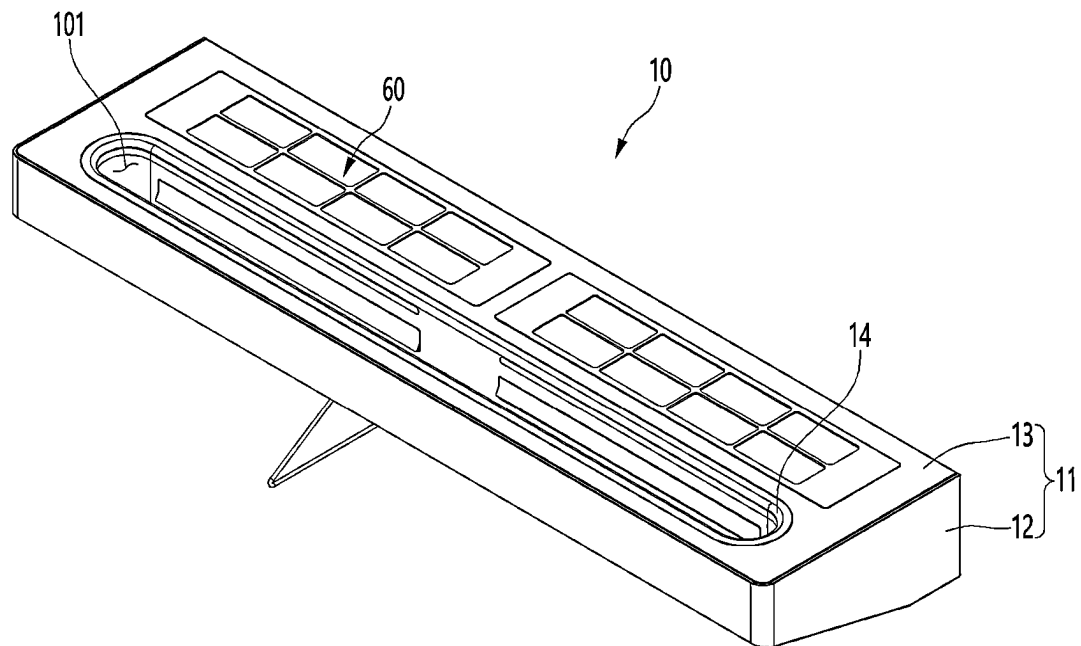


FIG. 2

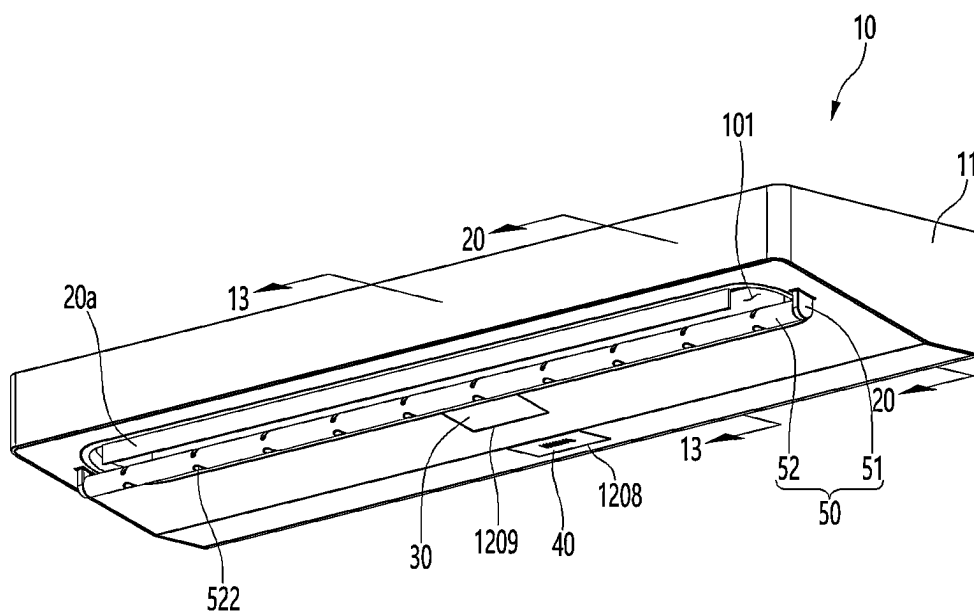


FIG. 3

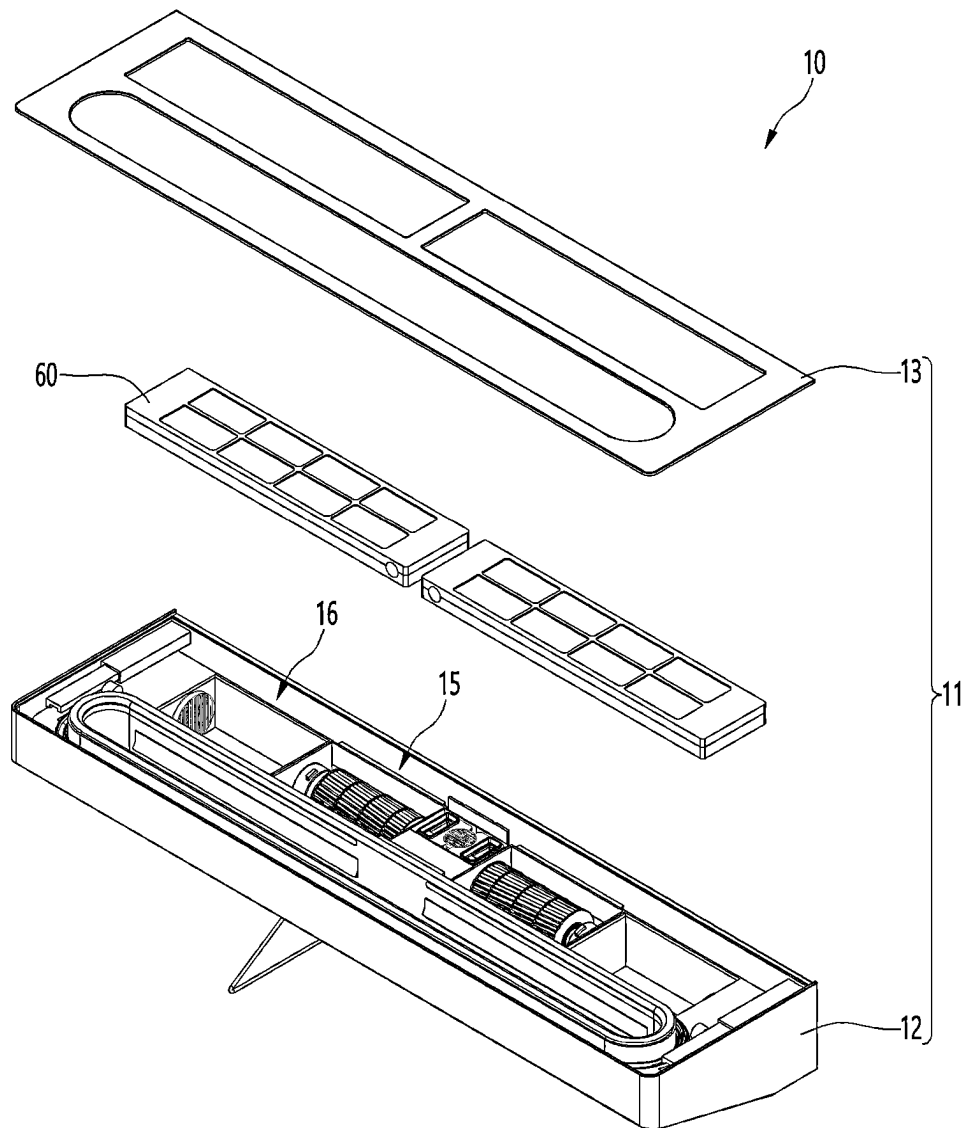


FIG. 4

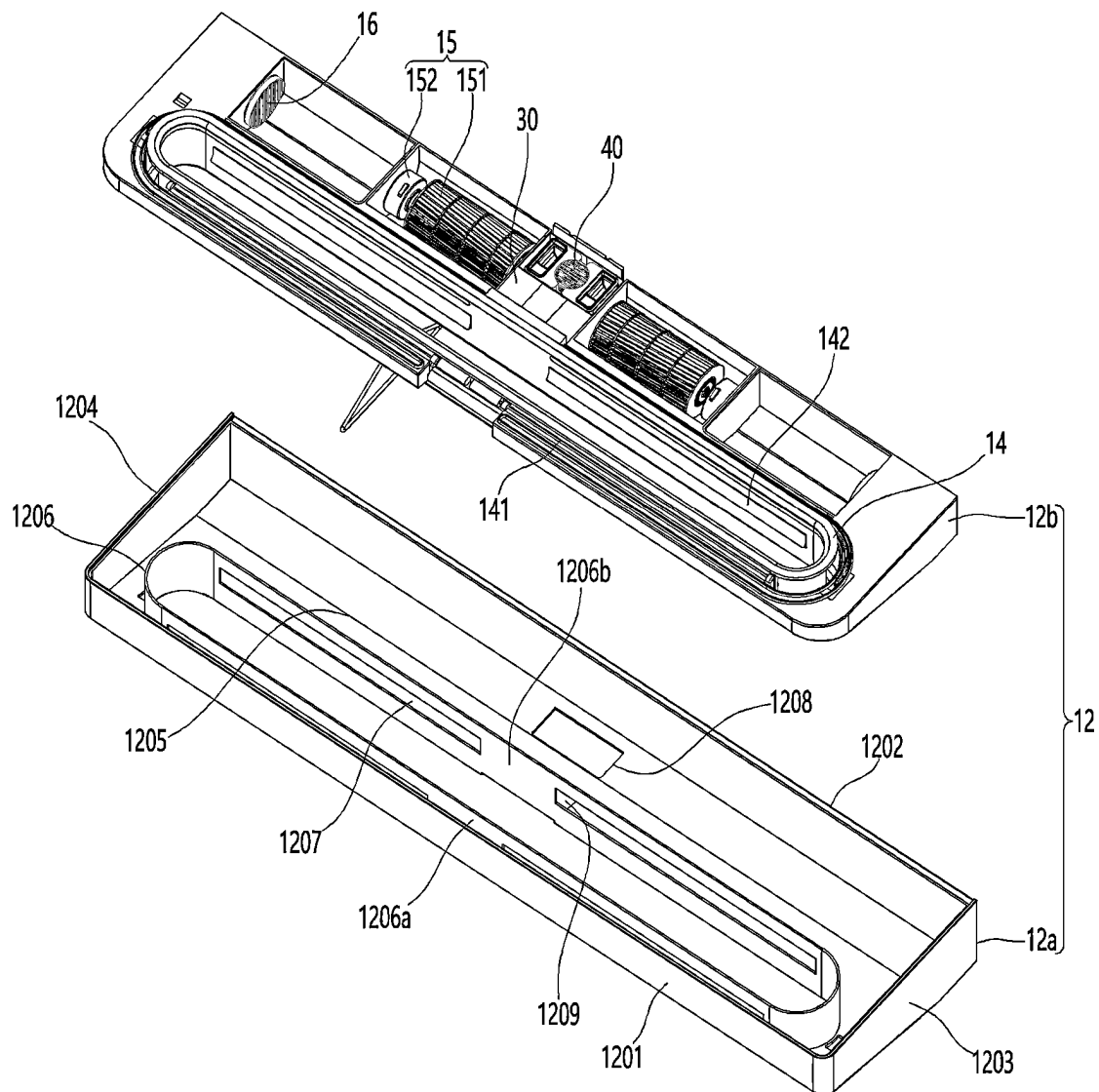


FIG. 5

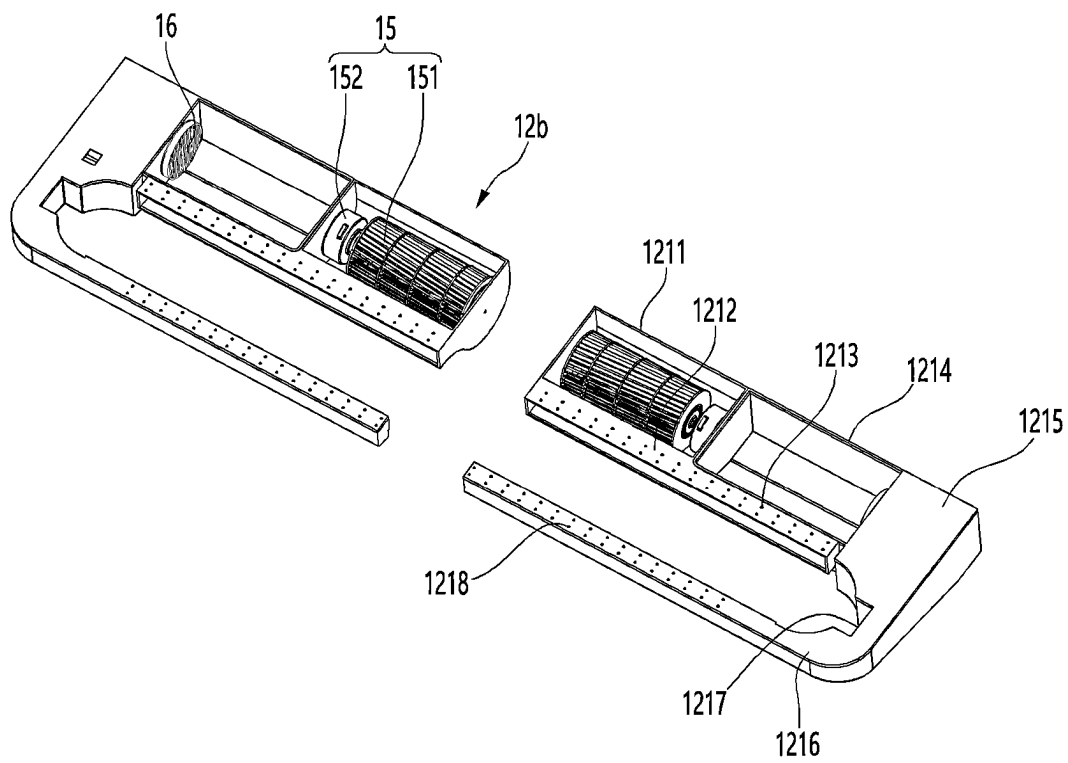


FIG. 6

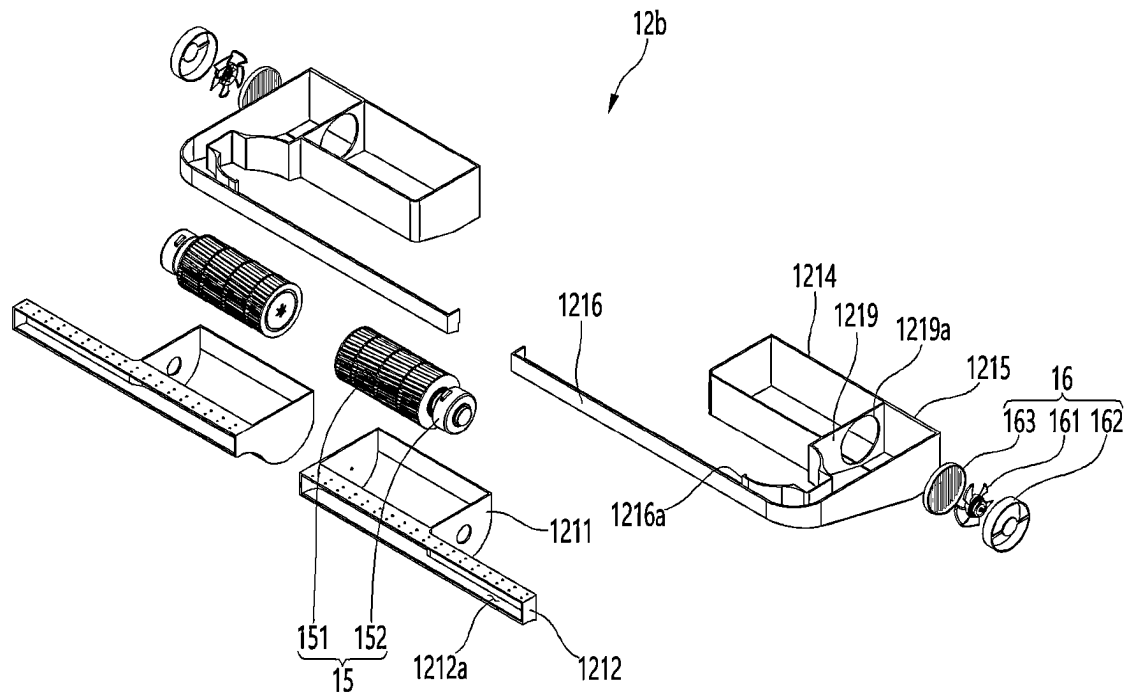


FIG. 7

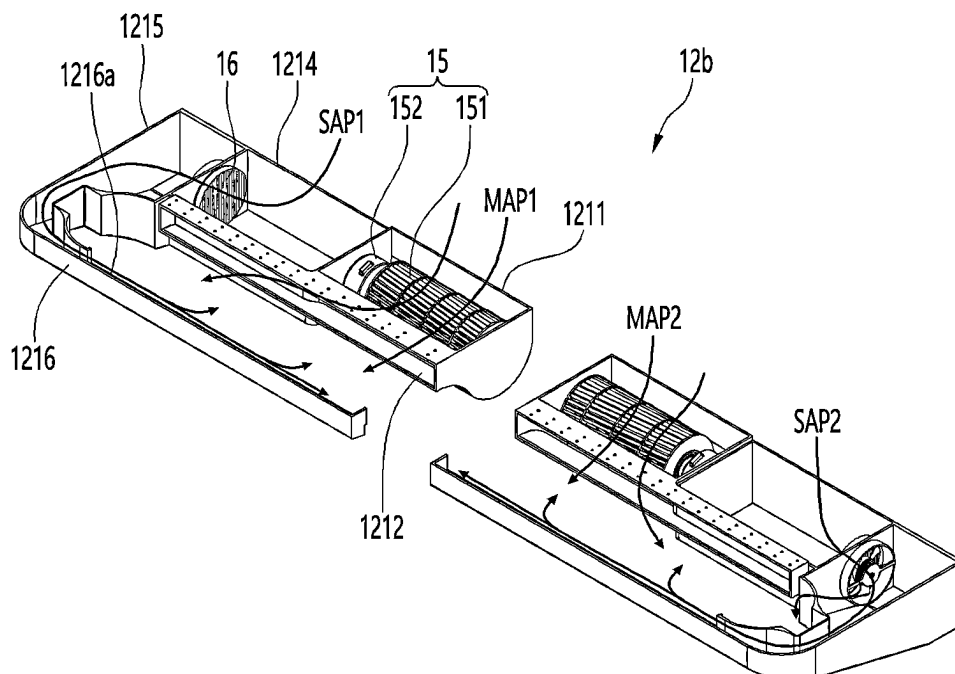


FIG. 8

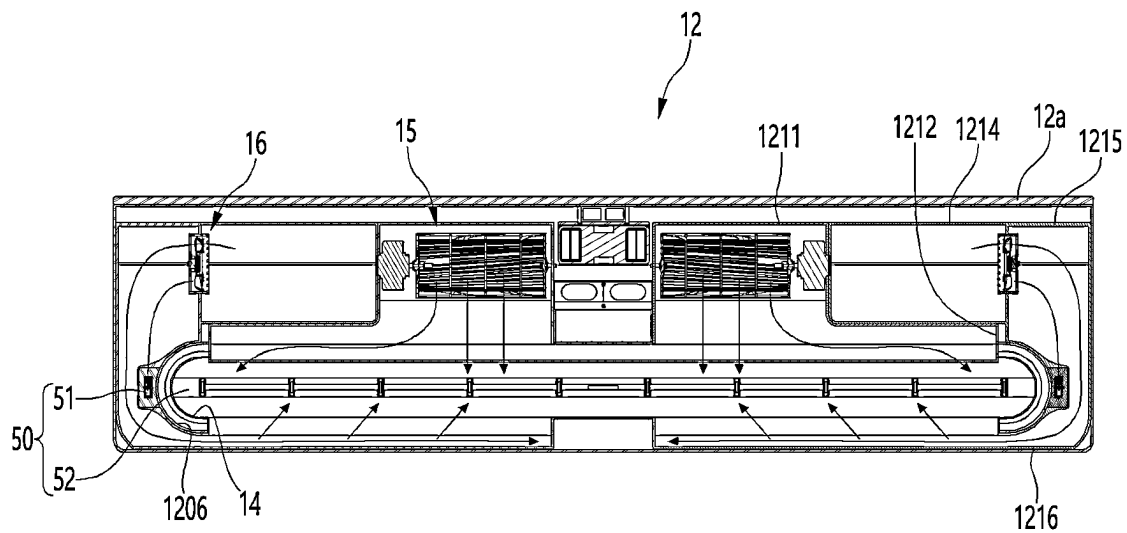


FIG. 9

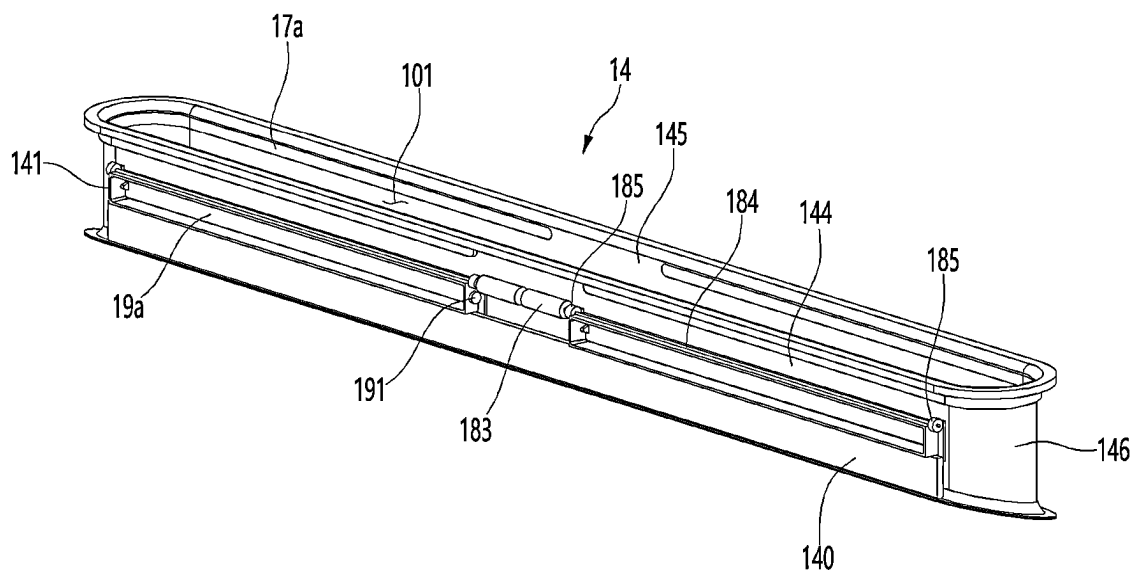


FIG. 10

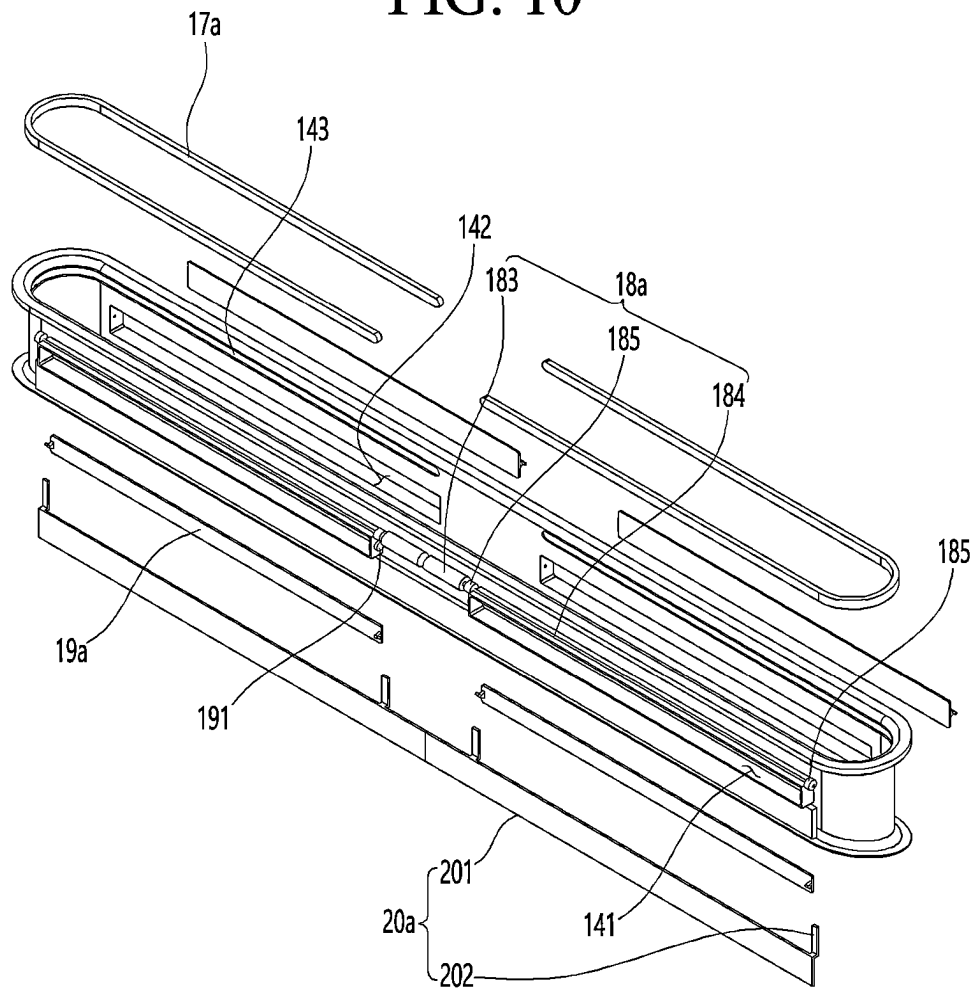


FIG. 11

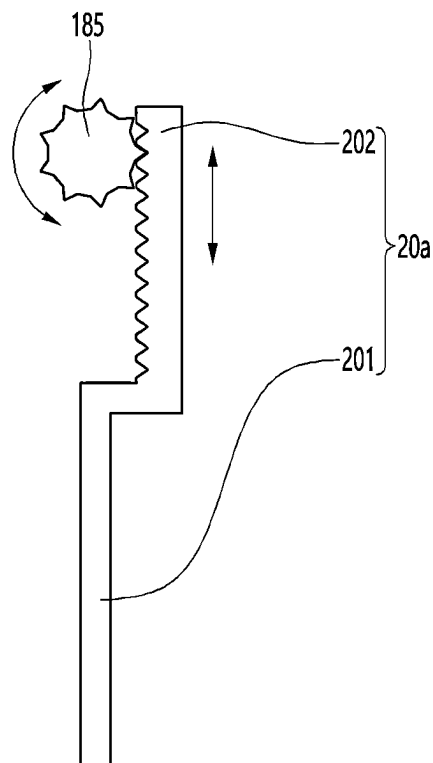


FIG. 12

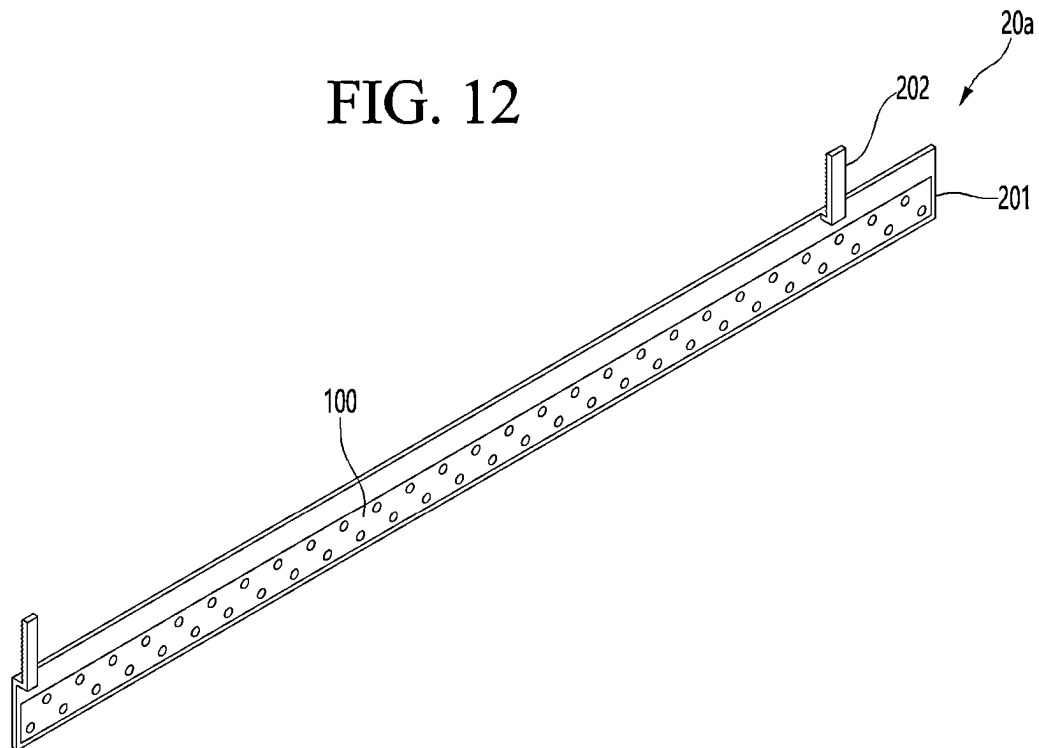


FIG. 13

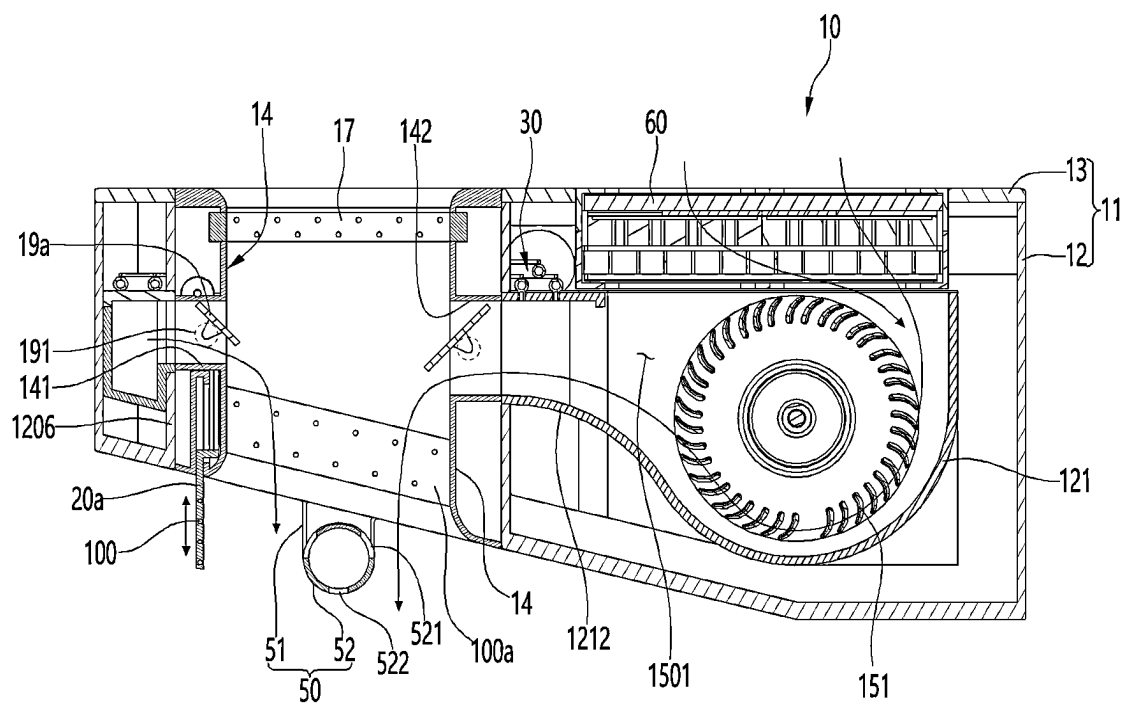


FIG. 14

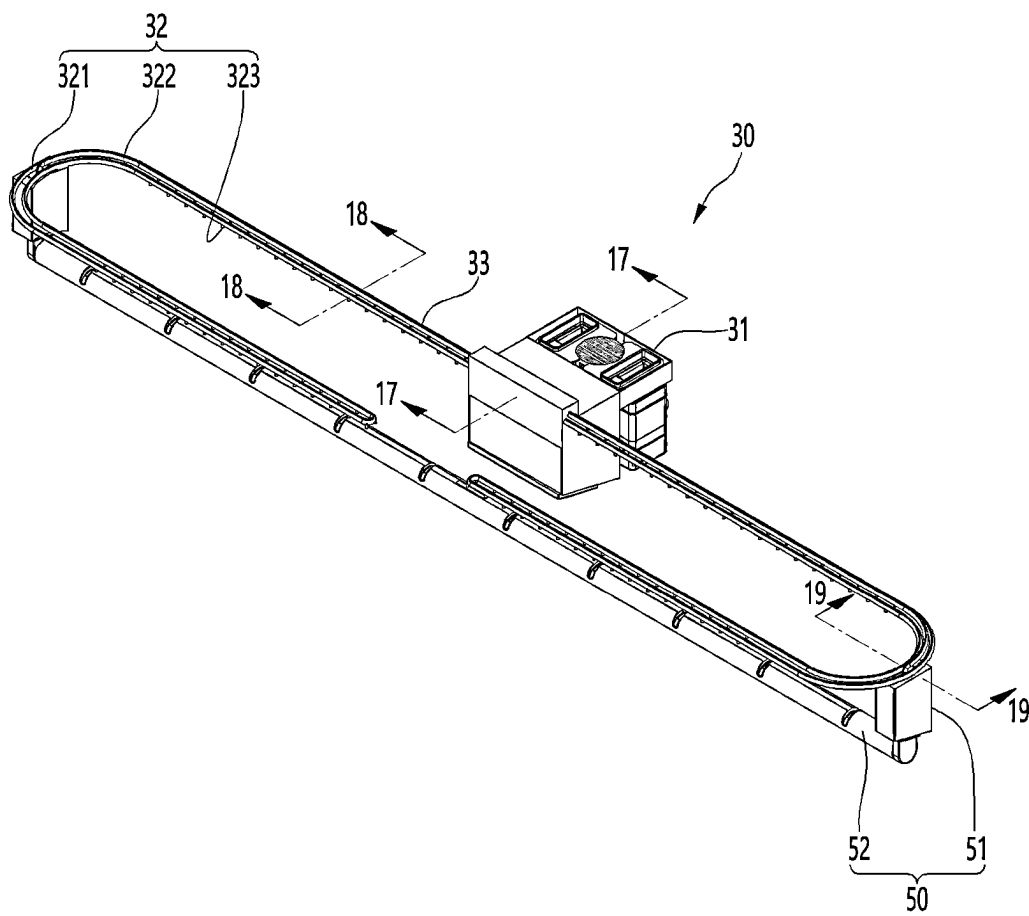


FIG. 15

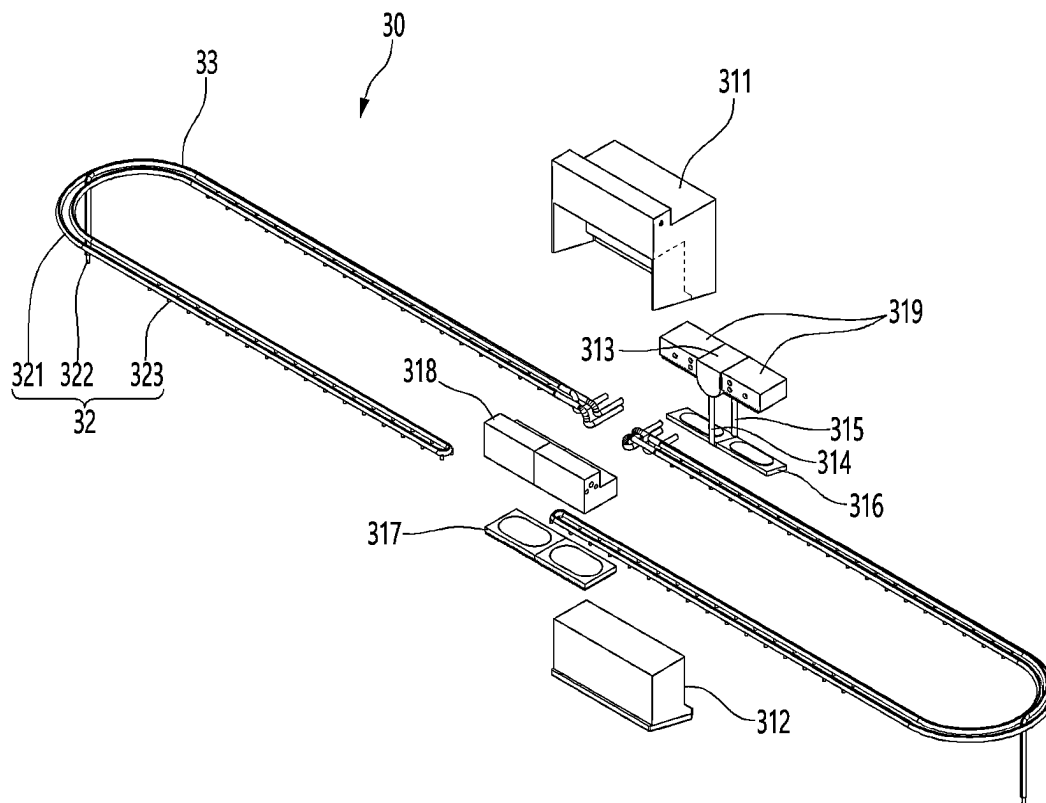


FIG. 16

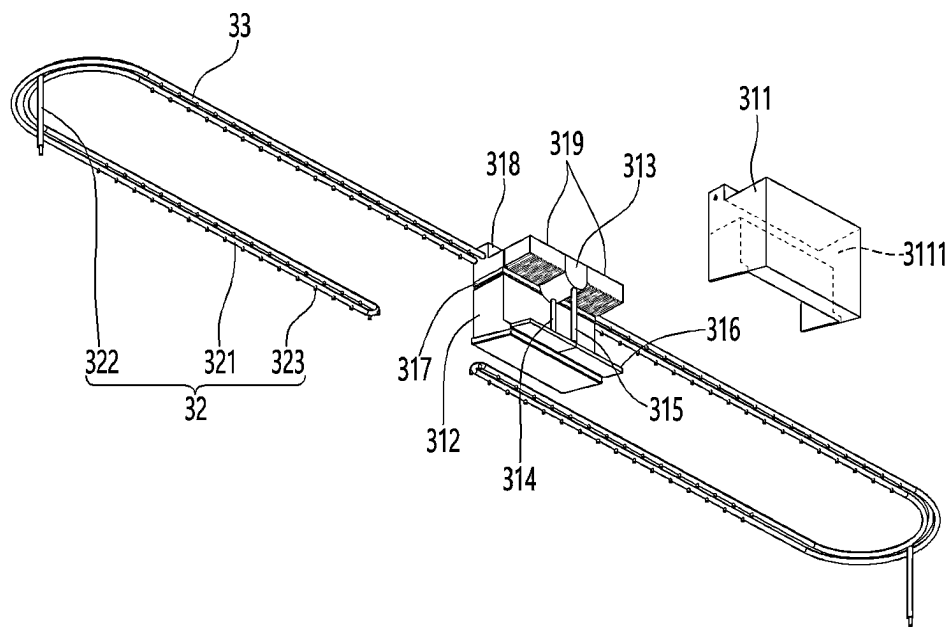


FIG. 17

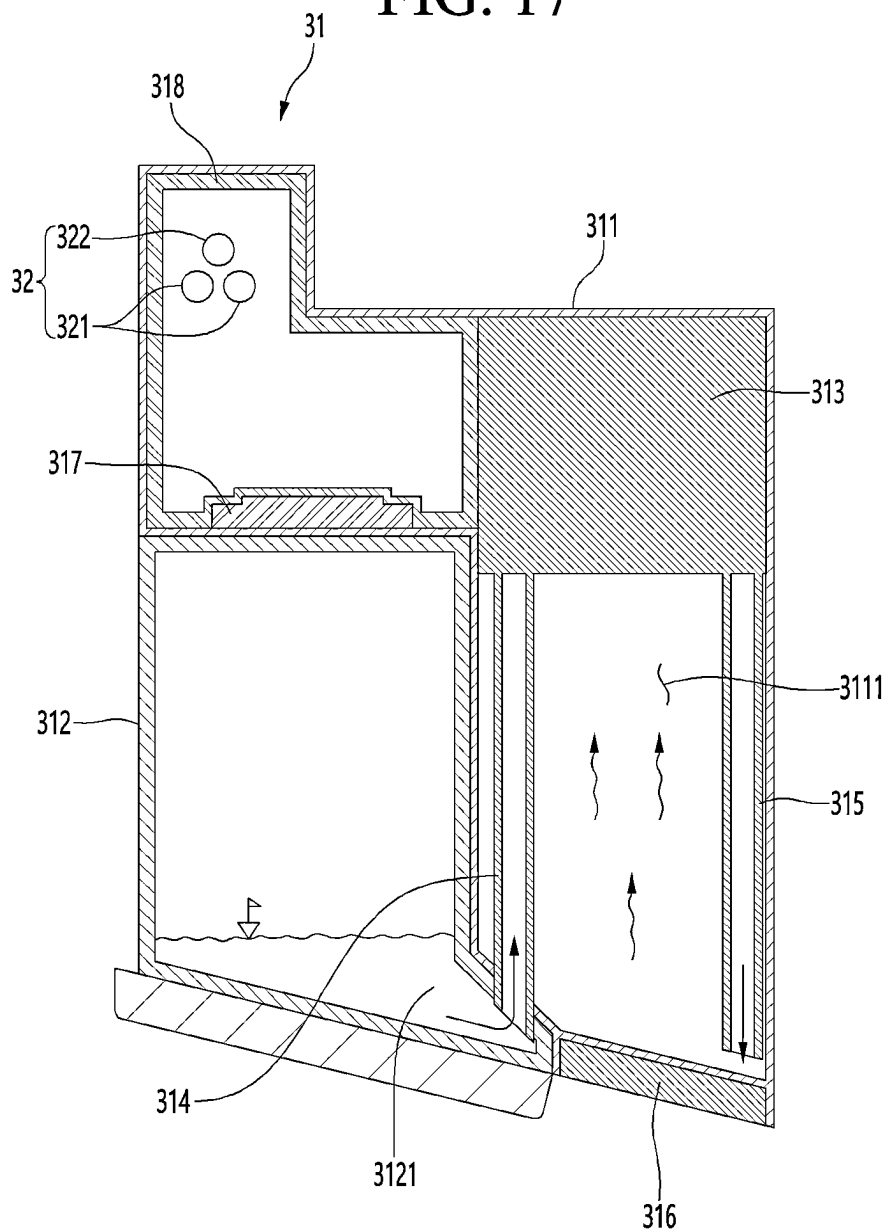


FIG. 18

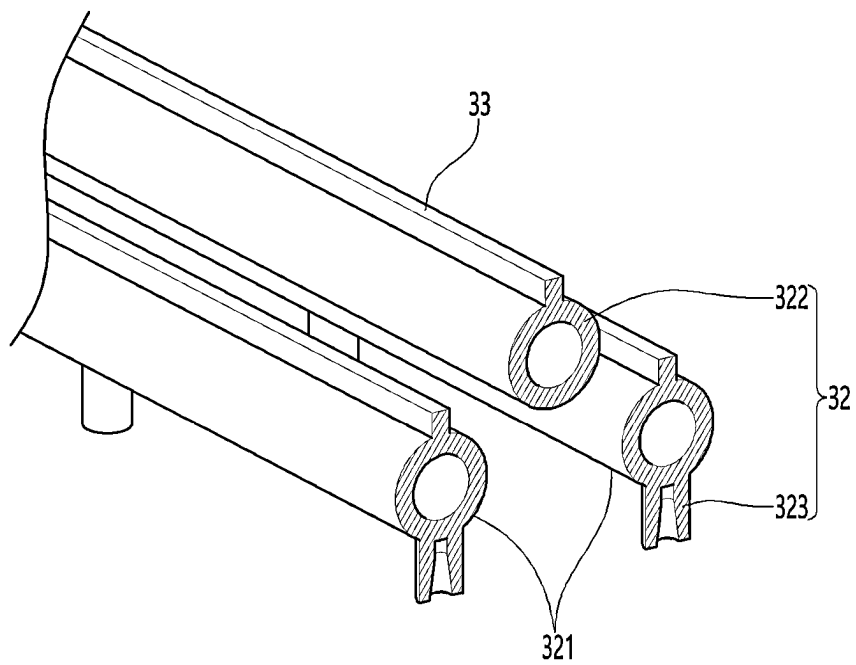


FIG. 19

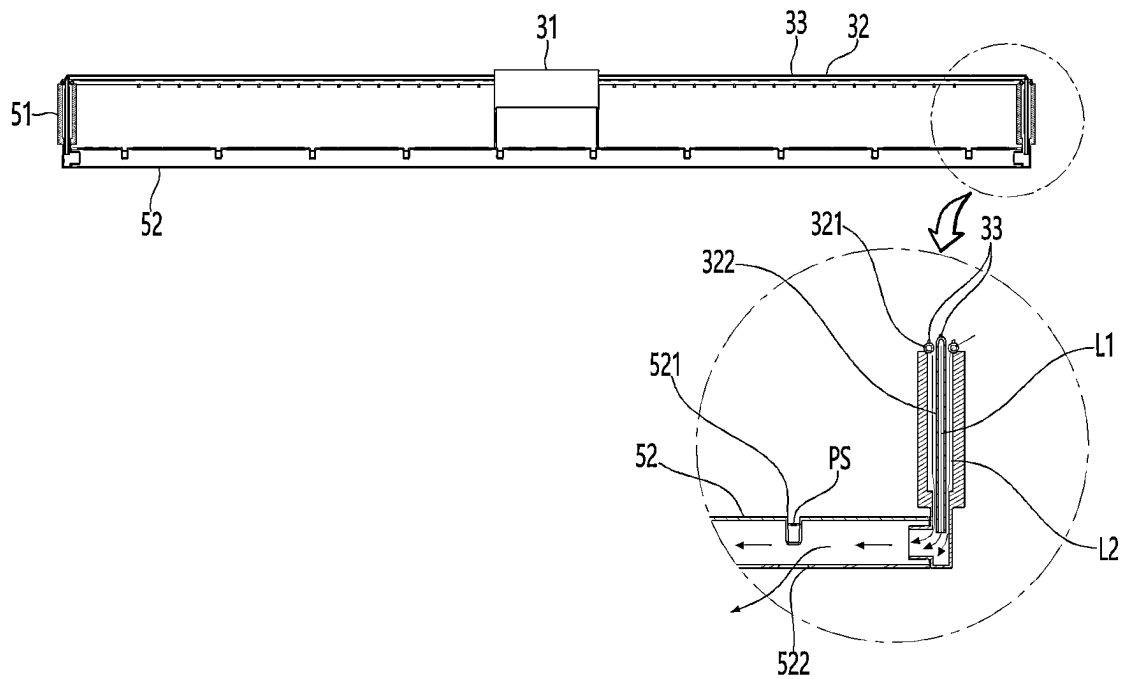


FIG. 20

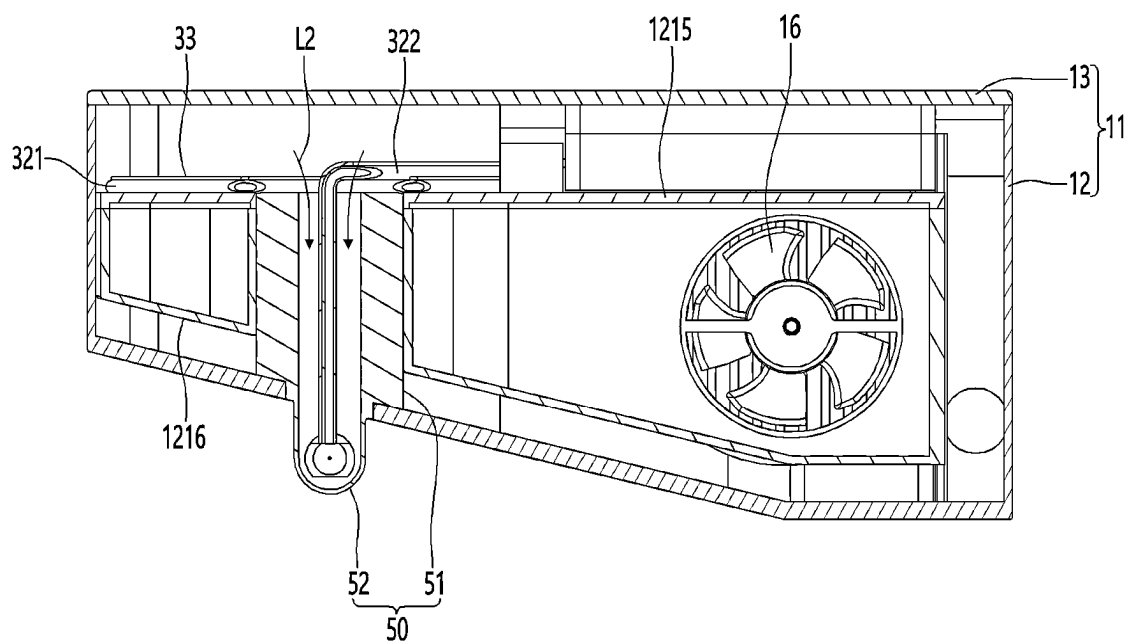


FIG. 21

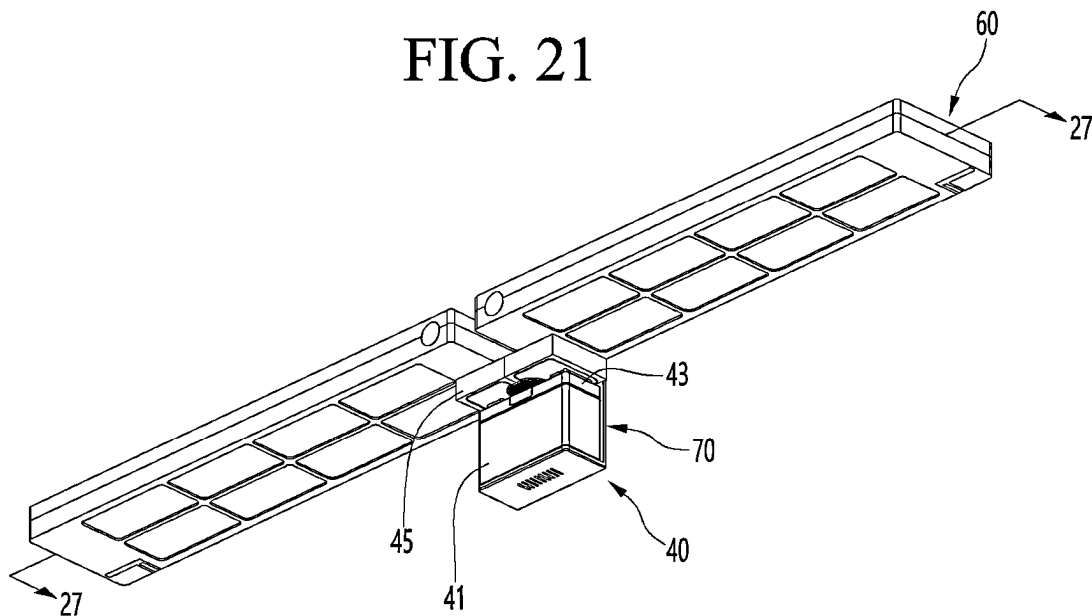


FIG. 22

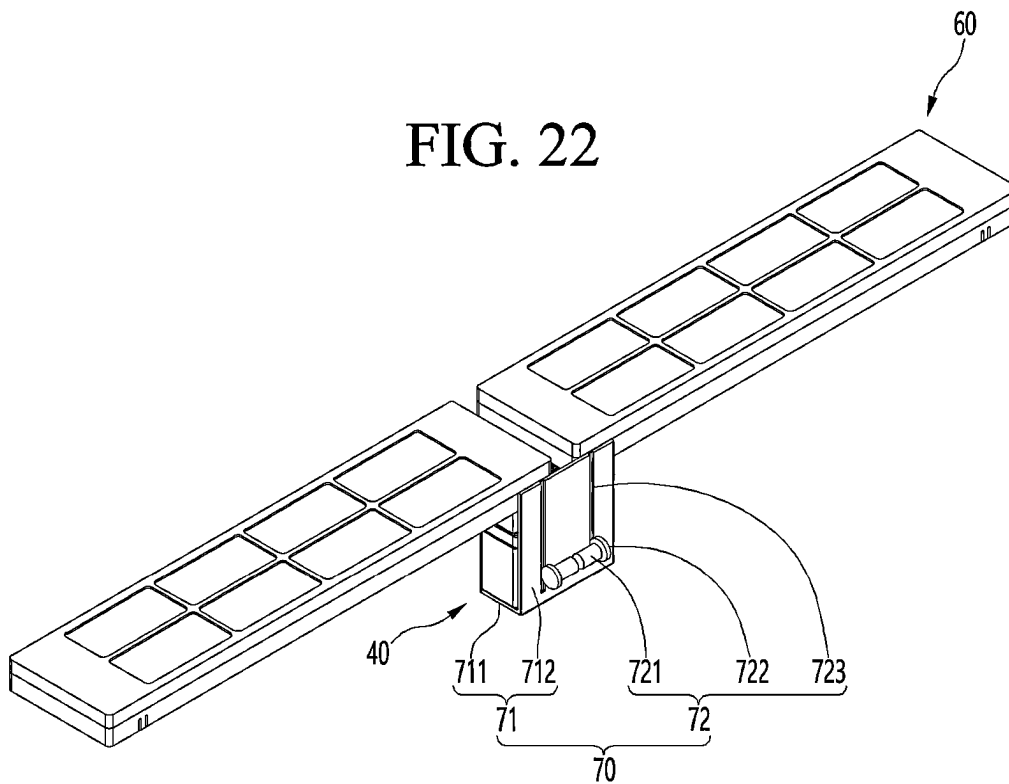


FIG. 23

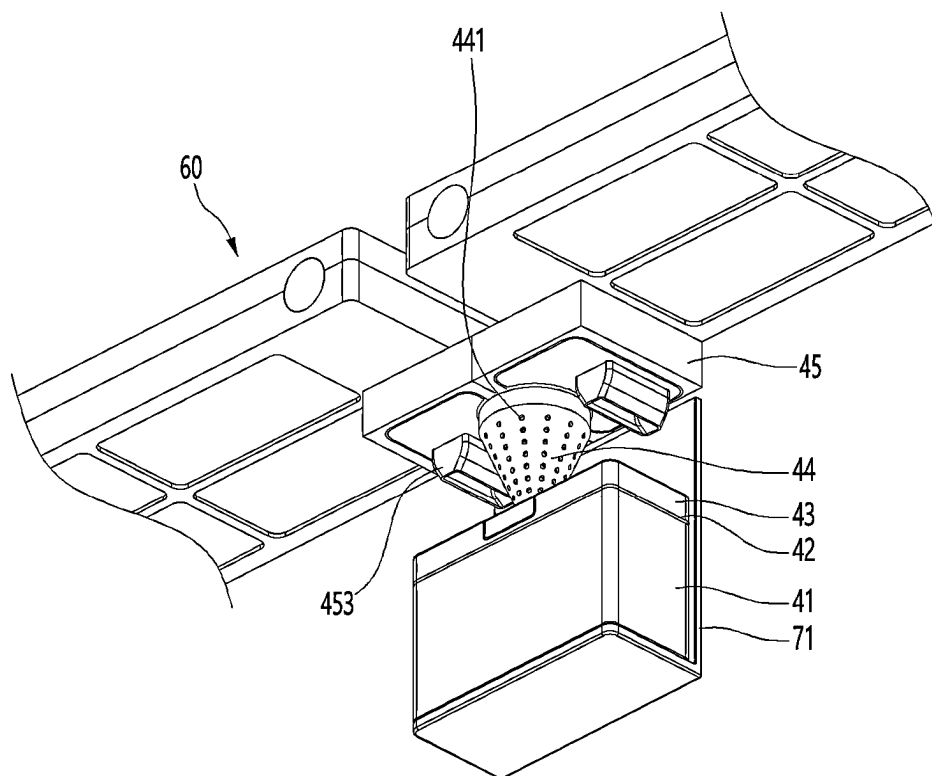


FIG. 24

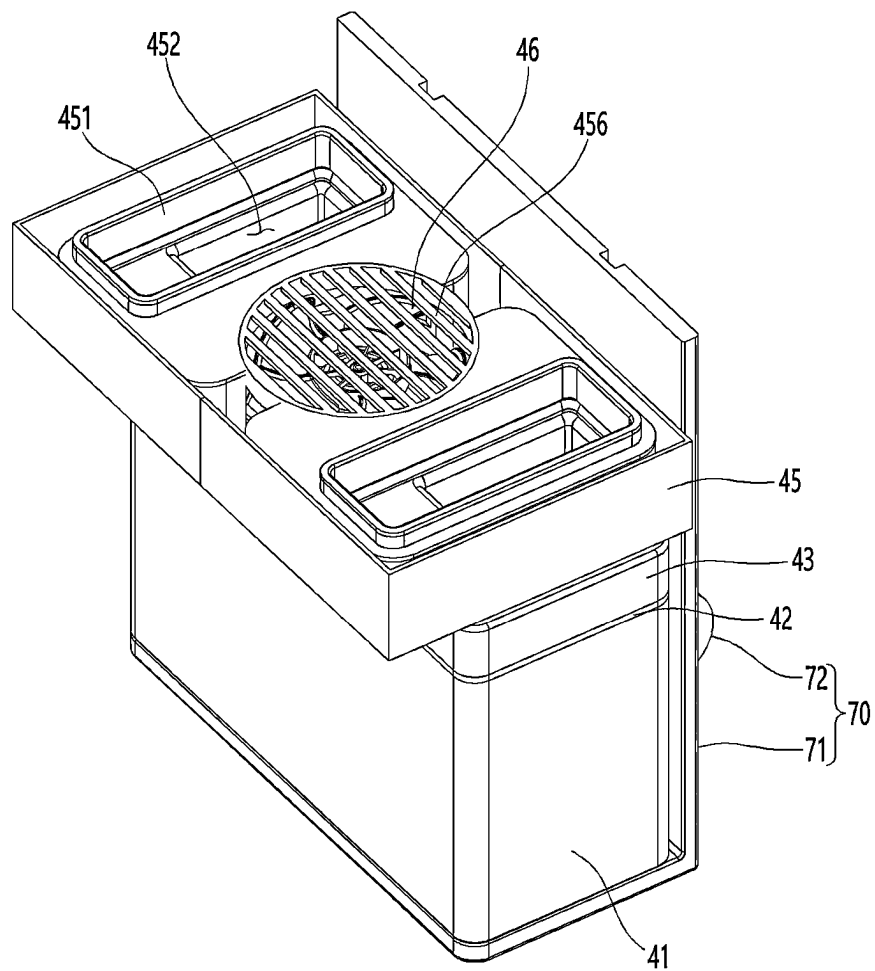


FIG. 25

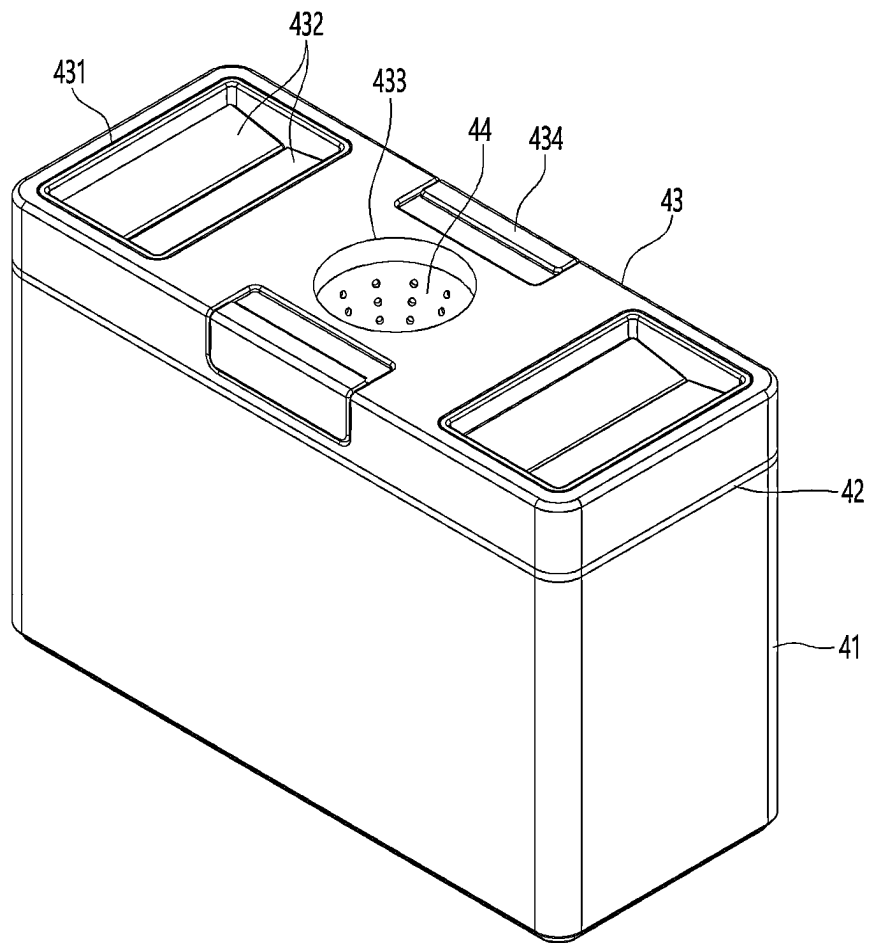


FIG. 26

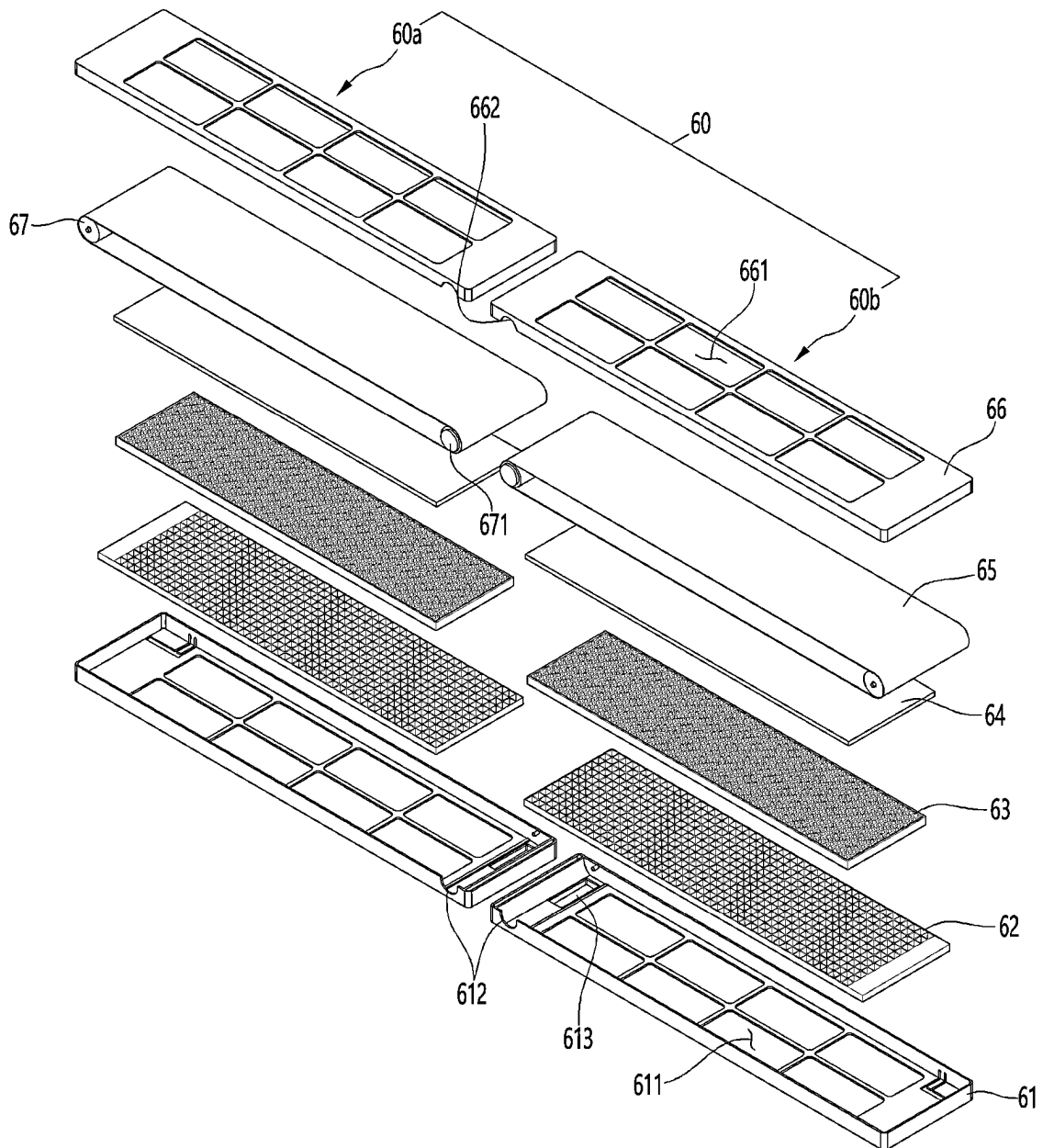


FIG. 27

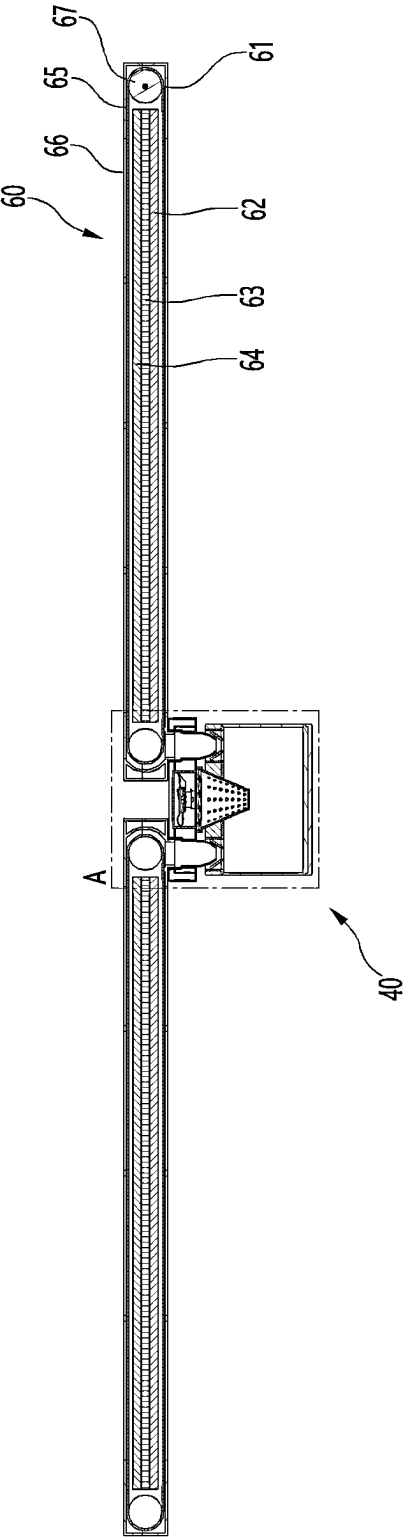


FIG. 28

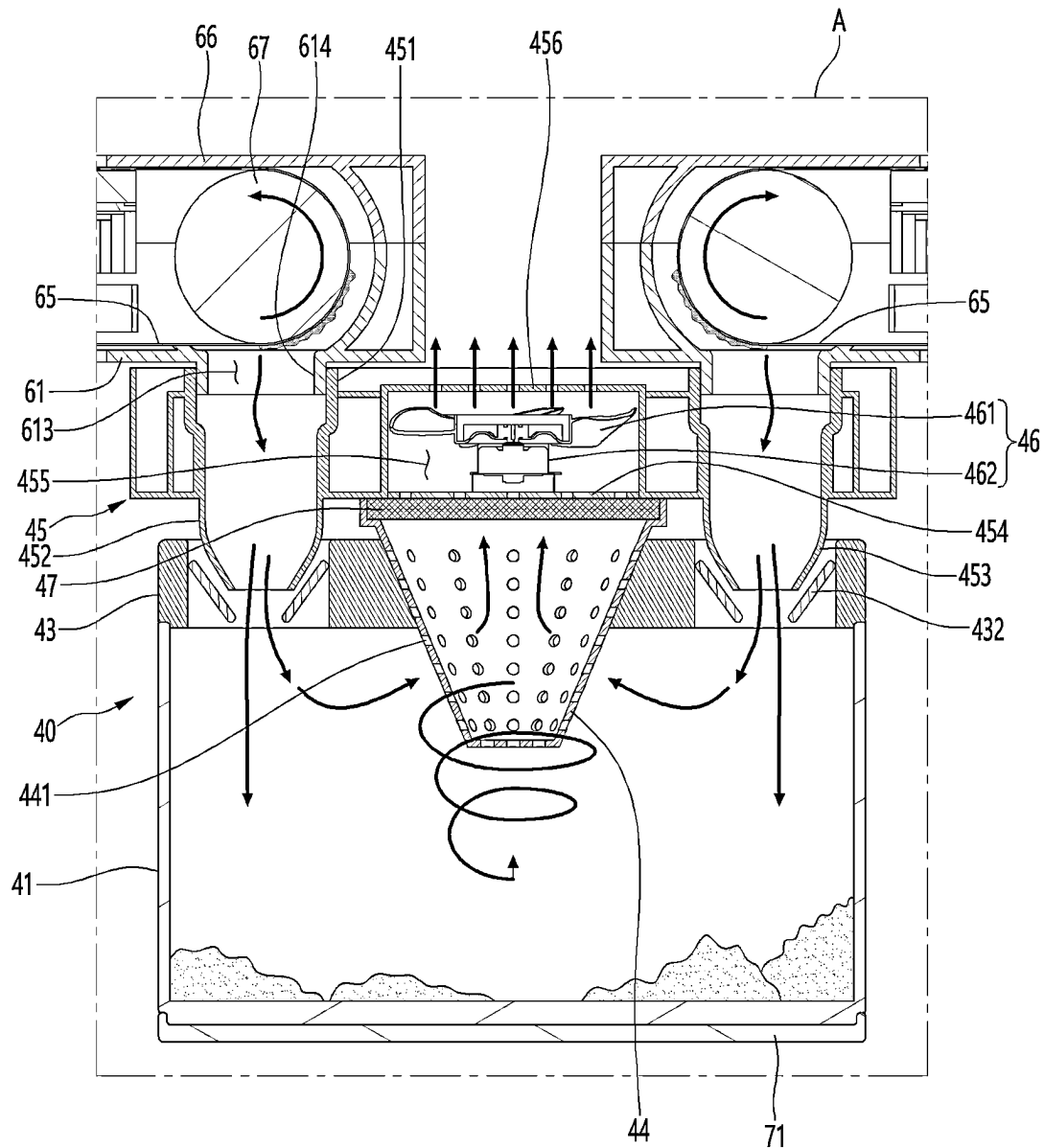


FIG. 29

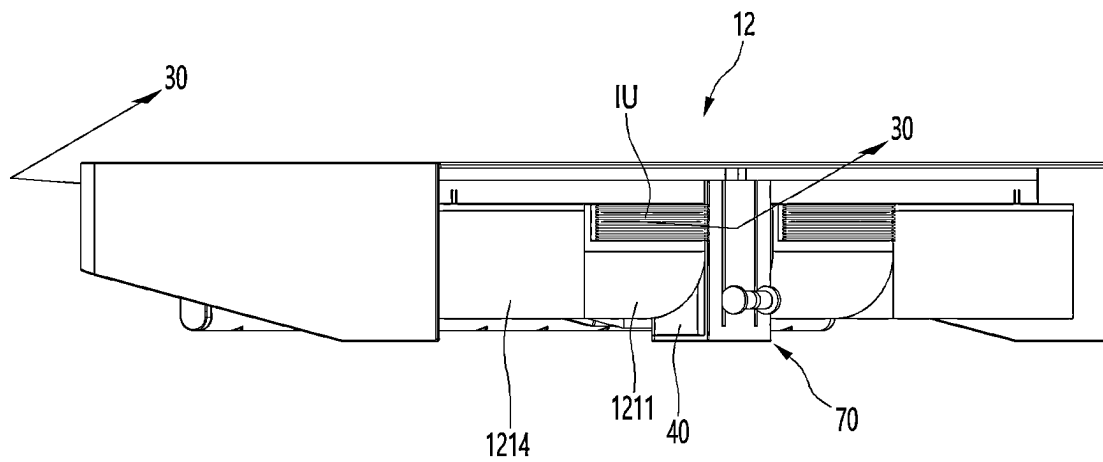


FIG. 30

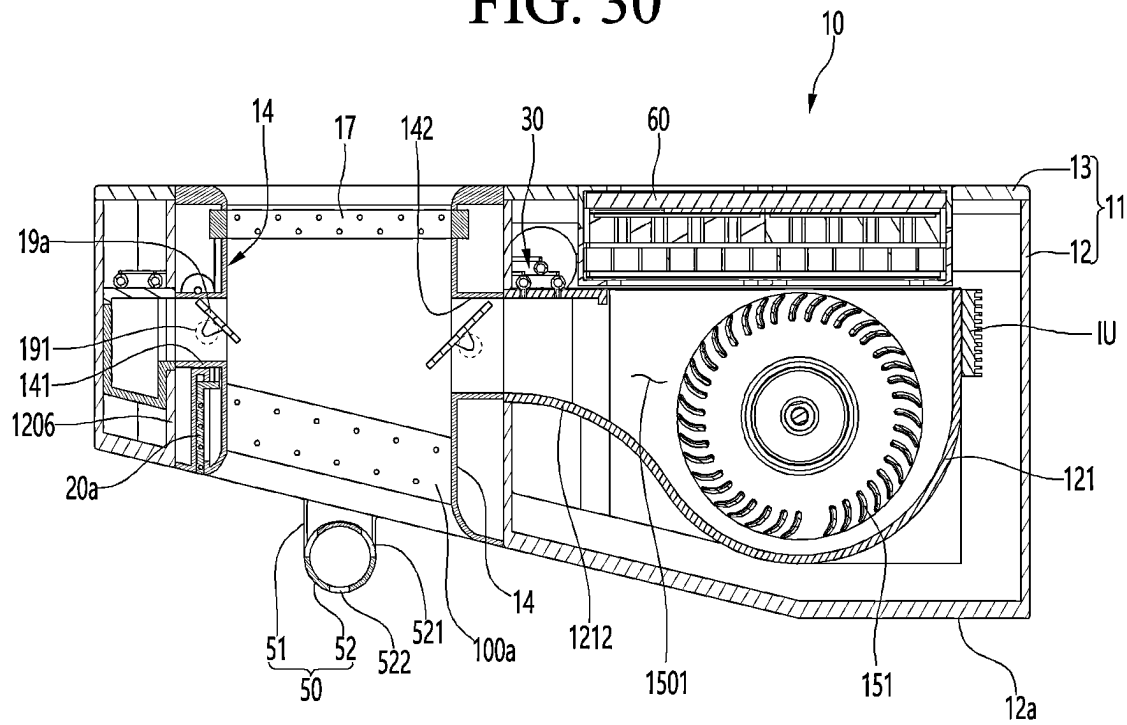


FIG. 31

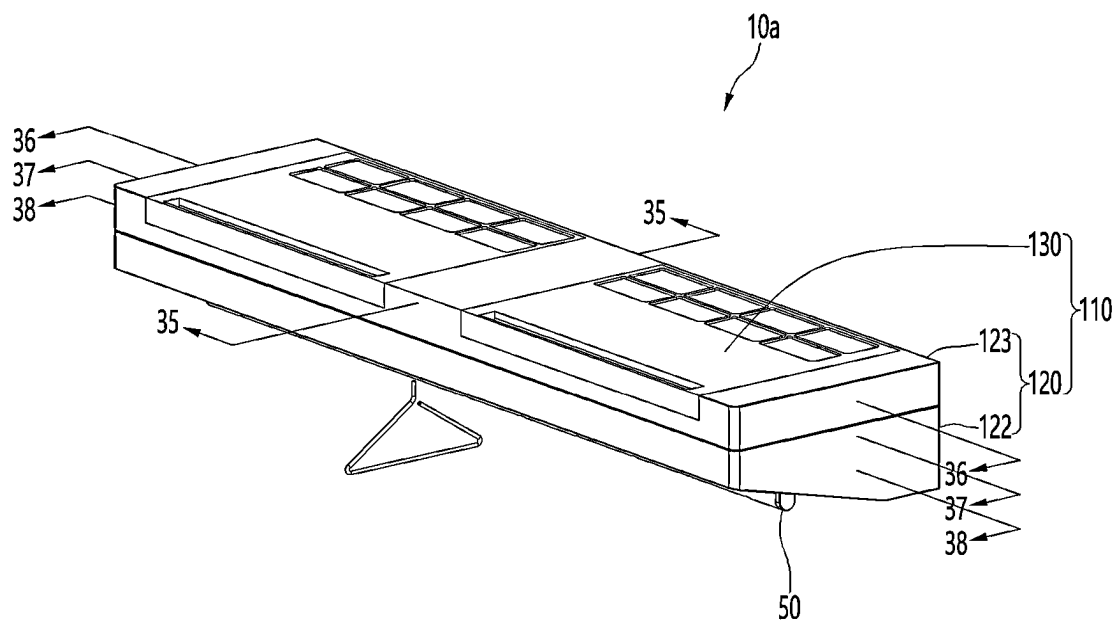


FIG. 32

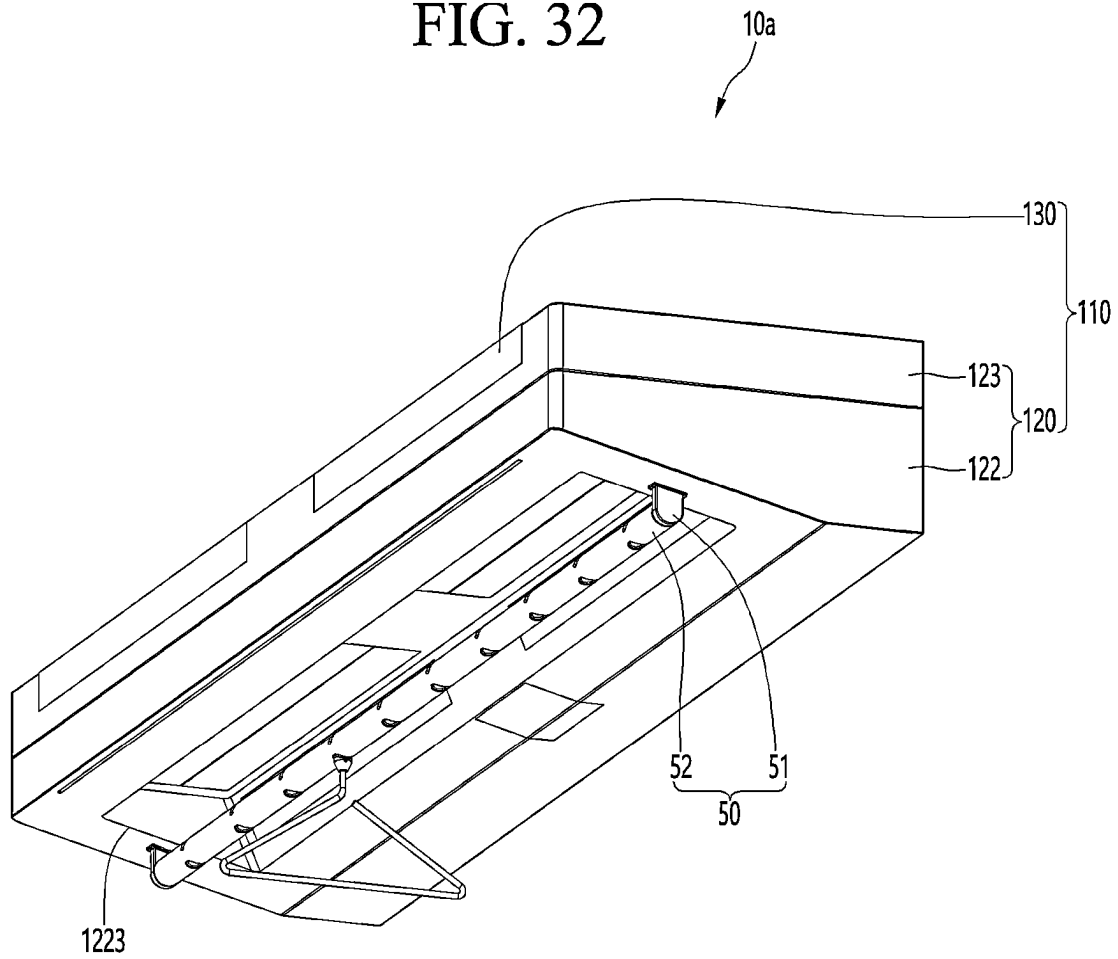


FIG. 33

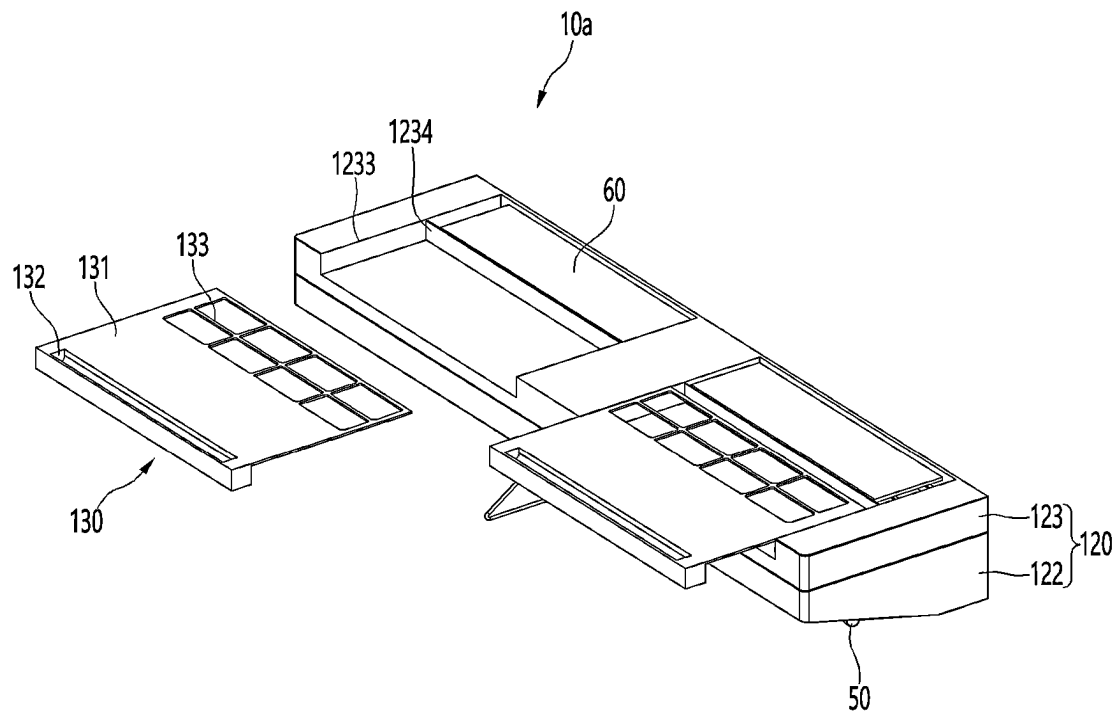


FIG. 34

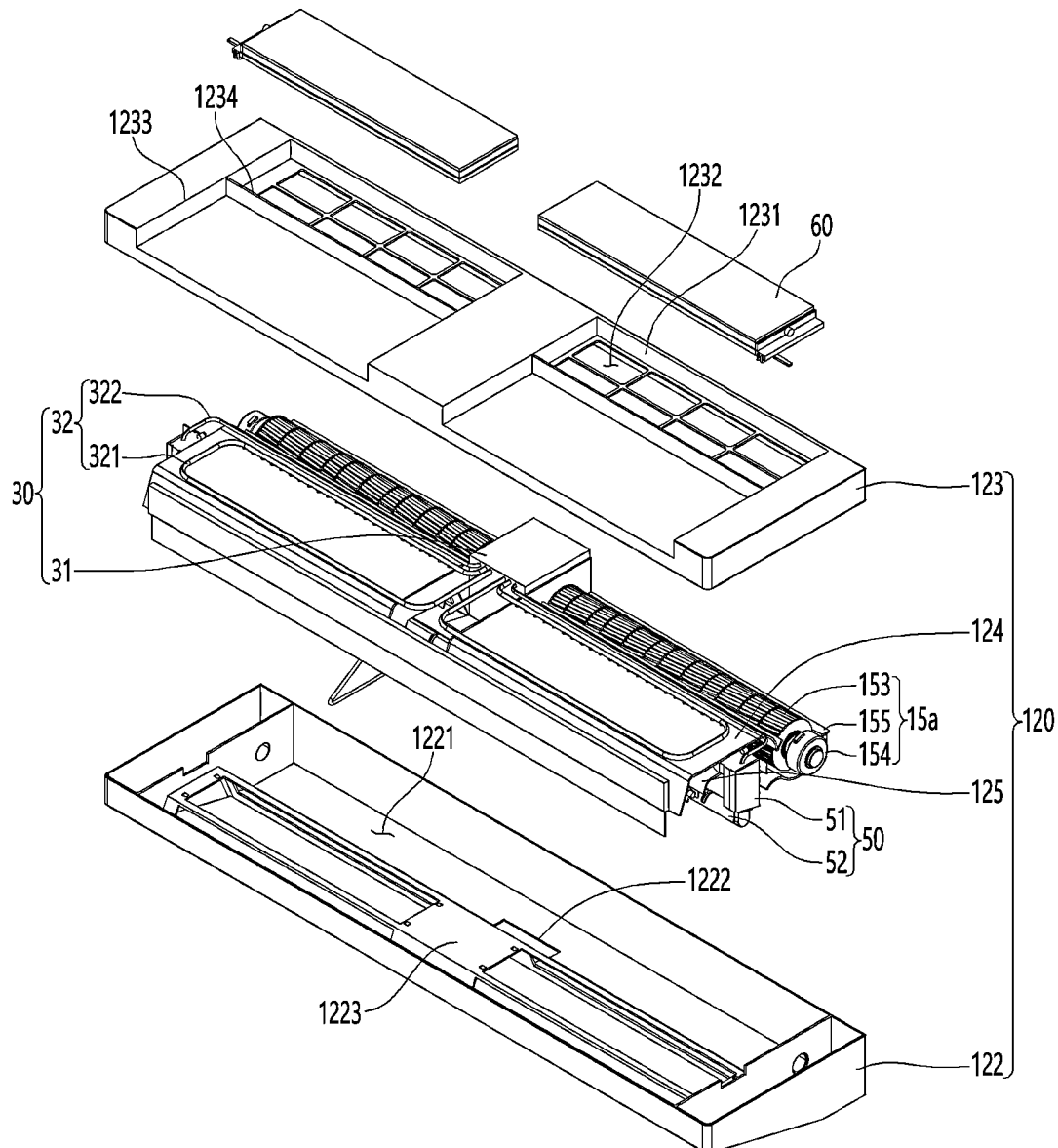
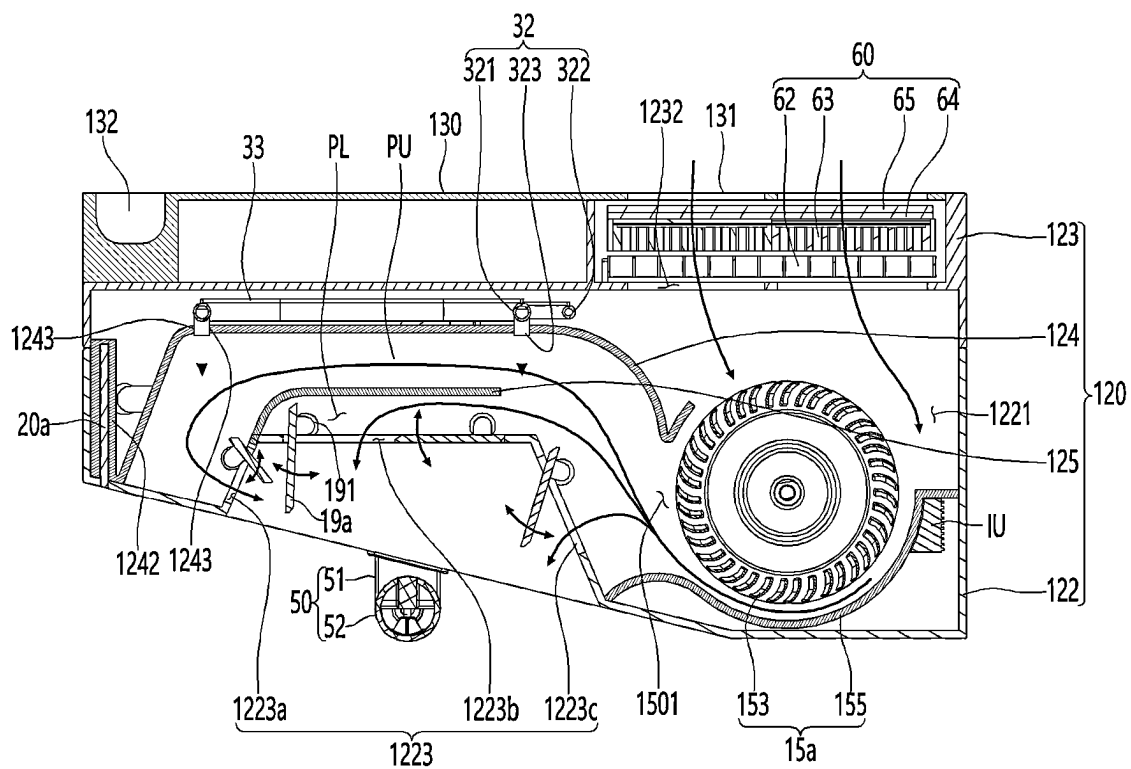


FIG. 35



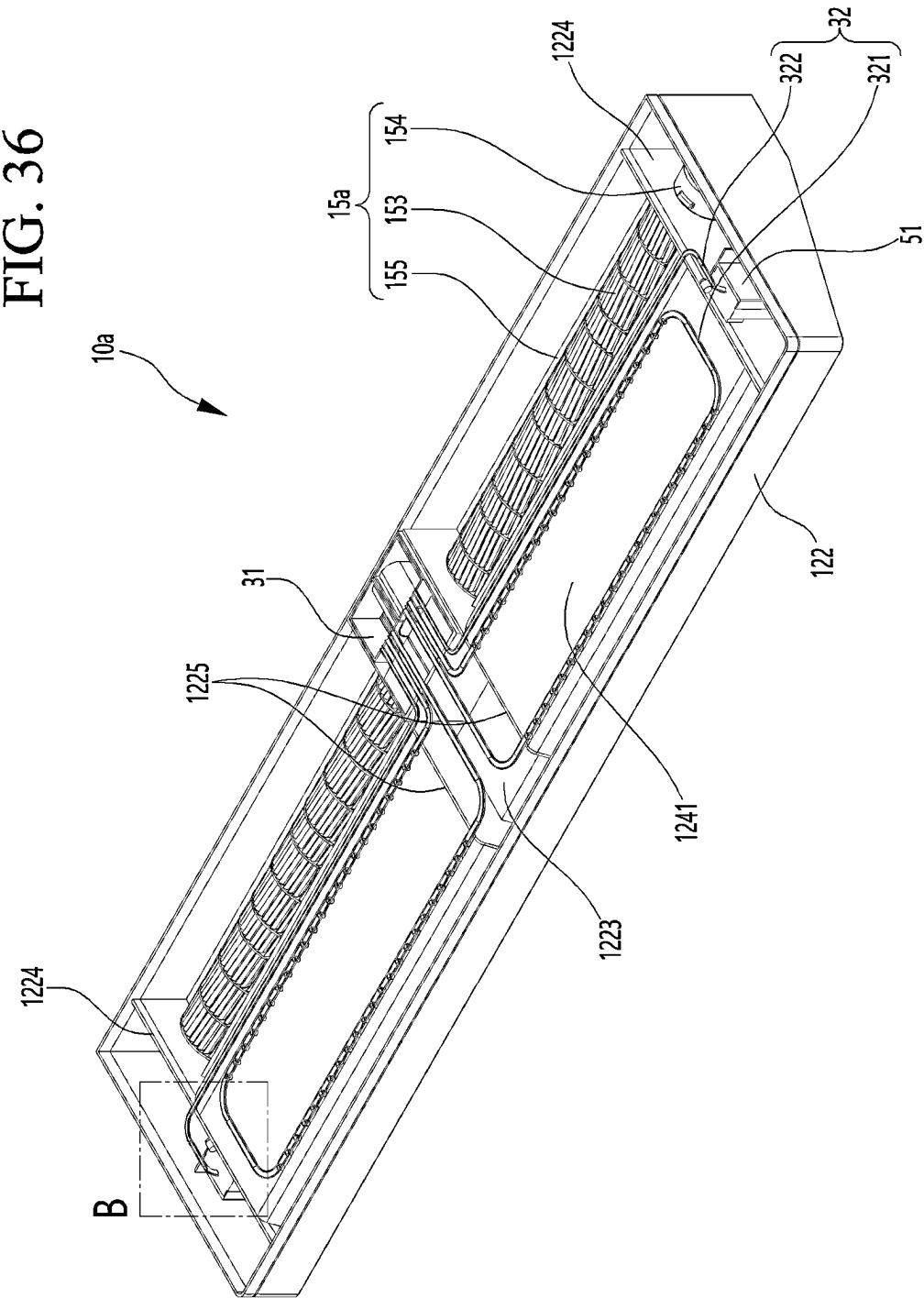


FIG. 38

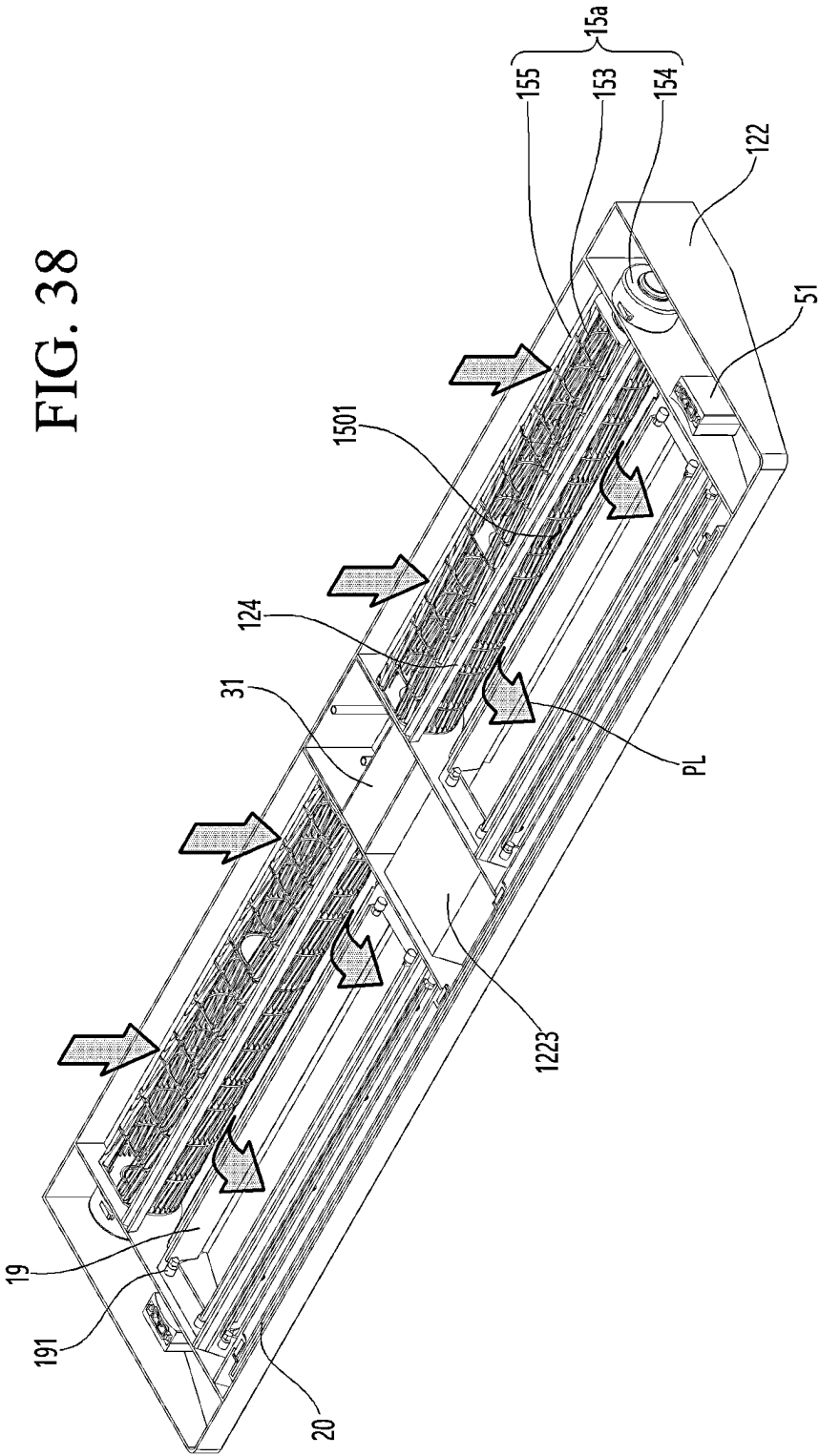


FIG. 39

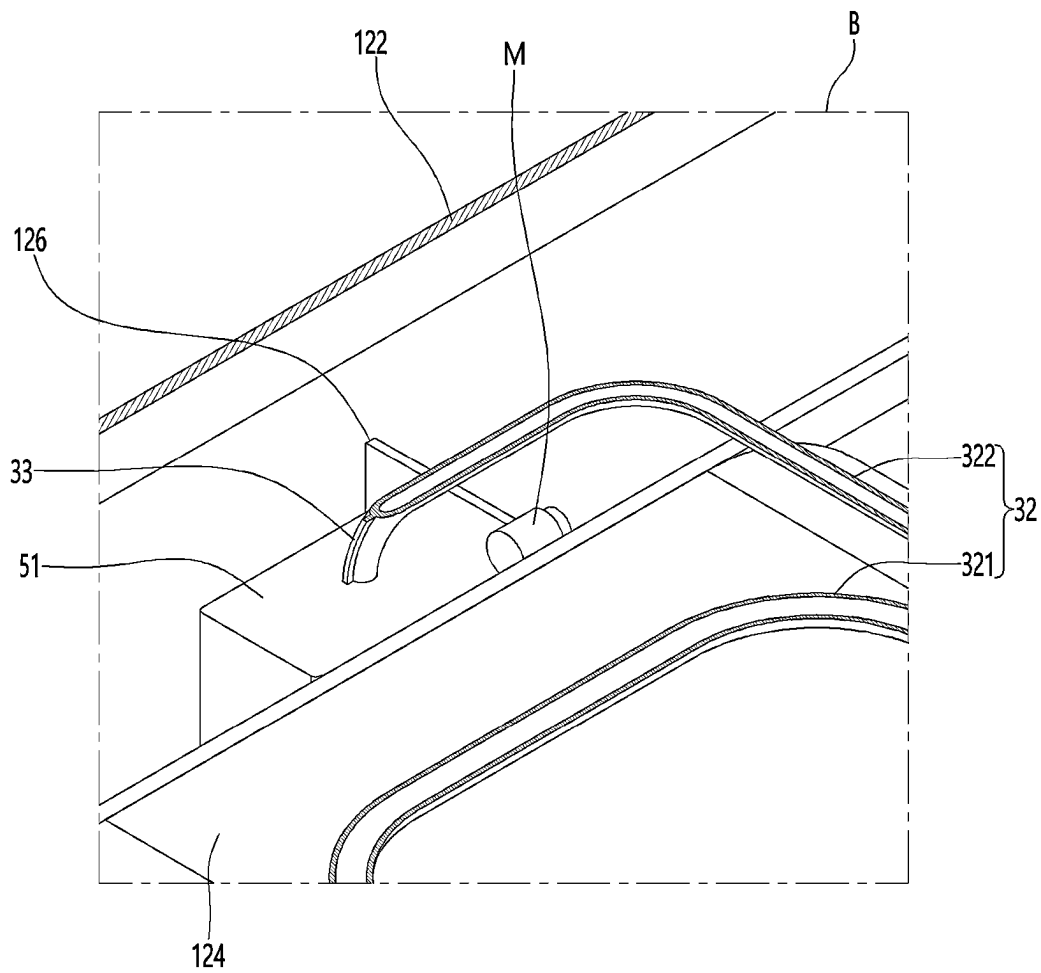


FIG. 40

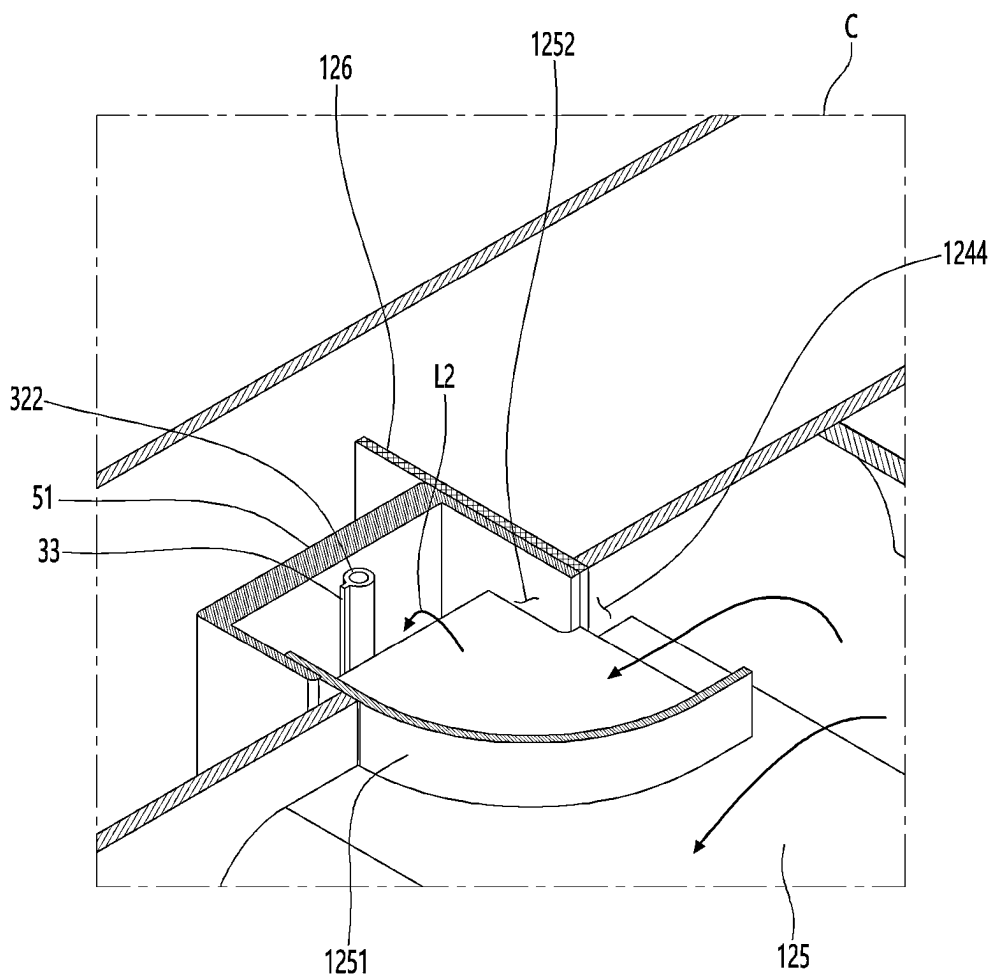


FIG. 41

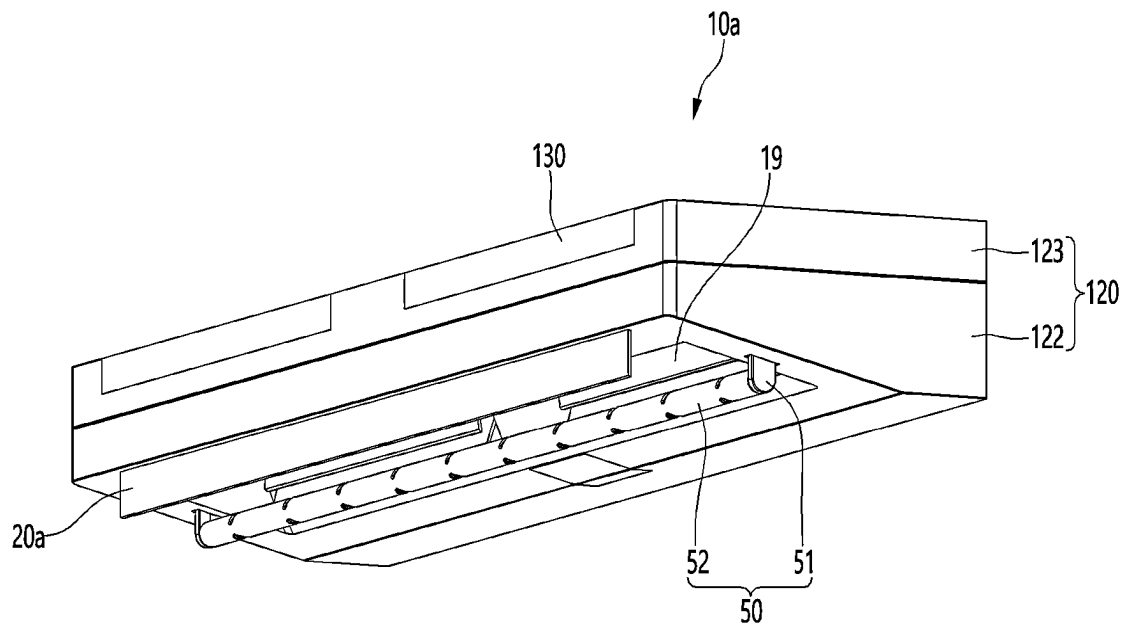


FIG. 42

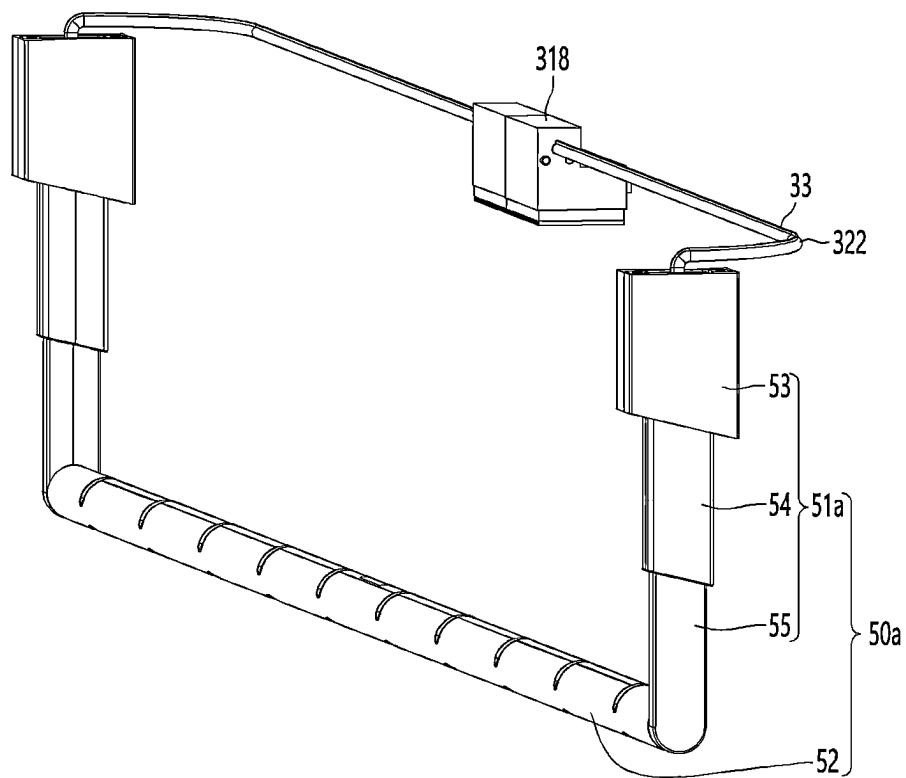


FIG. 43

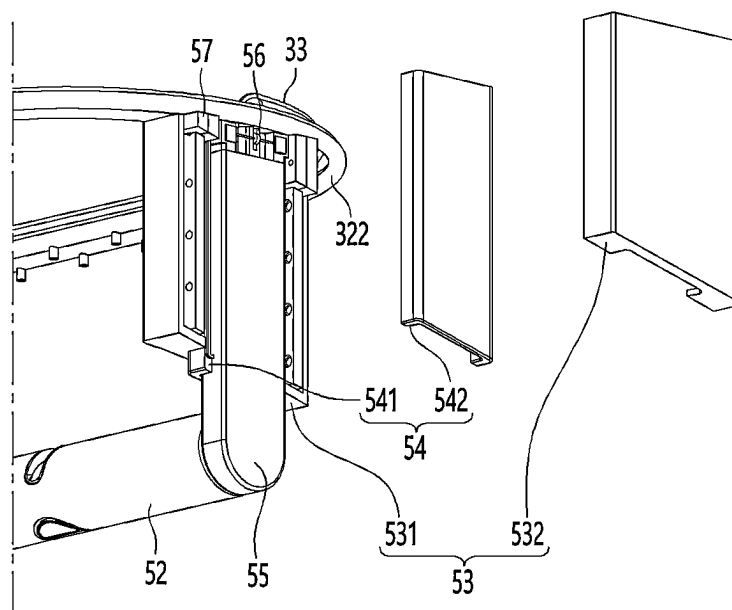


FIG. 44

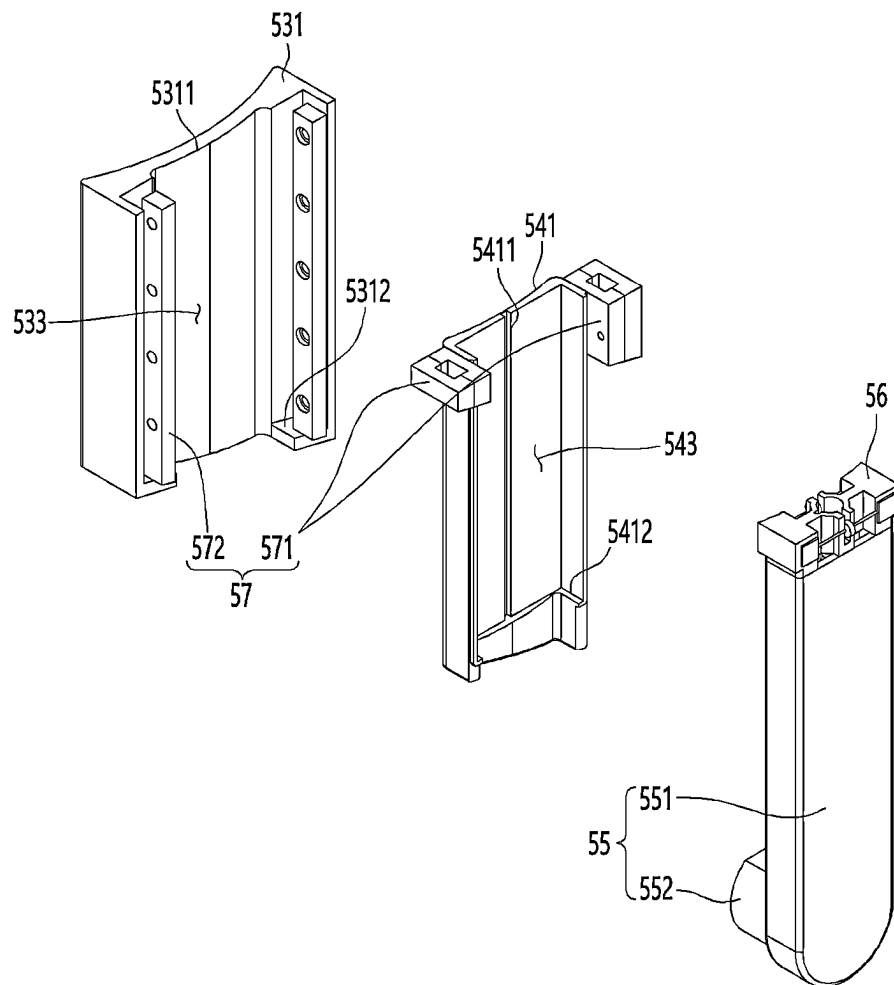


FIG. 45

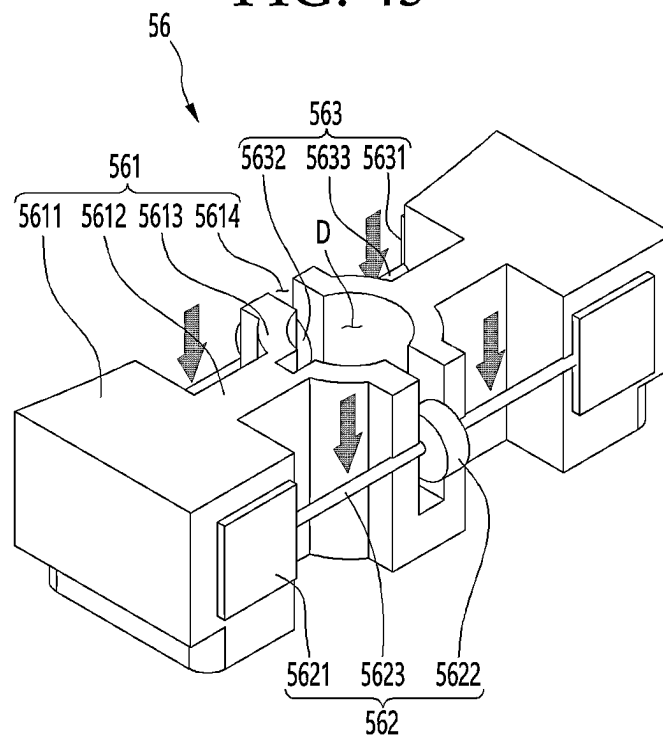


FIG. 46

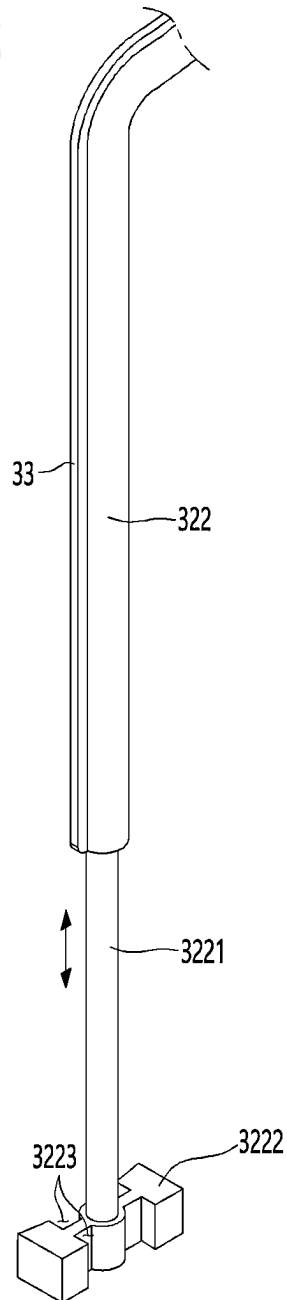


FIG. 47

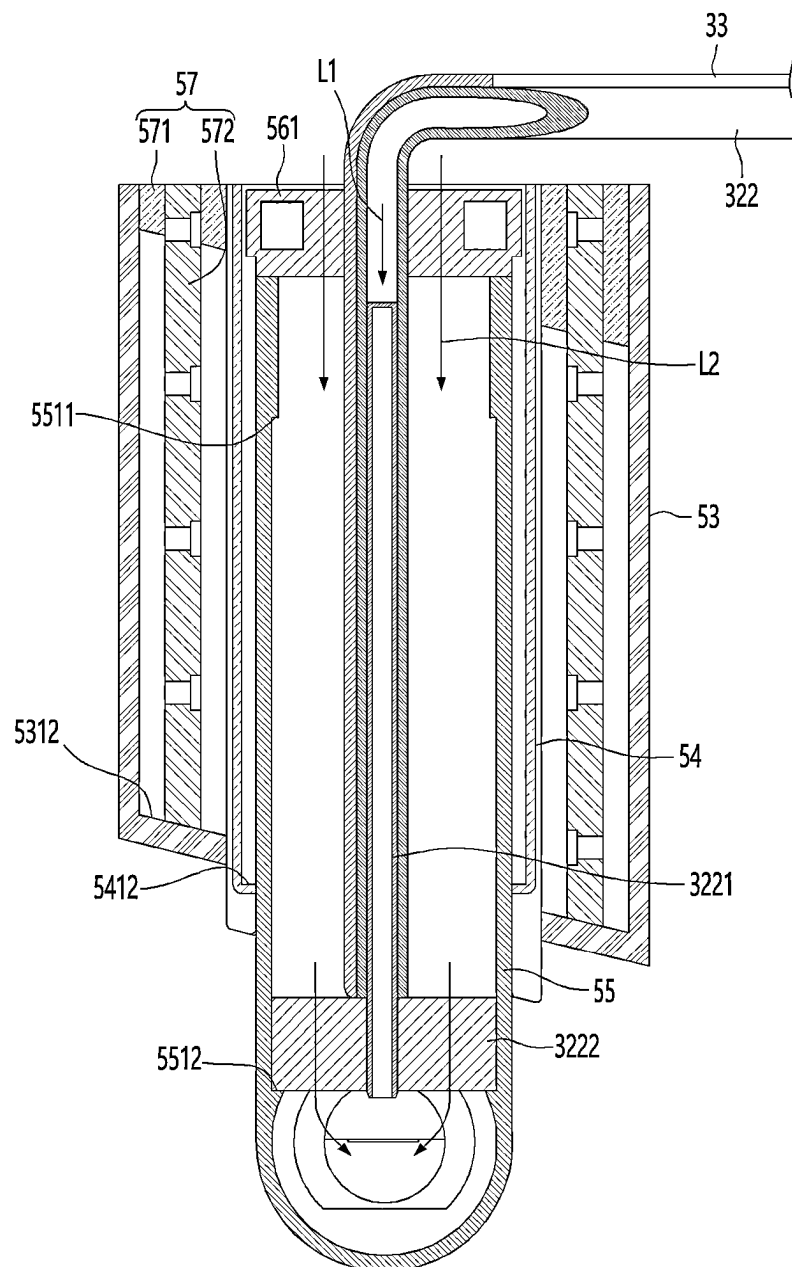


FIG. 48

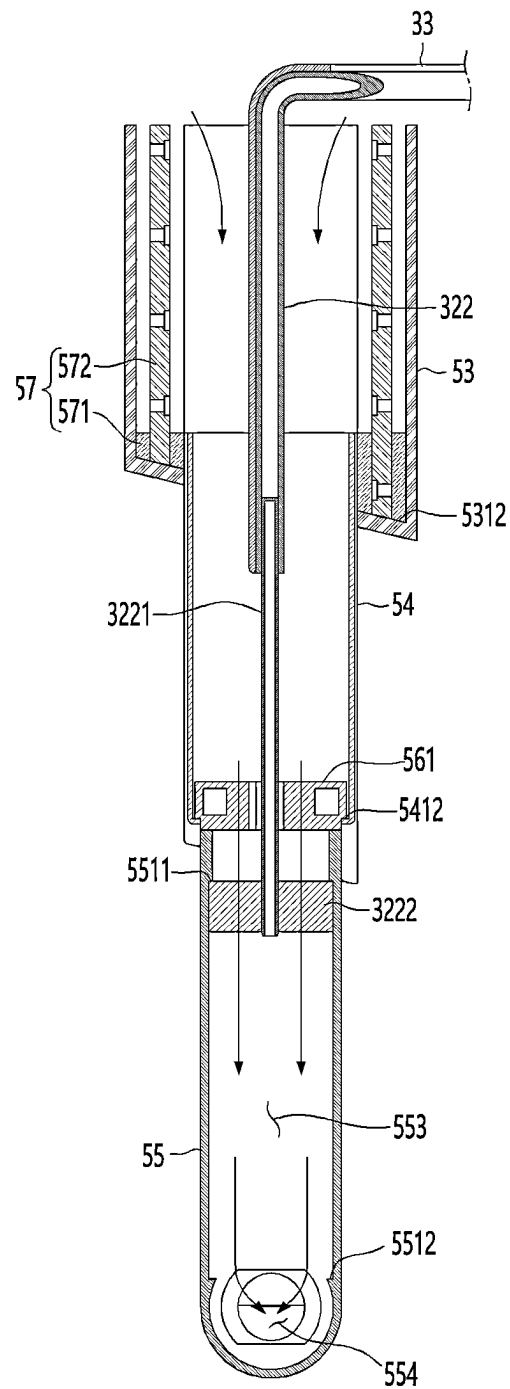


FIG. 49

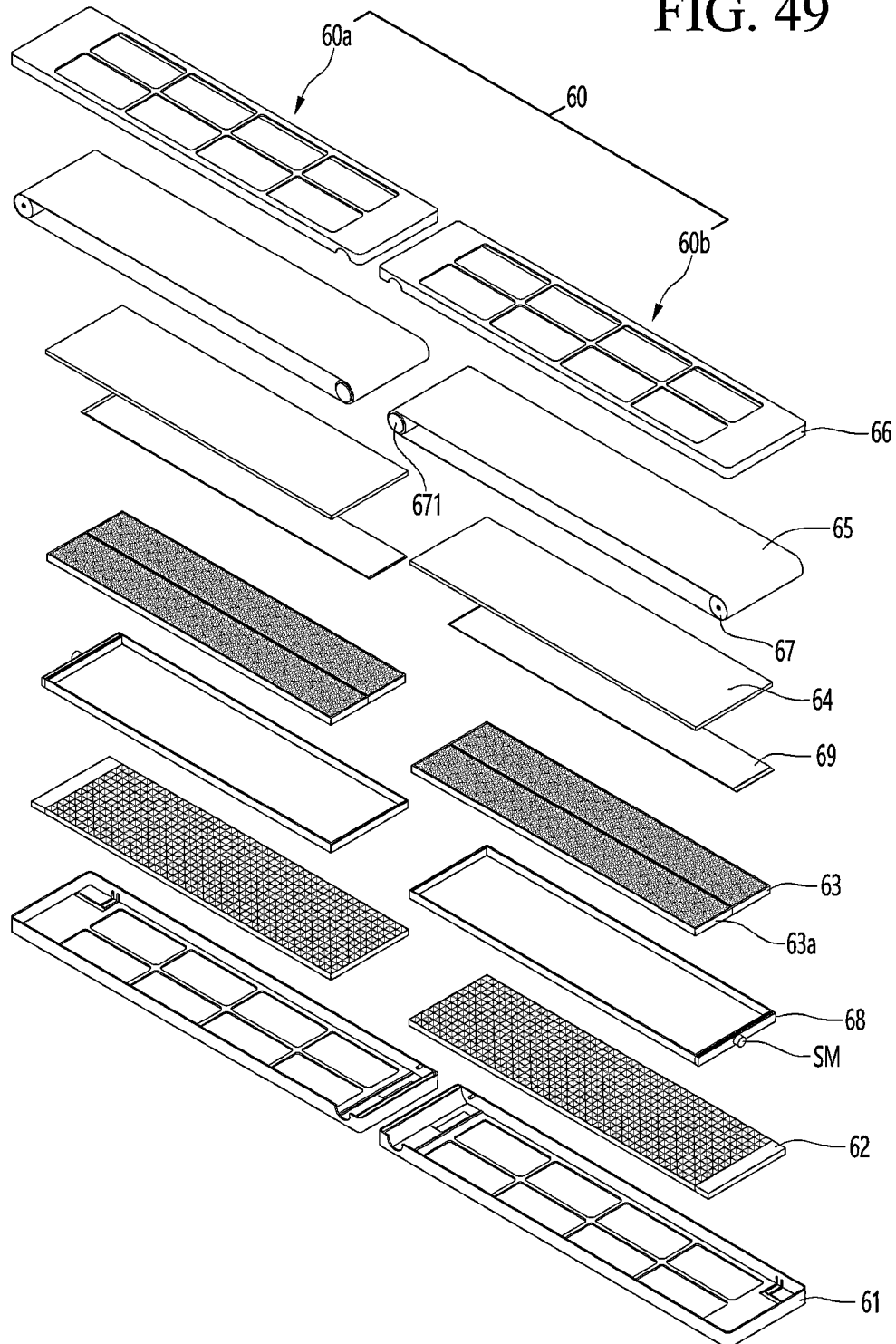


FIG. 50

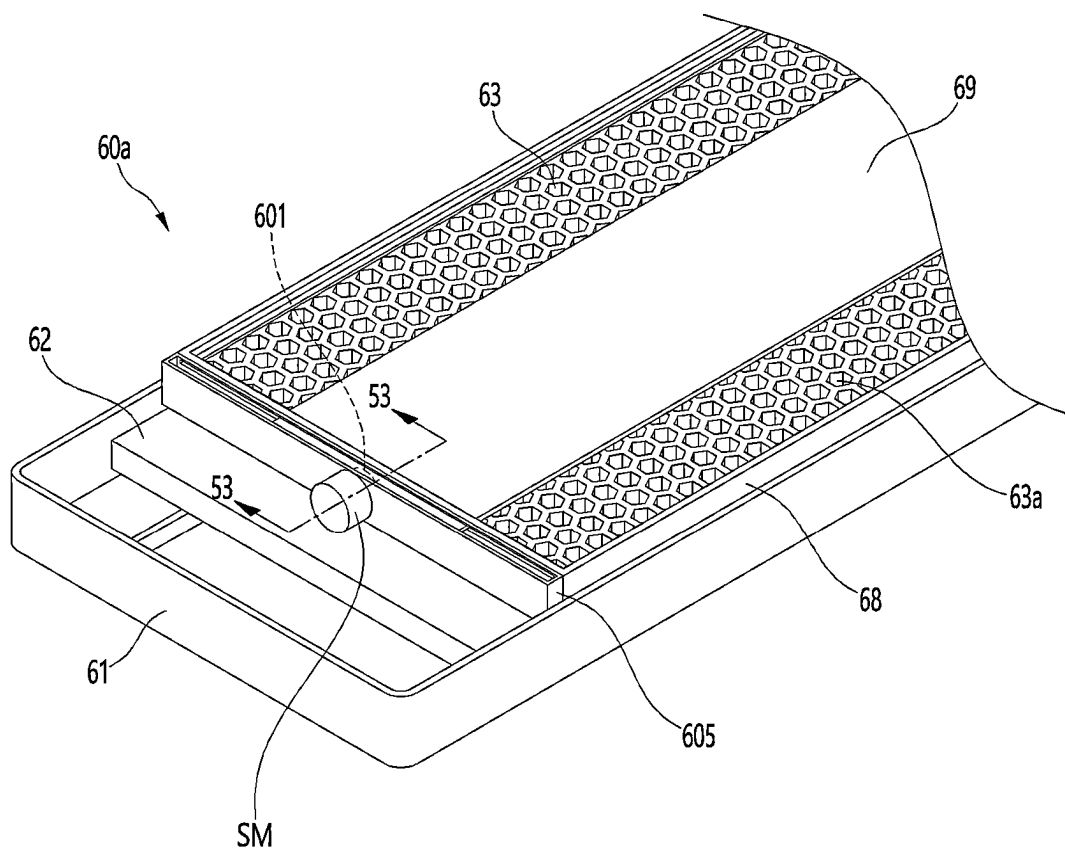


FIG. 51

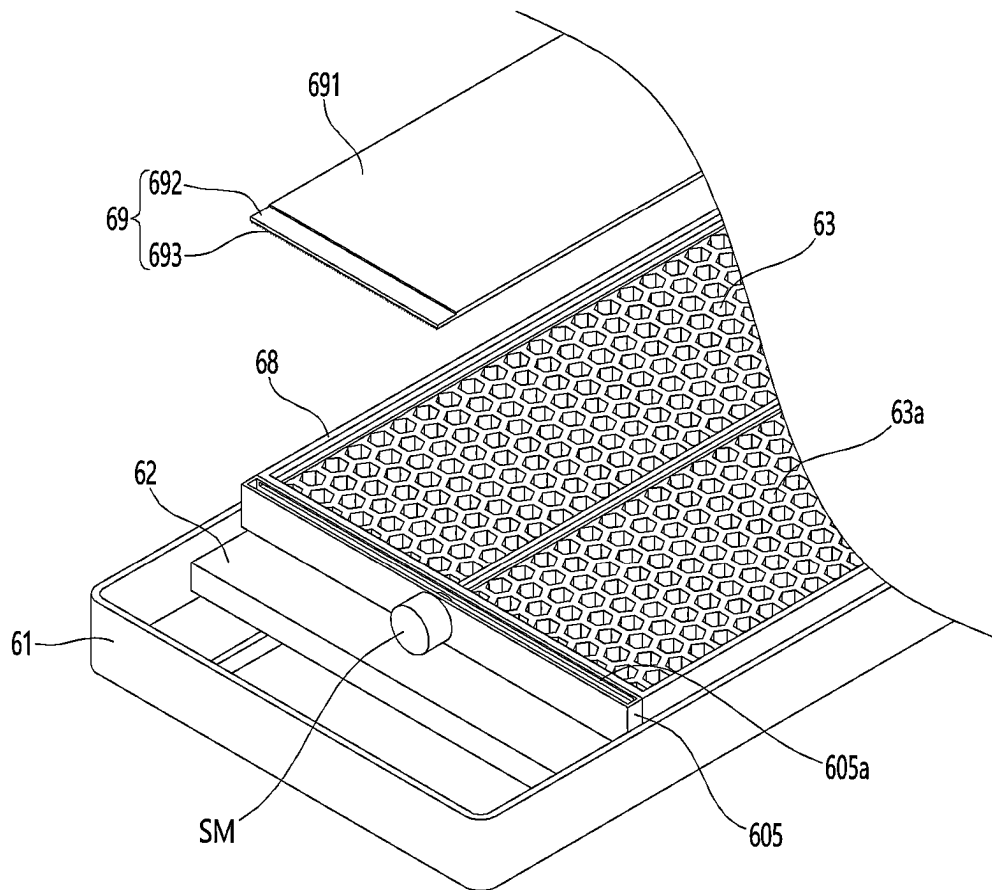


FIG. 52

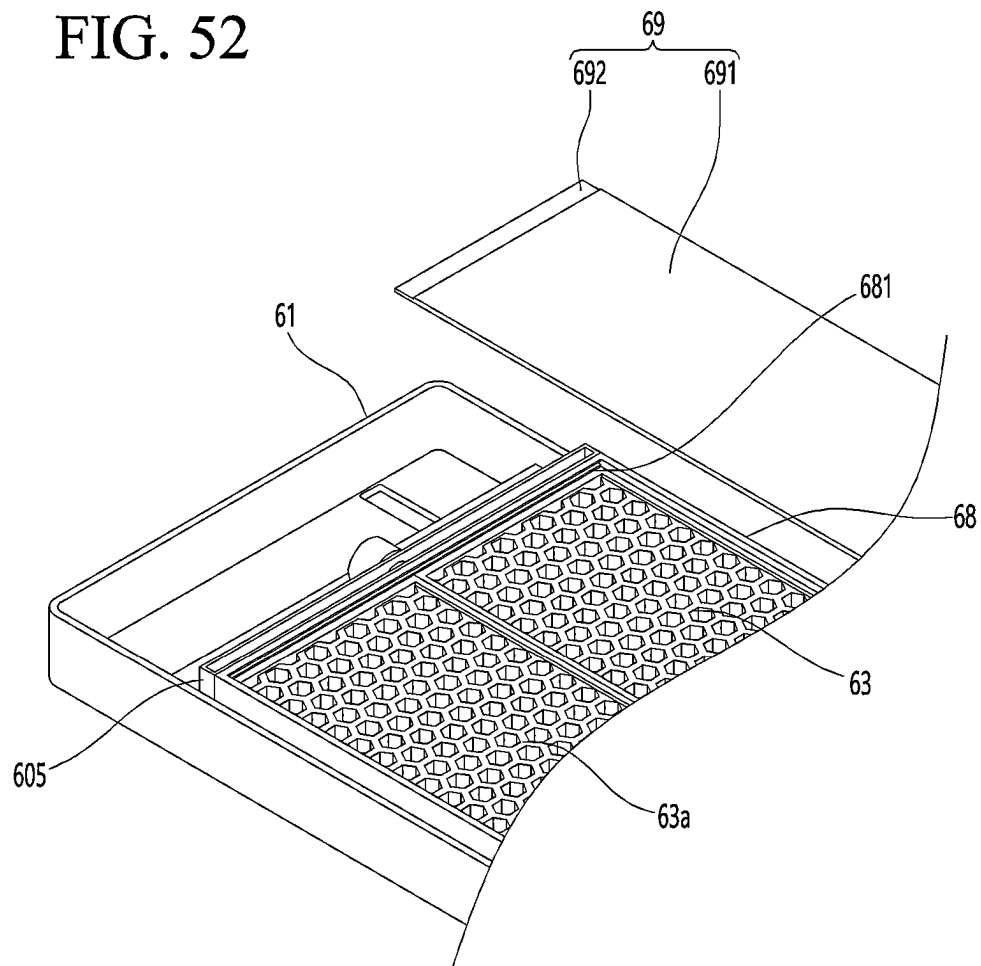


FIG. 53

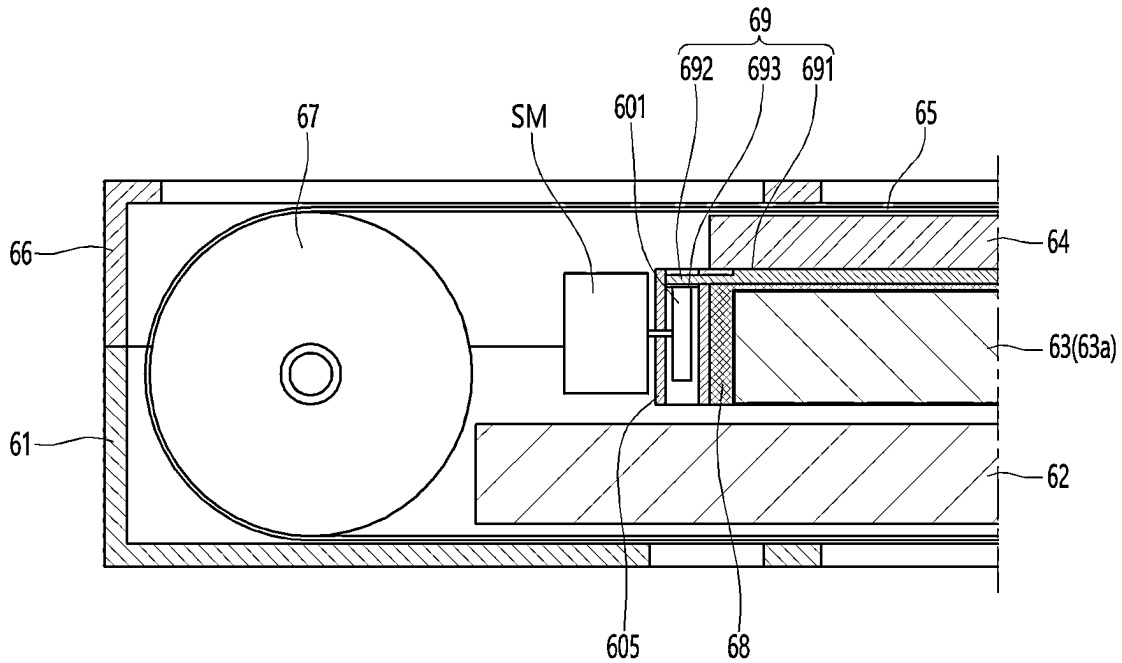


FIG. 54

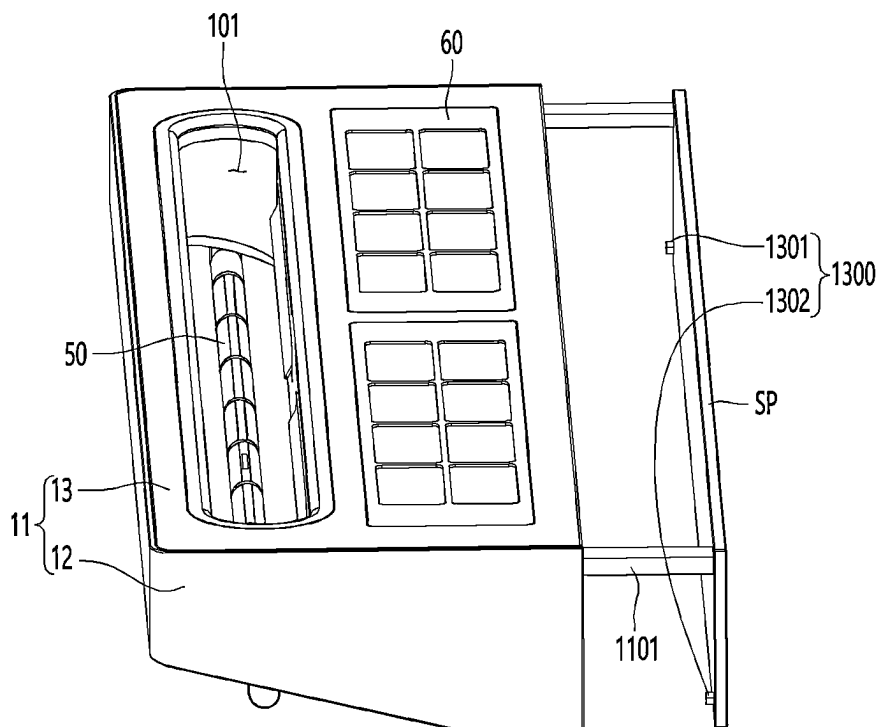


FIG. 56

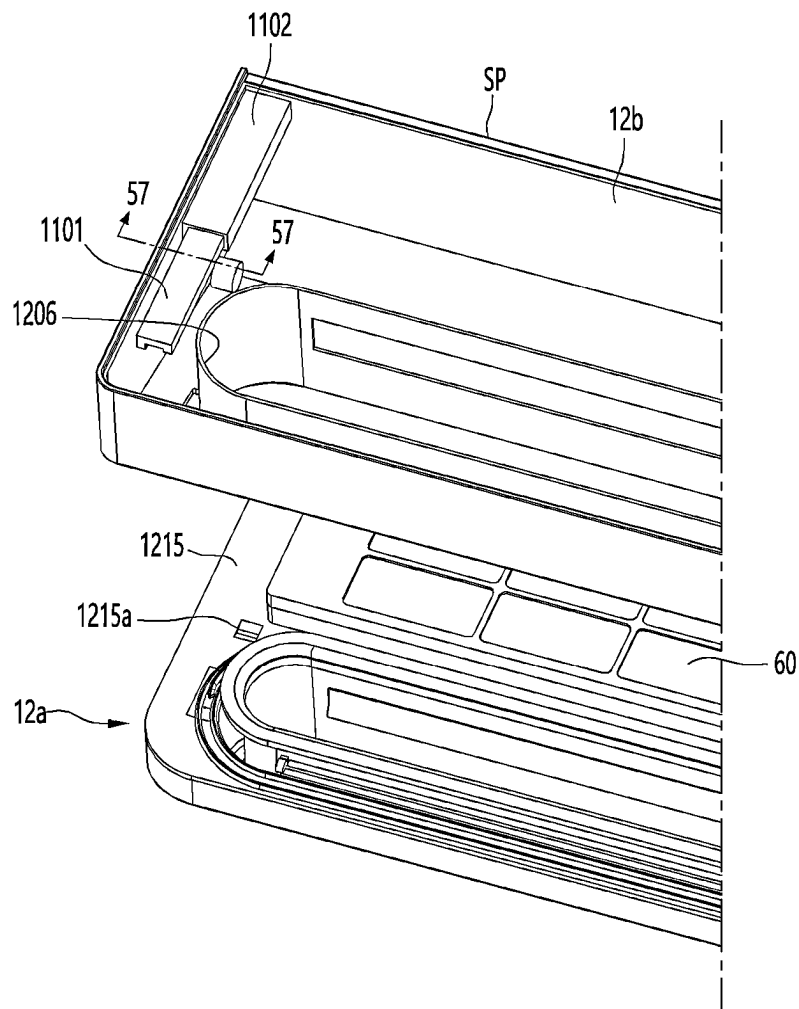


FIG. 57

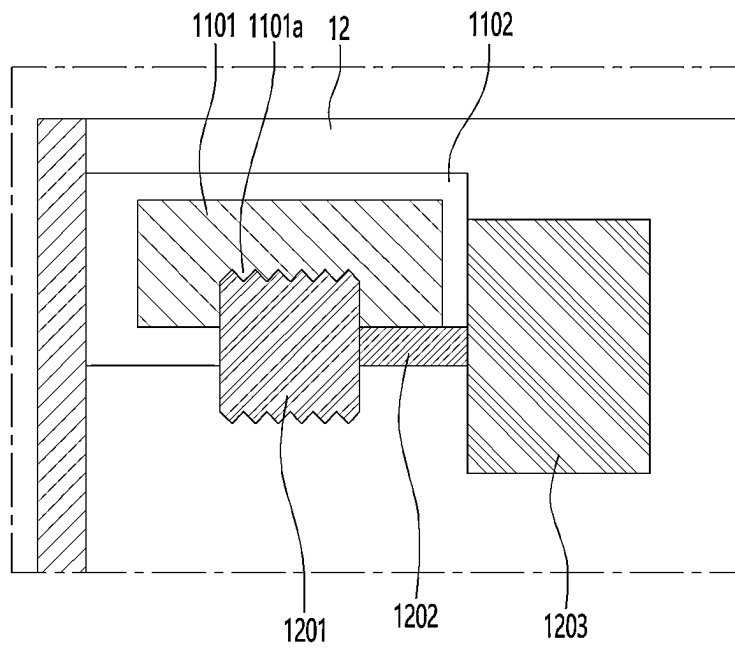


FIG. 58

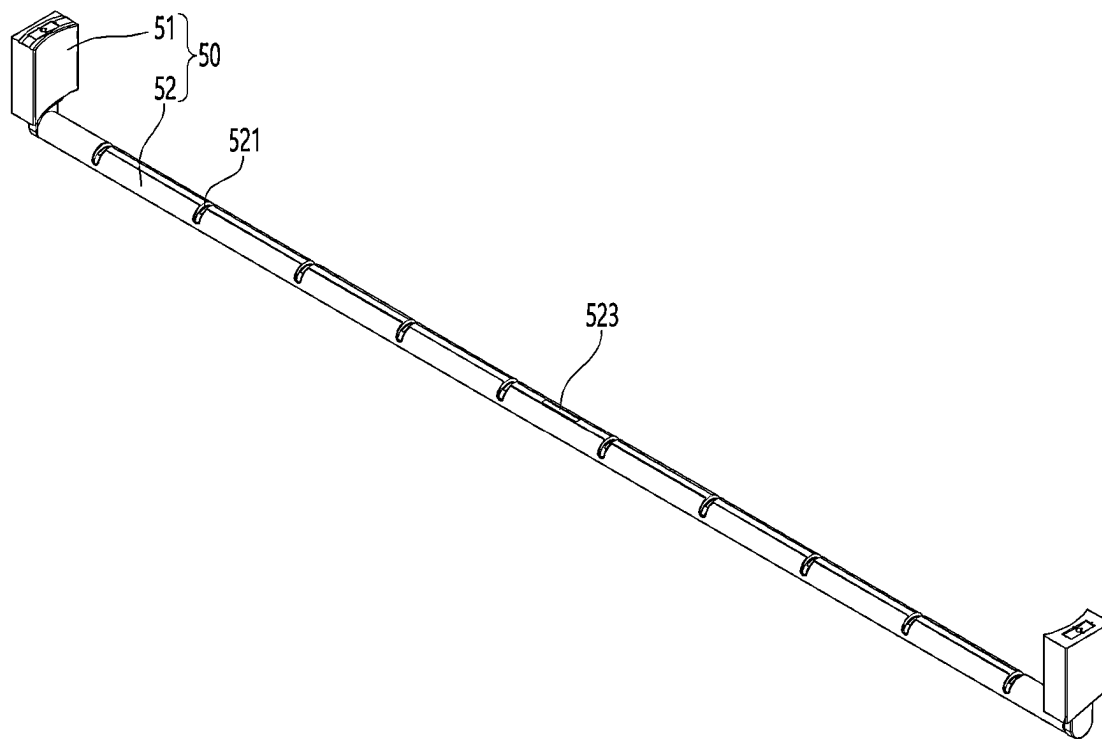


FIG. 59

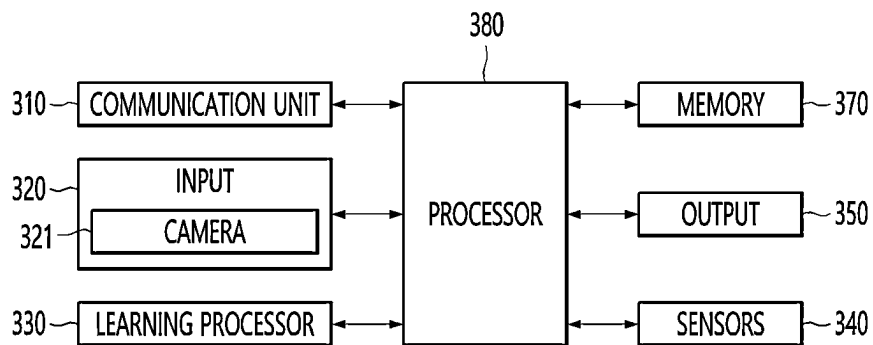


FIG. 60

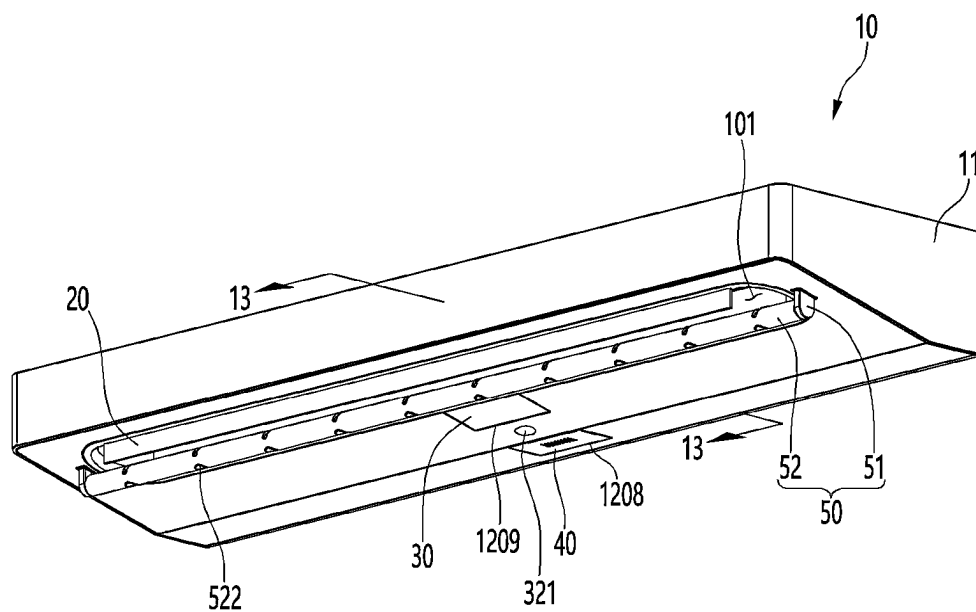


FIG. 61

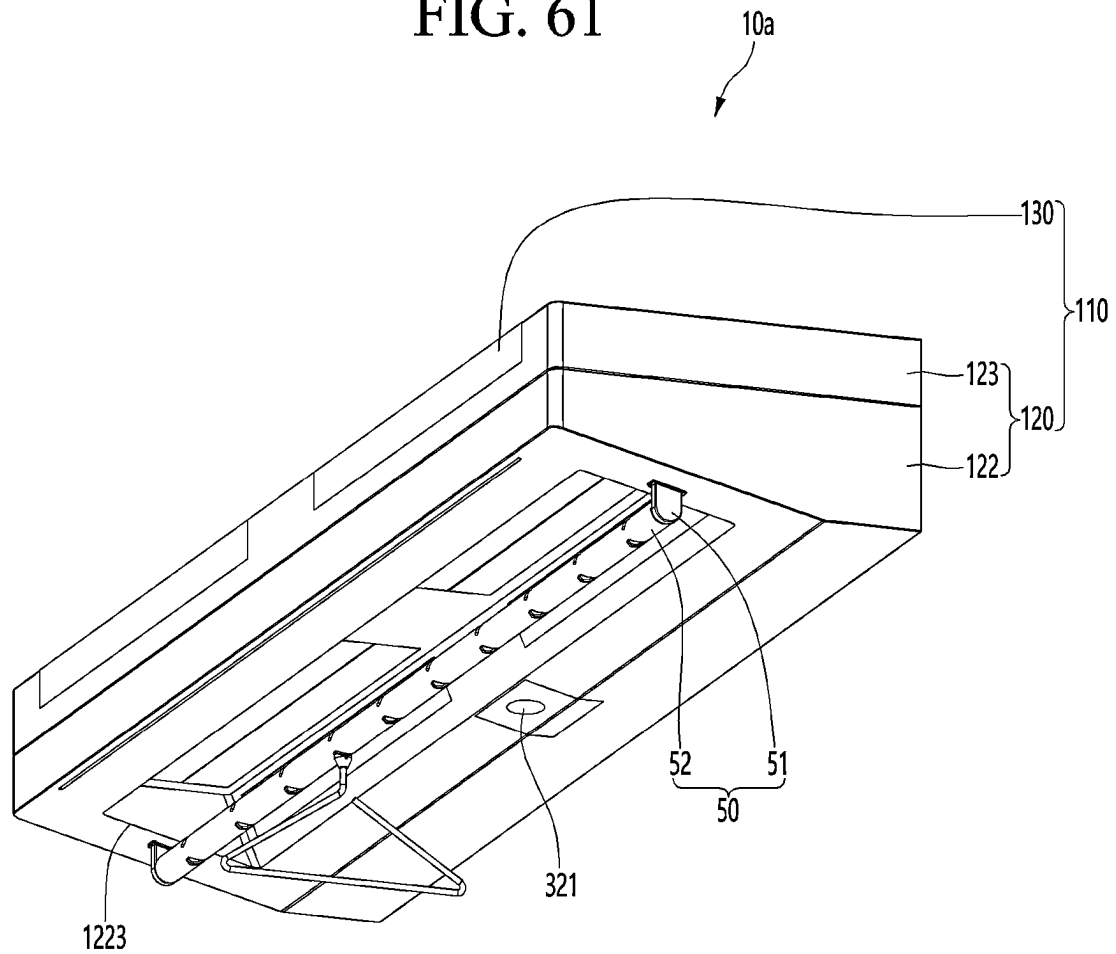


FIG. 62

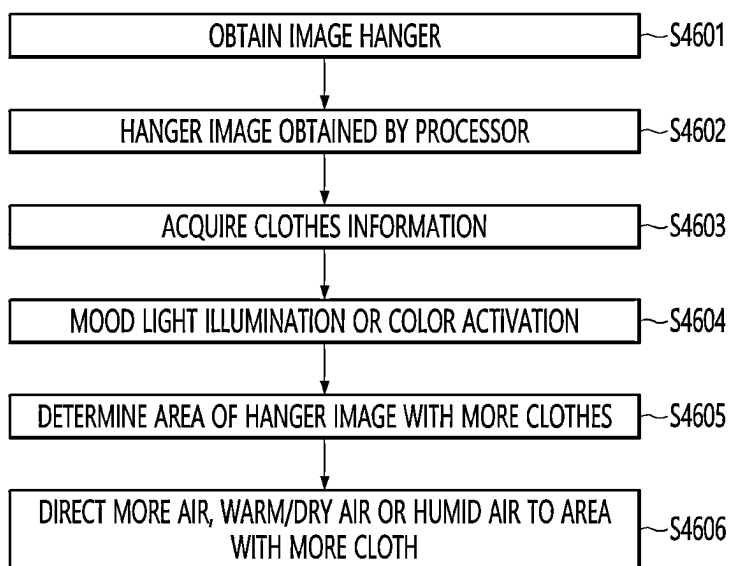


FIG. 63

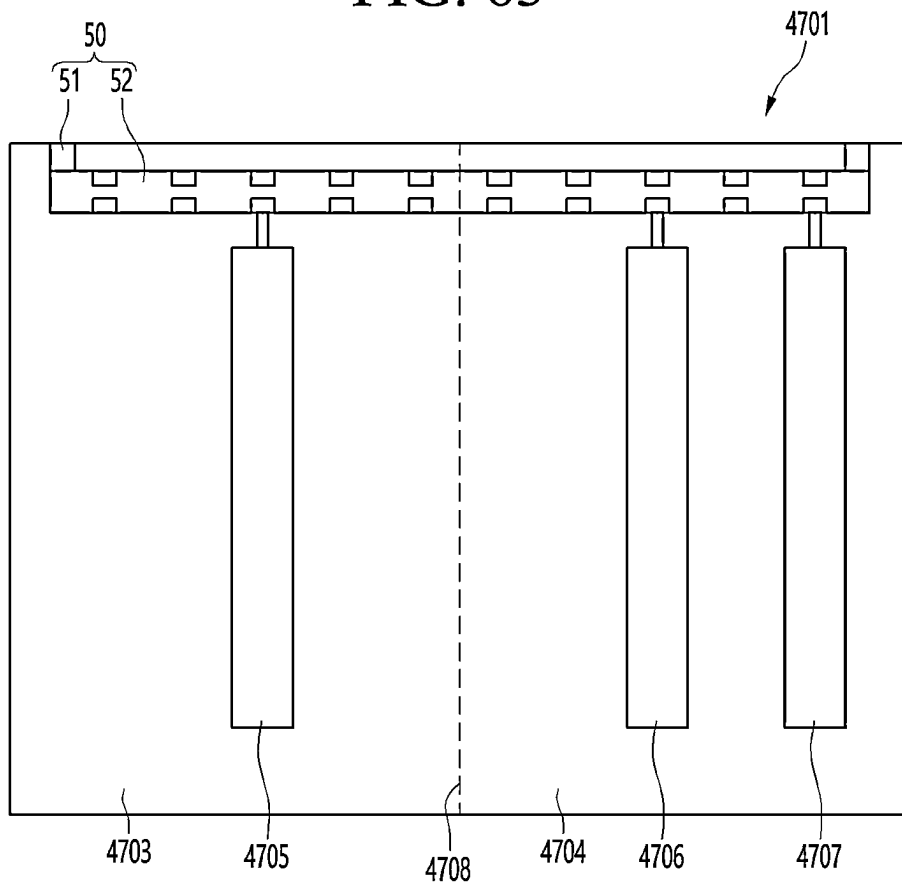


FIG. 64

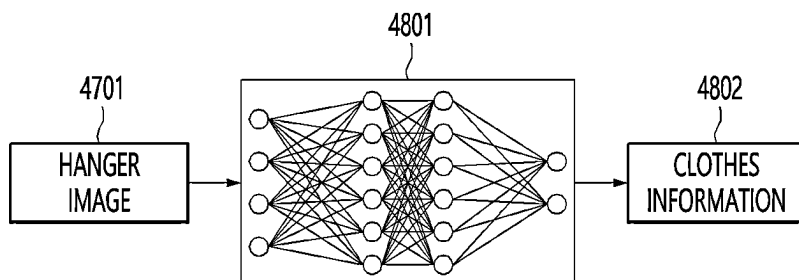


FIG. 65

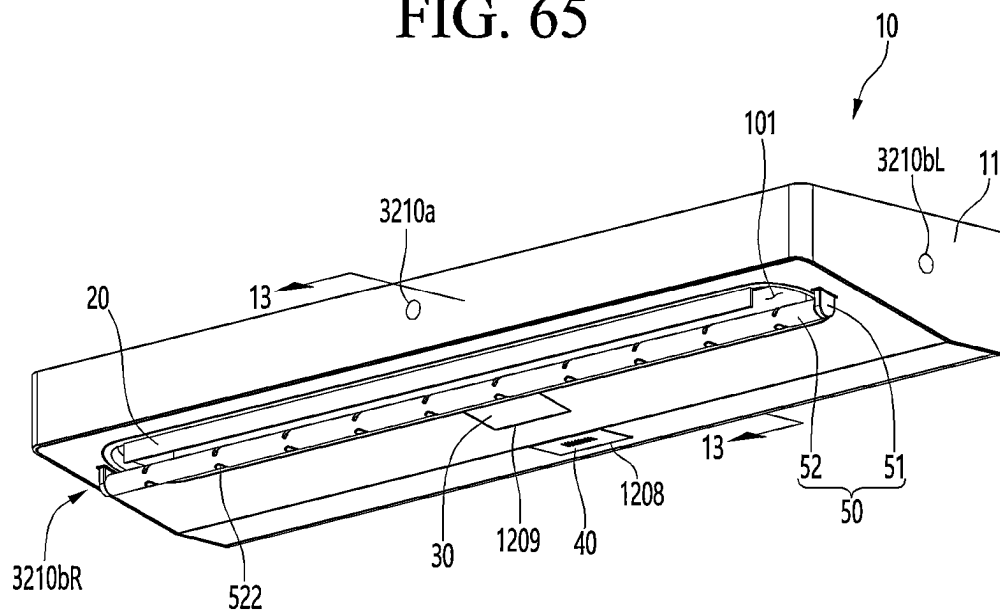


FIG. 66

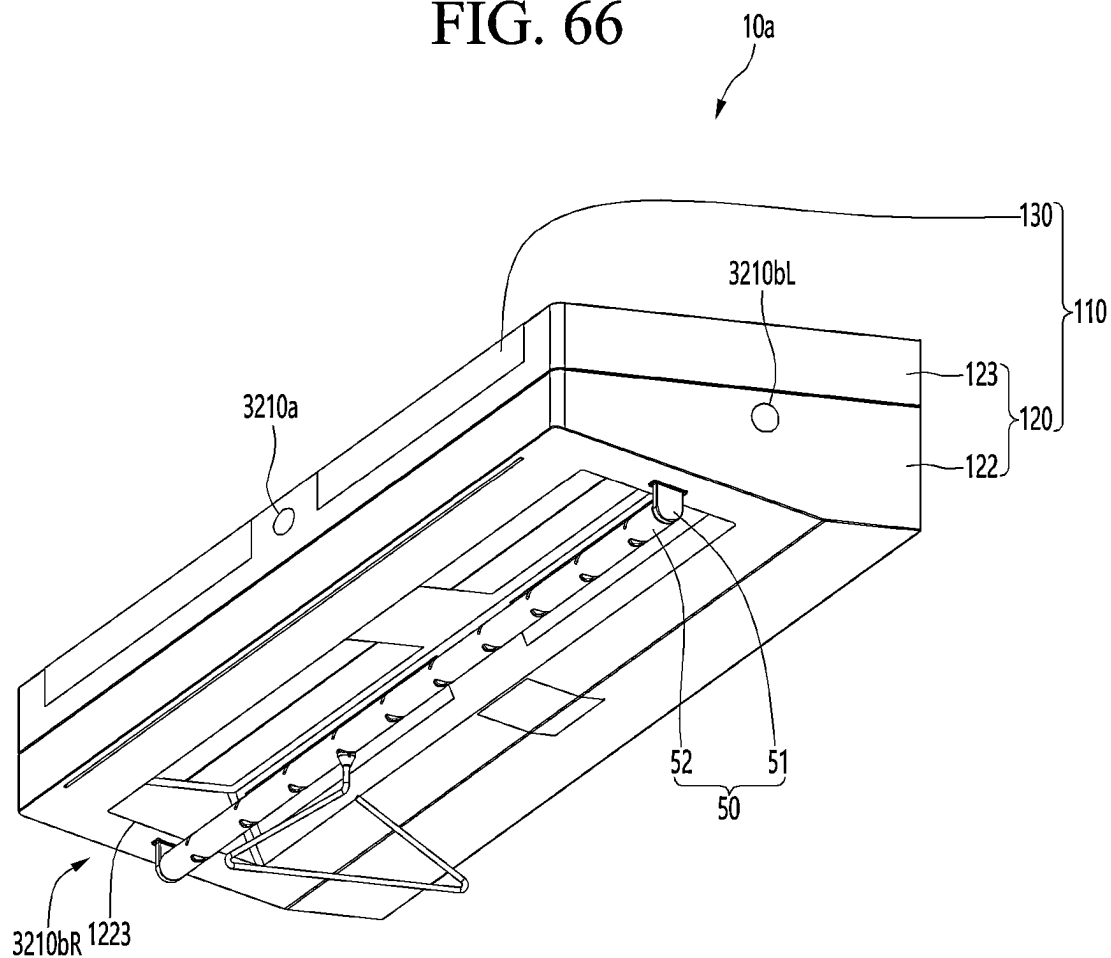


FIG. 67

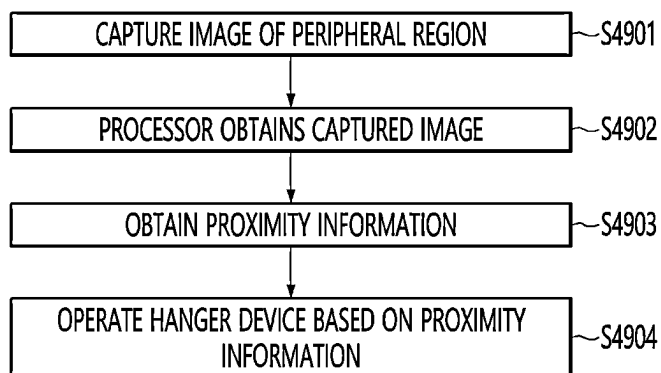


FIG. 68

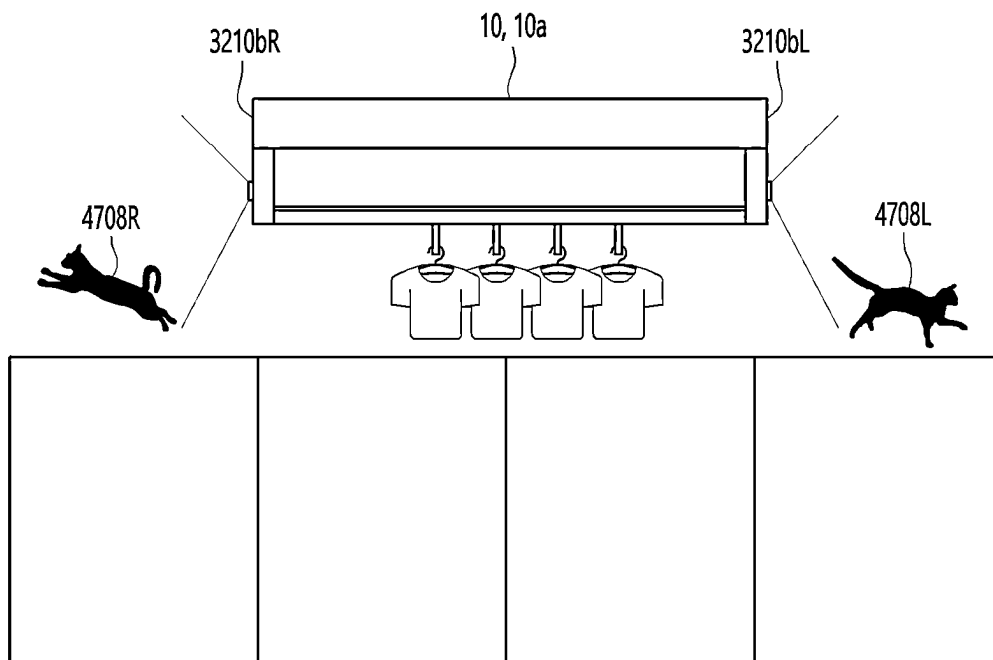


FIG. 69

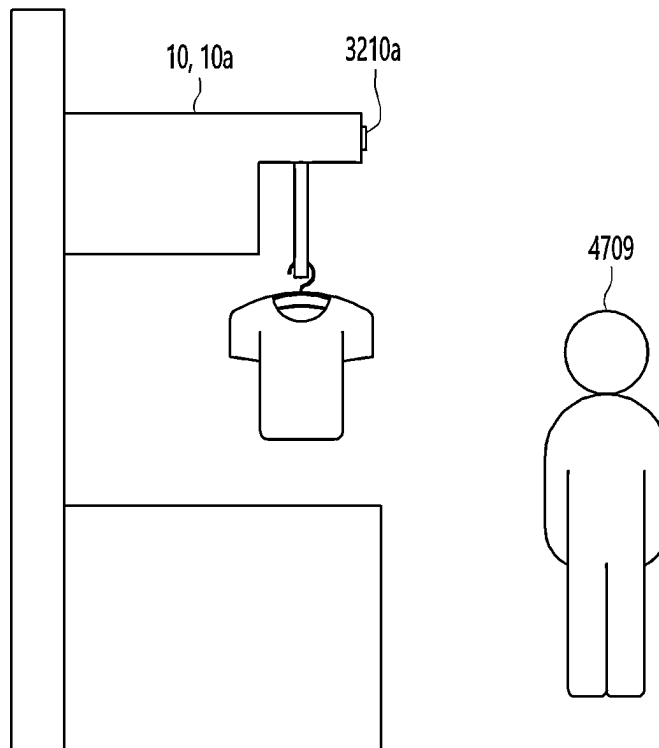
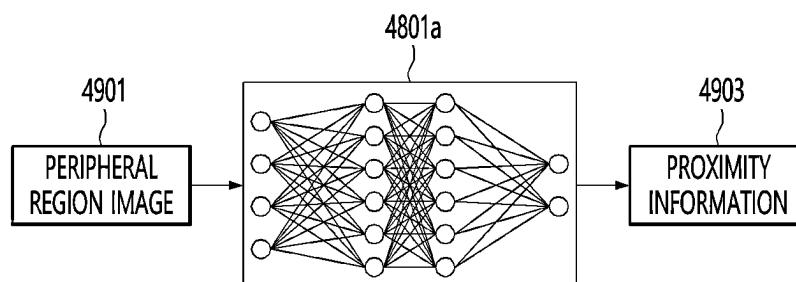


FIG. 70



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HANGER DEVICE

CROSS-REFERENCE TO RELATED APPLICATION(S)

This application claims priority under 35 U.S.C. § 119 to Korean Application Nos. 10-2022-0000952; 10-2022-0000954; 10-2022-0000961; 10-2022-0000965; 10-2022-0000976; 10-2022-0000981; 10-2022-0000985; 10-2022-0000988; 10-2022-0000993; 10-2022-0001153; and 10-2022-0001155; all filed on Jan. 4, 2022, whose entire disclosure(s) is/are hereby incorporated by reference.

BACKGROUND

1. Field

The present disclosure relates to a hanger apparatus.

2. Background

In general, a hanger is used for hanging various items, e.g., clothes, requiring hanging. In addition, a hanger device disclosed in the prior art, Korea Patent Publication No. 2018-0128746 (Dec. 4, 2018), is configured to blow air to dry hung clothes. Recently, there is an increasing consumer demand for a hanger device capable of performing additional functions.

SUMMARY OF THE DISCLOSURE

The present disclosure is directed to a hanger device, which may comprise: a main body having a suction unit configured to suction air and a discharge unit through which the suctioned air is discharged; a fan module provided inside the main body; a filtering module for removing foreign substances contained in the air suctioned in by the fan module; a discharge duct forming a guide passage for guiding the air passing through the filtering module to the discharge unit; and a hanging unit provided at a lower portion of the main body.

The hanging unit may include: a pair of supports extending from the bottom of the main body, the length of the supports being adjusted in the vertical direction; a hanging bar connecting the ends of the pair of supports; and a lifting unit configured to change the length of the pair of supports. The pair of supports communicates with the guide passage, and the hanging bar communicates with the pair of supports. Each of the pair of supports may include a fixed arm fixed to the body, at least one intermediate arm configured to move up and down while connected to the fixed arm, and a movable arm capable of moving up and down while connected to the intermediate arm.

The hanging unit may further include a first elevating unit for moving the movable arm up and down along the intermediate arm, and a second elevating unit for elevating the intermediate arm along the fixed arm. At least one of the first elevating unit or the second elevating unit may include a drive means which may be achieved by a rack engaged with a pinion, and a motor for rotating the pinion. The drive means may be also a linear motor actuators having a lead screw to achieve movement of the intermediate arm.

The clothes hanger device according to the present disclosure may further includes a steam supply device for supplying steam into the guide passage. The steam supply device may include a steam generator, a main supply pipe extending from the steam generator, at least a portion of

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which extends along the guide passage, a sub-supply pipe extending into the steam generator and extending into the pair of supports; and an extension pipe movable in the vertical direction while connected to the sub-supply pipe. A coupler is mounted at the lower end of the extension tube and is accommodated in the moving arm such that the coupler moves as one body with the moving arm.

The filtering module may include a dust filter configured to remove dust contained in the air sucked in through the suction unit, a dehumidifying filter for absorbing moisture contained in the air that has passed through the dust filter, a deodorizing filter for purifying or deodorizing the air that has passed through the dust filter, and at least one of a heater for heating the air that has passed through the dust filter.

The hanger device according to at least one of the embodiments disclosed herein may provide at least one of the following effects.

The clothes or items hung or provided on the hanging unit may be dried by circulating the indoor air, and further, the air circulation may decrease the drying time of the clothes. The drying time may be further shortened by a heater provided in the filtering module to generate warm air.

By allowing the air sucked into the hanger device by the fan module to pass through the filtering module, ambient indoor air of a room where the hanger device is situated may be purified.

By allowing the air sucked through the hanger device by the fan module to pass through the dehumidifying filter, indoor humidity may be controlled. For example, the humidity of a room where the hanger device is situated may be reduced based on the operation of the fan module during a humid environment or a season having a relatively higher humidity, e.g., summer.

By allowing the steam generated from the steam supply device provided inside the hanger device to be sprayed or propagated toward the clothes hung on the hanging unit, items with wrinkles may be straightened. Further, the generated steam may be used to increase the humidity of indoor air where there may be a lack of humidity indoors, e.g., when heaters are operated indoors during winter months, the indoor air is dry.

When the sterilization module installed around the outlet of the hanger device is turned on, harmful viruses contained in the air passing through the filtering module are sterilized, and indoor air quality is increased.

Because the height of the hanging bar may be adjusted according to the length of the item to be hung on the hanger device and/or the height of the person who hangs the item, user convenience may be increased.

When a shielding plate is provided to cover at least one of a dehumidifying filter or a deodorizing filter, dehumidification and deodorizing functions may be selectively performed or simultaneously performed.

When the width of the clothes hanging on the hanger is wide, the hanger device may be configured to move forward from the installation wall to prevent the clothes or items from touching the wall.

When a gyro sensor indicates which side of the hanging bar has more clothes or items, more air, warm dry air or steam may be supplied to the side where more clothes or items are hung.

Based on the image captured by a camera indicating which side of the hanging bar has more clothes or items, more air, warm dry air or steam can be supplied to the side where more clothes or items are hung.

It is possible to automatically control the operation of the hanger device may automatically control the operation by

identifying an object and distance of approaching object based on an image from at least one camera of the surrounding area.

BRIEF DESCRIPTION OF THE DRAWINGS

The embodiments will be described in detail with reference to the following drawings in which like reference numerals refer to like elements wherein:

FIG. 1 is a perspective view showing front, right side and top of a hanger device according to an embodiment of the present disclosure;

FIG. 2 is a perspective view showing front, right side and bottom of the hanger device according to an embodiment of the present disclosure;

FIG. 3 is an exploded view of the hanger device illustrated in FIG. 1;

FIGS. 4 to 6 are exploded views of the hanger device illustrated in FIG. 2;

FIGS. 7 and 8 illustrates various air flow occurring inside the hanger device;

FIG. 9 is a perspective view of a discharge tube provided in the hanger device;

FIG. 10 is an exploded view of the discharge tube illustrated in FIG. 9;

FIG. 11 is an cross-sectional view illustrating an operation of a wind guard shown in FIG. 10;

FIG. 12 is a rear perspective view of the wind guard;

FIG. 13 is a longitudinal cross-sectional view of the hanger device taken along line 13-13 of FIG. 2;

FIG. 14 is a perspective view illustrating a steam supply device of the hanger device according to an embodiment of the present disclosure;

FIGS. 15 and 16 are exploded views of the steam supply device illustrated in FIG. 14;

FIG. 17 is a side cross-sectional view of the steam generator along line 17-17 of FIG. 14;

FIG. 18 is a longitudinal cross-sectional view of the steam supply pipe along line 18-18 of FIG. 14;

FIG. 19 is a longitudinal sectional view of the hanging unit taken along line 19-19 of FIG. 14;

FIG. 20 is a side cross-sectional view of the hanger device taken along 20-20 of FIG. 2;

FIG. 21 is a front-bottom perspective view showing a coupling state of a filter cleaner and a filtering module according to an embodiment of the present disclosure;

FIG. 22 is a rear-top perspective view illustrating a coupling state between the filter cleaner and the filtering module;

FIG. 23 is a view showing a state in which the dust bin is separated from the dust connector by the operation of the elevating driver;

FIG. 24 is a view showing a coupling state of a dust connector and a dust container;

FIG. 25 is an external perspective view of the dust container;

FIG. 26 is an exploded view of a filtering module according to an embodiment of the present disclosure;

FIG. 27 is a longitudinal cross-sectional view of the filter cleaner and the filtering module along line 27-27 of FIG. 21;

FIG. 28 is an enlarged view of section A of FIG. 27;

FIG. 29 is a rear-side perspective view of a hanger device according to an embodiment of the present disclosure;

FIG. 30 is a side cross-sectional view of the hanger device along line 30-30 of FIG. 29;

FIG. 31 is a perspective view showing front, side and top of a hanger device according to another embodiment of the present disclosure;

FIG. 32 is a bottom perspective view showing front, bottom and side of the hanger device according to another embodiment of the present disclosure;

FIG. 33 is a perspective view of the hanger device showing a state in which the upper cover is separated from the cabinet;

FIG. 34 is an exploded view of the hanger device shown in FIG. 31;

FIG. 35 is a side cross-sectional view of the hanger device taken along line 35-35 of FIG. 31;

FIG. 36 is a cross-sectional cut-away perspective view of the hanger device taken along lines 36-36 of FIG. 31;

FIG. 37 is a cross-sectional cut-away perspective view taken along line 37-37 of FIG. 31;

FIG. 38 is a cross-sectional cut-away perspective view taken along line 38-38 of FIG. 31;

FIG. 39 is an enlarged view of section B of FIG. 36;

FIG. 40 is an enlarged view of section C of FIG. 37;

FIG. 41 is a bottom perspective view of a hanger device according to an embodiment of the present disclosure showing a state in which the wind guard is lowered;

FIG. 42 is a perspective view of a hanging unit according to another embodiment of the present disclosure;

FIGS. 43 and 44 are exploded views of a hanging unit shown in FIG. 42;

FIG. 45 is a perspective view of a moving arm lifting unit;

FIG. 46 is an enlarged view showing the structure of a sub-supply pipe;

FIG. 47 is a longitudinal cross-sectional view of the support showing the state before the movable arm is withdrawn;

FIG. 48 is a longitudinal cross-sectional view of the support showing the maximum withdrawal of the movable arm and the intermediate arm;

FIG. 49 is an exploded perspective view of a filtering module according to another embodiment for allowing a selective dehumidification function;

FIG. 50 is an enlarged view showing an internal configuration of a filtering module shown in FIG. 49;

FIGS. 51 and 52 are enlarged perspective views of the filtering module shown with a shielding plate separated from a shield frame;

FIG. 53 is a longitudinal cross-sectional view taken along line 53-53 of FIG. 50;

FIG. 54 is a top perspective view of the hanger device according to another embodiment where the hanger device is movable from the installation wall;

FIG. 55 is a bottom perspective view of the hanger device shown in FIG. 54;

FIG. 56 is an exploded perspective inside view of the main body showing details to allow movement from and away from the installation wall;

FIG. 57 is a longitudinal cross-sectional view taken along lines 57-57 of FIG. 56;

FIG. 58 is a perspective view of a hanging unit having an eccentricity detection function;

FIG. 59 shows a schematic of the electronics components or module for the various described embodiments;

FIGS. 60 and 61 illustrate a camera arrangement position on the hanger device of FIG. 1 and FIG. 31 embodiments, respectively;

FIG. 62 is a flow diagram for a method of operating a hanger device shown in FIGS. 60 and 61;

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FIG. 63 is an exemplary view an image captured by a camera; and

FIG. 64 is a diagram for describing an image recognition model;

FIGS. 65 and 66 illustrate camera arrangement position according to another embodiment;

FIG. 67 is a flow diagram of a method of operating a hanger device based on at least one image captured by the camera arrangement positions of FIG. 65 or 66;

FIG. 68 is an exemplary diagram schematically illustrating a situation of a hanger device to describe an image of a side peripheral region captured by at least one camera of FIG. 65 or 66;

FIG. 69 is an exemplary diagram schematically illustrating a situation of a hanger device to describe an image of a front peripheral region captured by at least one camera of FIG. 65 or 66; and

FIG. 70 is a view for explaining an image recognition model based on the at least image captured by at least one camera of FIG. 65 or FIG. 66.

DETAILED DESCRIPTION OF THE DISCLOSURE

A hanger device 10 according to an embodiment of the present disclosure, as illustrated in FIGS. 1-3, includes a main body 11 forming an external shape or appearance and a hanging unit 50 connected to the main body. The main body 11 includes a cabinet 12 and a cover 13 covering an interior of the cabinet 12, and a discharge tube 14 mounted vertically across the body 11 to form a through hole 101, which penetrates from an upper surface to a lower surface of the main body 11. The cross-sectional shape of the through hole 10, for example, may be an extended opening in a shape of a track or a rectangular shape such that one end of the through hole 101 is adjacent to the left edge of the main body 11, and the other end thereof is adjacent to the right edge of the main body 11. Both ends of the through hole 101 may be rounded in an arc shape, but is not limited thereto.

The hanging unit 50 includes a pair of supports 51 extending downward from the main body, and a hanging bar or tube 52 connected between the pair of supports 51. The hanging bar 52 is placed horizontally at a position spaced a predetermined distance downward from the bottom surface of the main body 11 by the support 51, and located in the center of the through hole 101. A plurality of outlets 522 are formed on the bottom surface of the hanging bar 52 to be spaced apart from each other by a predetermined interval.

A wind guard 20a (e.g., an air deflector or an air louver) may be accommodated in the inside of the main body 11 corresponding to the front of the through hole 101. The wind guard 20a may be movable in the vertical direction such that the wind guard 20a may descend to be exposed below the lower side of the main body 11. When the wind guard 20a is slidably drawn out to the lower side of the main body 11, a display screen may be activated on the front surface of the wind guard 20a, and various information may be displayed on the display screen. For example, a text or a video indicating the current driving or operation mode, the current indoor temperature, humidity, pollution level, etc. may be provided on the display screen.

A steam supply device 30 and an filter cleaner 40 may be mounted inside the main body 11. A liquid tank 312 (see, e.g., FIG. 15) for storing liquid, e.g., water, for the steam supply device 30 may be access through a liquid tank access opening 1209, and a dust container 41 (see, e.g., FIG. 23) for the filter cleaner 40 may accessed through a dust container

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access opening 1208. Accordingly, a user may remove the liquid tank for re-fill of liquid, remove the dust container to empty the dust, and mount them back into the body 11.

The hanger device 10 may also include a filtering module 60, a fan module including a main fan module 15 and a sub-fan module 16 installed inside the main body 11. When the modules 15 and 16 are activated, the air outside the main body 11 passes through the filtering module 60, and the filtered air is discharged to the outside of the main body 11 via the through hole 101.

FIGS. 4 to 6 are exploded views of some of the components inside the cabinet 12 of the hanger device 10. The cabinet 12 may include a lower cabinet 12a and an upper cabinet 12b. The fan modules 15 and 16, the steam supply device 30, and the filter cleaner 40 are mounted on the upper cabinet 12.

The lower cabinet 12a may have a hexahedral shape with an open top, and the upper cabinet 12b may have a structure configured to fit into and/or to cover the top of the lower cabinet 12a. The lower cabinet 12a may include a front part 1201 (e.g., front wall), a rear part 1202 (e.g., a rear wall), a right side part 1203 (e.g., a right sidewall), a left side part 1204 (e.g., left sidewall), and a bottom part 1205 (e.g., bottom panel). The surface of the bottom part 1205 is formed to be inclined upward toward the front, resulting in a height of the rear portion 1202 to be larger than the height of the front portion 1201, but the present disclosure is not limited thereto.

The interior of the lower cabinet 12a may be divided includes a first region and a second region disposed in front of the first region. The filter cleaner 40 and the steam supply device 30 may be located in the first region, and a tube guide 1206 for coupling with the discharge tube 14 may be located in the second region. The dust container access opening 1208 and the liquid tank access opening 1209 may be formed in the bottom part 1205 of the lower cabinet 12a corresponding to the first region.

The tube guide 1206 may extend upward a predetermined length from the surface of the bottom part 1205 and may have substantially the same cross-sectional shape of the discharge tube 14. A plurality of discharge ports 1207 may be formed on the front section 1206a and rear section 1206b of the tube guide 1206. For example, two or more discharge ports may be formed on each of the front and rear sections 1206a, 1206b of the tube guide 1206. Alternatively, a single discharge port may be formed on the front section 1206a and the rear section 1206b of the tube guide 1206.

The upper cabinet 12b includes the discharge tube 14, which define the shape of the through hole 101. The discharge tube 14 is fitted inside the tube guide 1206 of the lower cabinet 12a. As shown in FIG. 4, the discharge tube 14 includes front discharge duct 141 and rear discharge duct 142 opening corresponding to the discharge ports 1207 formed on the front and rear sections 1206a, 1206b of the tube guide 1206. See also FIG. 10.

As illustrated in FIGS. 5 and 6, the upper cabinet 12b may include a left upper cabinet and a right upper cabinet, and may be configured to have a shape symmetrical to each other based on a vertical plane dividing the cabinet 12 into left and right halves. The steam supply device 30 and the filter cleaner 40 may be disposed between the left upper cabinet and the right upper cabinet, as shown in FIG. 4.

Each of the upper left cabinet and the upper right cabinet includes a main flow path assembly and a sub flow path assembly coupled to a side of the main flow path assembly. The main flow path assembly includes a main fan housing 1211 on which the main fan module 15 is placed, and a main

discharge duct **1212** extending from the front end of the main fan housing **1211** in the width direction of the cabinet **12**. The sub flow path assembly includes a suction guide **1214**, a discharge guide **1215** provided on a side surface of the suction guide **1214**, and a sub discharge duct **1216**, which is bent to extend in an L shape from the front end of the discharge guide **1215**. The suction guide **1214** is in close contact with the side surface of the main fan housing **1211**.

A sub-fan mounting hole **1219a** is formed in the partition wall **1219** that divides the suction guide **1214** and the discharge guide **1215**, and a sub-fan module **16** is installed in the sub-fan mounting hole **1219a**. The suction guide **1214** functions as a suction guide for guiding the air passing through the filtering module **60** to the sub fan module **16**. The sub fan module **16** forces the air into the discharge guide **1215**, and the forced air is guided to the sub-discharge duct **1216**.

The upper cabinet **12b** includes the discharge tube **14**, and the discharge tube **14** is fitted inside the tube guide **1206** of the lower cabinet **12a**. The sub discharge duct **1216** provided in each of the upper left cabinet and the upper right cabinet is inserted into the space between the side surfaces **1203** and **1204** and the front part **1201** of the lower cabinet **12a**. The sub discharge duct **1216** includes a side sub discharge duct that extends from the front surface of the discharge guide **1215**, extends along the side surface of the tube guide **1206**, and is bent at the front end. As a result, the sub discharge duct **1216** extends alongside and between the front part **1201** and the front section **1206a** of the tube guide **1206**.

The discharge port **1216a** of the sub discharge duct **1216** is configured to communicate with the discharge port **1207** formed on the front section **1206a** of the tube guide **1206**. The main discharge duct **1212** is in close contact with the back surface of the rear section **1206b** such that the discharge port **1212a** of the main discharge duct **1212** communicates with the discharge port **1207** provided on the rear section **1206b** of the tube guide **1206**. The discharge port **1216a** of the sub discharge duct **1216** communicating with the discharge port **1207** formed on the front section **1206a** may be referred to as a front discharge port. The discharge port **1212a** formed in the main discharge duct **1212** and communicating with the discharge port **1207** formed in the rear section **1206b** may be referred to as a rear discharge port.

A recess inside a side of the sub discharge duct **1216** forms a support mounting groove **1217**. The pair of supports **51** of hanging unit **50** is accommodated in the support mounting groove **1217** provided on left and right sides of the cabinet **12**. A plurality of steam holes **1213** are formed on the upper surface of the main discharge duct **1212** and a plurality of steam holes **1218** are formed on an upper surface of the sub discharge duct **1216**.

The main discharge duct **1212** may extend to the discharge guide **1215** and extend past the front surface of the suction guide **1214**. As shown in FIG. **5**, the width of the main discharge duct **1212** may extend from one end of the main fan housing **1211** to the discharge guide **1215**. One side end of the main fan housing **1211** is close to the center of the lower cabinet **12a**, and the other end of the main fan housing **1211** is in close contact with the suction guide **1214**.

The main fan module **15** includes a main fan **151** and a main fan motor **152** for driving the main fan **151**, and the sub-fan module **16** includes a sub-fan **161**, a sub-fan motor **162** for driving the sub-fan **161**, and a suction grill **163** disposed the suction side of the sub-fan **161**. The main fan **151** may be a cross-flow fan, and the sub-fan **161** may be an axial fan or a centrifugal fan.

With reference to FIGS. **7** and **8**, when the main fan module **15** and the sub-fan module **16** operate, indoor or ambient air is suctioned through the filtering module **60**, and the filtered air flows into the cabinet **12** of the hanger device **10**. A portion of the filtered air is introduced into the main fan housing **1211**, discharged through the discharge port **1212a** of the main discharge duct **1212**, and exhausted at the rear discharge duct **142** of the discharge tube **14** (corresponding to a rear side surface of the through hole **101**). This main air passage is indicated as MAP1 and MAP2 for the left and right sides in FIG. **7**. The remaining air suctioned by the sub-fan **161** flows into the suction guide **1214**, the discharge guide **1215** and the sub-discharge duct **1216** and is discharged through the discharge port **1216a** and the front discharge duct **141** of the discharge tube **14** (corresponding to the front side surface of the through hole **101**). The secondary air passage is indicated as SAP1 and SAP2 for the left and right sides.

A portion of the air forced toward the sub-discharging duct **1216** is supplied to the inside of the pair of supports **51** of the hanging unit **50**. The supplied air flows along an inside of the hanging bar **52** and is discharged toward the clothes or items hung on the hanger through the plurality of outlets **522** formed in the hanging bar **52**.

For example, the upper surface of the support **51** may be opened and designed to communicate with the flow path inside the sub-discharge duct **1216** such that a portion of the air flowing along the inside of the sub-discharge duct **1216** may flow into the supports **51**. As another example, an air outlet may be formed on the side of the sub discharge duct **1216** defining the support mounting groove **1217**, and an air inlet, communicating with the air outlet, is formed on the side of the support **51**. Other configurations are possible to allow a portion of the air flowing along the sub discharge duct **1216** to be introduced into the support **51**.

With reference to FIGS. **9** to **12**, the discharge tube **14** according to an embodiment of the present disclosure includes a front section **144** extending a predetermined length along the width direction of the hanger device **10**, a rear section **145** extending a predetermined length along the width direction of the hanger device **10** and a pair of side sections **146** connecting both ends of the front section **144** and the rear section **145**. The pair of side sections **146** may be rounded with a predetermined curvature. The upper surface of the side sections **146** may be flat, but the lower surface of the side sections **146** may be inclined toward the rear of the discharge tube **14**. As a result, a height of the rear section **145** may be greater than a height of the front section **144**.

One or more discharge ducts may be formed on each of the front section **144** and the rear section **145**. For example, a pair of front discharge ducts **141** may protrude from the front surface of the front section **144**, and a pair of rear discharge ducts **142** may protrude from the rear surface of the rear section **145**. The protruding ends of the front discharge duct **141** and the rear discharge duct **142** are respectively coupled to the discharge ports **1207** formed on the front and rear sections **1206a**, **1206b** of the tube guide **1206** to form an air discharge passage at front and rear of the through hole **101**. A vane or a damper **19a** may be rotatably coupled to each of the front discharge duct **141** and the rear discharge duct **142** by a vane motor **191**, which allows selective control of air discharged to the through hole **101**.

A wind guard cover **140** is formed to protrude from a front surface of the front section **144** so that a space is formed to accommodate the wind guard **20a** between the wind guard cover **140** and the front section **144**. The wind guard cover

140 is provided between the lower end of the front discharge duct **141** and the lower end of the front section **144**. An opening is formed between the lower surfaces of the wind guard cover **140** and the front section **144** such that the wind guard **20a** may descend past the opening to be exposed to the outside, as shown in FIG. 2.

A pair of wind guard driving units **18a** for lowering or raising the wind guard **20a** are provided above the front discharge duct **141**. Each of the pair of wind guard driving units **18a** includes a wind guard shaft **184**, a pair of pinions **185** connected to both ends of the wind guard shaft **184**, and a wind guard motor **183** configured to rotate the wind guard shaft **184** and the pair of pinions **185**.

The wind guard **20a** includes a guard body **201** and a guard rack **202** extending from the upper end of the guard body **201**. Based on the number of pinions of the wind guard driving unit **18a**, the same number of guard racks **202** extend from the guard body **201**. In this embodiment, a pair of guard racks **202** extend from the upper left and right sides of the guard body **201** to engage with the pair of pinions **185** provided at both ends of one wind guard shaft **184**. Gear teeth formed on the front surface of the guard rack **202** engage with the pinion **185**. See FIG. 11.

As previously discussed, a display screen may be provided on the front surface of the guard body **201**, and various information may be output through the display screen. As shown in FIG. 12, a sterilization module **100** may be mounted on the rear surface of the guard body **201**. The sterilization module **100** may be composed of a PCB substrate and a sterilization LED mounted on the PCB substrate to emit ultraviolet rays to sterilize air and/or steam discharged through the through hole **101**. For example, when the hot air supply mode or the steam supply mode is turned on, the wind guard **20a** descends to be exposed outside of the hanger device **10**, and in this state, air and/or steam descending through the through hole **101** may be sterilized when the sterilization module **100** is turned on. In an alternative embodiment, the irradiation direction of the ultraviolet rays emitted from the sterilization module **100** may be angled to be directed toward the hanger such that the clothes or items hung on the hanger are sterilized. In an alternative embodiment, the sterilization module **100a** may be also mounted on the lower inner peripheral surface of the discharge tube **14**.

A mood light **17a** may be mounted on the discharge tube **14**. For example, a mood light hole or groove **143** may be formed at an upper end of the discharge tube **14**, and the mood light **17a** may be mounted in the mood light hole **143**. The mood light **17a** may be mounted at any point between the upper end and the lower end of the discharge tube **14**, and may be formed in a closed loop shape along the discharge tube **14**. Alternatively, as shown, a pair of mood lights **17a** having a U-shape may be disposed at positions symmetrical to each other.

When power is applied to the hanger device **10** or a hot air supply mode (or drying mode) or steam supply mode (humidification mode or wrinkle removal mode) is turned on, the mood light **17a** may be turned on, and light may be emitted softly through the through hole **101**. The mood light **17a** may be also controlled to change the illuminance or color of the mood light **17a** according to the indoor temperature, indoor humidity, indoor pollution, or the degree of drying of clothes. The mood light **17a** may be an LED module and a surface light emitting unit including a light guide plate for diffusing light emitted from the LED module. Further, a mood light may be mounted on the rear surface of the guard body **201** corresponding to the upper side of the sterilization module **100**.

As shown in FIG. 13, when the main fan module **15** and the sub-fan module **16** are turned on, indoor air passes through the filtering module **60**, and filtered air discharged to the hanging unit **50** through the front discharge duct **141** and the rear discharge duct **142**. The air discharge direction may be controlled by the rotation angle of the vane **19a** in the vertical direction. Further, the wind guard **20a** is lowered to be exposed to the outside for minimizing a phenomenon in which the air discharged from the discharge ducts **141** and **142** is diffused toward the front of the hanger device **10**. The wind guard **20a** allows air to be discharged toward the clothes or items hung on the hanger device **10**, and at the same time, the air is sterilized by the sterilization module **100**.

As shown in FIGS. 14 to 18, the steam supply device **30** according to the embodiment of the present disclosure includes a steam generator **31** for generating steam, a steam supply pipe **32** extending from the steam generator **31**, and a heater **33** attached to the outer circumferential surface of the steam supply pipe **32** along the steam supply pipe **32**. The steam generated by the steam generator **31** may flow along the steam supply pipe **32** by the pressure of the generated steam. In an operation mode in which the steam supply device **30** operates to supply steam, the fan modules **15** and **16** may maintain a stopped state to prevent conflict between the dehumidification function and the steam supply function.

The steam supply pipe **32** includes a main supply pipe **321** extending adjacent to an outer edge of the upper end of the discharge tube **14**, a sub-supply pipe **322** extending downwards within a support **51** of the hanging unit **50**, and a plurality of discharge nozzles **323** extend from a bottom surface of the main supply pipe **321**. The plurality of discharge nozzles **323** align with the steam holes **1213** formed on an upper surface of the main discharge duct **1212** and the steam holes **1218** formed on the upper surface of the sub discharge duct **1216**. See FIG. 5. The steam discharged through the discharge nozzles **323** is supplied to the main discharge duct **1212** and the sub discharge duct **1216**.

The heater **33** is configured to be in close contact with the surface of the steam supply pipe **32** to prevent the steam flowing along the steam supply pipe **32** from being cooled and condensed. The heat generated by the heater **33** is transferred to the steam supply pipe **32** by heat conduction such that the steam flowing along the inside of the steam supply pipe **32** is prevented from being condensed due to heat loss.

The steam generating unit **31** may include a steam case **311** to house a pump **313**, suction pipe **314**, discharge pipe **315**, induction heaters **316**, **317** (or other types of heater, e.g. a sheath heater or a planar heater), a heating tank **318**, a steam chamber **319**, and the steam generating chamber **3111**. The heating tank **318** is coupled to the front upper side of the steam case **311**, and the pump **313** is coupled to the rear upper side of the steam case **311**. The liquid tank **312** is detachably coupled underneath the heating tank **318**. A steam chamber **319** is provided on the left and right sides of the pump **313**. A steam generating chamber **3111** is formed inside the steam case **311** corresponding to the lower side of the pump **313**.

An induction heater **316** is mounted on the bottom of the steam generating chamber **3111** to heat the water supplied to the steam generating chamber **3111** to a vapor state. The liquid tank **312** is disposed in front of the steam generating chamber **3111**, and the bottom of the liquid tank **312** is inclined downward toward the steam generating chamber **3111**. The rear lower end edge of the liquid tank **312** may

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extend further to the rear to allow a residual water reservoir 3121 to be formed such that the residual water reservoir 3121 is configured to be located below the front lower end of the steam generating chamber 3111. the liquid tank 312 is accessible via liquid tank access opening 1209 to fill or re-fill the liquid tank.

As shown in FIG. 17, the lower end of the steam case 311 defining the lower front end of the steam generating chamber 3111 is configured to be stacked above or overlapped over the residual water reservoir 3121. A suction pipe 314 and a discharge pipe 315 extend vertically from the bottom of the pump 313. The end of the suction pipe 314 passes through a bottom of the steam case 311 into the liquid tank 312 and is configured to end near a bottom of the residual water reservoir 3121. With this structure, majority of liquid stored the liquid tank 312 may be suctioned by the pump 312 through the suction pipe 314, and discharged into the steam generation chamber 3111 by the discharge pipe 315 for vaporization by the induction heater 316. The discharge pipe 315 may extend vertically and spaced apart from the suction pipe 314 to the rear of the steam case 311.

The heating tank 318 may be disposed above the liquid tank 312, and the induction heater 317 may be mounted at the bottom of the heating tank 318. The steam chamber 319 communicates with the heating tank 318, and the steam generated in the steam generating chamber 3111 is supplied into the heating tank 318 through the steam chamber 319.

The steam supply pipe 32 and the heater 33 are connected to the left and right sides of the heating tank 318. The main supply pipe 321 and the sub supply pipe 322 are connected to the left side and the right side of the heating tank 318. One end of the sub supply pipe 322 is connected to the side of the heating tank 318 and the other end extends into the support 51. Both ends of the main supply pipe 321 are connected to the side surfaces of the heating tank 318 to form a loop such that majority of the steam is supplied to the front and rear discharge ducts 141 and 142, but a part of the steam flowing along the main supply pipe 321 may return to the heating tank 318.

As the heating tank 318 is heated by the induction heater 317, the steam flowing into the heating tank 318 through the steam chamber 319 is mixed with the steam returning to the heating tank 318, which further prevents liquid condensation. Because the steam pressure inside the heating tank 318 is maintained by the induction heater 317, the pressure drop inside the steam supply pipe 32 may be minimized such that a spray distance (or a discharge distance) of the steam discharged through the discharge nozzle 323 may be sufficiently maintained.

With reference to FIGS. 19 and 20, one end of the sub-supply pipe 322 of the steam supply pipe 32 extends into the support 51 such that a part of the steam supplied from the steam supply device 30 is directed into the hanging bar 52. One end of the sub-supply pipe 322 may extend to a point adjacent to the lower end of the support 51 to which the hanging bar 52 is connected. An internal flow path of the sub supply pipe 322 accommodated in the support 51 may be defined as a steam flow path L1.

Further, a part of the air introduced into the cabinet 12 by the sub-fan module 16 is supplied into the support 51 of the hanging unit 50. As the air flows into the filtering module 60, foreign substances may be filtered out, and the air may be heated to a temperature higher than ambient temperature prior to entering the support 51, which will be described hereinafter with reference to FIG. 26. Accordingly, a flow path including the inside of the support 51 and the inside of the hanging bar 52 may be defined as a dry air flow path L2.

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As shown, the hanger bar 52 is hollow to allow communication with the steam flow path L1 and dry air flow path L2. A plurality of hanger grooves 521 may be formed to be spaced apart from each other at a predetermined distance on top of the hanging bar 52. The hanger groove 521 may be recessed a predetermined depth from the outer circumferential surface of the hanging bar 52, and may extend a predetermined length in the circumferential direction of the hanging bar 52. A plurality of outlets 522 may be formed on the bottom of the hanging bar 52. Accordingly, the dry air provided via dry air flow path L2 or the steam provided via steam flow path L2 may be discharged to the outside of the hanging bar 52 through the plurality of outlets 522.

A pressure sensor PS may be mounted on the bottom of the hanger groove 521. When a hook of a hanger is placed into the hanger groove 521, the pressure sensor PS detects the load of the cloth or item on the hanger. Based on the position information of the pressure sensor PS that detects the load on the clothes hanger and the load information of the clothes detected by the pressure sensor PS, the discharge of dry or steam may be selectively supplied to the location where the clothes or items are hung.

For example, when clothes are eccentrically hung on the left or right side of the hanging bar 52, i.e., the clothes are hung on the left and right sides of the cabinet 12, more dry air or steam may supplied to the side where the clothes are hung. When more items or heavier items are hung on the right side, the main fan module 15 and the sub-fan module 16 may be operated alone or at higher speed compared to the left side such that greater amount of dry air is provided to the front discharge duct 141, rear discharge duct 142 and the dry air flow path located on the right side of the cabinet 12. Similar operation of the steam supply device may be applied by providing left and right valves to main steam supply pipes 322 and sub supply pipes 323 coupled to the right and left side of the heating tank.

With reference to FIGS. 21 and 22, the hanger device 10 includes a filtering module 60 for filtering or removing foreign substances contained in indoor or ambient air, an filter cleaner 40 for further cleaning the foreign substances in the suctioned air, and a lift 70 for raising or lowering a dust container 41 of the filter cleaner 40. The filter cleaner may be used clean the filtering module 60 by removing dust (foreign substances) accumulated inside the filters of the filtering module 60. The filter cleaner 40 may include the dust container 41 for collecting foreign substances separated from suctioned air, a top cover 43 coupled to an open upper end of the dust container 41, and a dust connector 45 coupled to the upper surface of the top cover 43.

The lift 70 includes a lift plate 71 on which the dust container 41 is seated, and a lift driver 72 for vertically moving the lift plate 71. The lifting plate 71 includes a horizontal portion 711 on which the dust container 41 is placed, and a vertical portion extending upward from the rear end of the horizontal portion 711 to support the rear surface of the dust container 41. One or more lift racks 723 extend a predetermined length in the vertical direction on the rear surface of the vertical portion 712. The lift driver 72 includes a lift pinion 722 configured to engage with the lift rack 723, and a lift motor 721 for rotating the lift pinion 722.

When the dust container 41 is filled with dust or foreign substances, a user may turn on the lift driver 72 to lower the dust bin 41 to the outside of the hanger device 10 through the dust container access hole 1208. When the dust container 41 descends together with the lift plate 71 and is completely exposed from the main body 11 through the dust container access hole 1208, the user may separate the dust container

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41 from the lift plate 71 to empty the dust or foreign substances from the dust container. After emptying, the dust container 41 is seated on the lift plate 71, and the lifting driver 72 is driven to raise the dust container 41 to sufficiently seal the dust container to the bottom surface of the dust connector 45.

In an alternative embodiment, when the lift plate 71 descends, the dust connector 45, the top cover 43, and the dust container 41 may be designed to be separated from the filtering module 60. Alternatively, when the lift plate 71 descends, the dust connector 45 and the dust collector 44 remain coupled to the filtering unit 60, and the top cover 43 and the dust container 41 may be designed to descend together with the lifting plate 71. Alternatively, when the lift plate 71 descends, only the dust container 41 may be designed to descend together with the lift plate 71.

The combination of the top cover 43 and the dust container 41 may be defined as a dust collecting device, and the combination of the dust collector 44 and the dust collecting filter 47 (FIG. 28) may be defined as a dust collecting device.

With reference to FIGS. 23 to 25, a dust collector 44 is mounted on the bottom surface of the dust connector 45, and a suction fan module 46 is installed inside the dust connector 45 corresponding to the upper side of the dust collector 44, and a discharge grill 456 is formed on the upper surface of the dust connector 45 corresponding to the upper side of the suction fan module 46. The dust collector 44 is formed in the shape of a cone whose width becomes narrower toward the bottom, and a plurality of filtering holes 441 are formed. An operation of removing the dust will be described hereinafter with reference to FIGS. 27 and 28.

A dust inlet 452 is formed at the left and right edges of the dust connector 45. A mounting sleeve 451 is provided on the upper surface of the dust connector 45 corresponding to the edge of the dust inlet 452 to form a protrusion. The mounting sleeve 451 may have a rectangular cross-sectional shape, but the shape is not limited thereto. A dust outlet 613 (see e.g., FIG. 28) communicating with the mounting sleeve 451 is formed on the bottom surface of the filtering module 60 to which the dust connector 45 is coupled. A number of filtering modules 60 may be based on a number of the mounting sleeves 451. In this instance, a pair of the mounting sleeves 451 are coupled to the pair of filtering modules 60. A suction duct 453 extends from the bottom surface of the dust connector 45, corresponding to the lower side of the pair of mounting sleeves 451.

The opened upper surface of the dust container 41 is shielded by the top cover 43 having a handle 434 at front and rear edges. A dust collector through hole 433 through which the dust collector 44 passes is formed in the center of the top cover 43. A duct insertion holes 431 may be formed on left and right edges of the top cover 43 and align with the left and right suction ducts 453. A pair of dampers 432 facing each other are provided inside the duct insertion hole 431. The pair of dampers 432 may be rotatably hinged and spring loaded in a closed position. When the dust container 41 is coupled to the dust connector 45, the suction duct 453 is inserted into the duct insertion hole 431 and push the pair of dampers 432 to rotate to an open position. When the dust container 41 is lowered to separate from the dust connector 45 by the lift 70, the suction duct 453 is pulled out from the duct insertion hole 431, and the pair of dampers 432 rotate to return to a closed position.

Referring to FIGS. 26 and 27, the filtering module 60 according to an embodiment of the present disclosure includes a first filtering module 60a connected to a mounting sleeve 451 on the left side of the dust connector 45, and a

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second filtering module 60b connected to the mounting sleeve 451 on the right side of the dust connector 45. Each of the first and second filtering modules 60a, 60b has an outer shape defined by a lower case 61 and an upper case 66, where a lower end of the upper case 66 is coupled to an upper end of the lower case 61.

An inlet 661 is formed on the upper surface of the upper case 66, and an outlet 611 is formed on the bottom of the lower case 61. A dust outlet 613 is formed at one end of the lower case 61, and the dust outlet 613 is configured to communicate with a dust inlet 452 formed on the upper surface of the dust connector 45. A sleeve 614 may extend along an edge of the dust outlet 613 and formed on the bottom surface of the lower case. The sleeve 614 may be configured to be in close contact with an inner circumferential surface or an outer circumferential surface of the mounting sleeve 451, and leaking of dust between the dust outlet 613 and the dust inlet 452 may be prevented. See also FIG. 28.

Each of the first and second filtering module 60a, 60b further includes a pre-filter 65 provided on the lower side of the upper case 66. The pre-filter 65 is configured to form a conveyor belt around a pair of rollers 67 and one of the rollers being driven by a belt motor 671. The pre-filter 65 is rotated in a prescribed direction by turning on the belt motor 671. The belt motor 671 is seated between motor accommodation grooves 612 and 662, which are formed in the lower case 61 and the upper case 66, respectively.

The upper and lower surfaces of the pre-filter 65 are spaced apart by a distance corresponding to the diameter of the pair of rollers 67. A HEPA filter 64 for filtering fine dust and a dehumidifying filter 63 disposed below the HEPA filter 64 may be stacked into the space provided between the upper lower surfaces of the pre-filter. The dehumidifying filter 63 may contain zeolite as a main component. If desired, a deodorizing filter may be further provided between the HEPA filter 64 and the dehumidifying filter 63. In certain cases, the HEPA filter 64 and the deodorizing filter may be provided as a single unit.

A heater 62 may be placed below the dehumidification filter 63 such that the air passing through the filtering module 60 is dehumidified and heated to a prescribed temperature to create dry air. The heater 62 may a planar heater or a sheath heater, but other types of heaters are also applicable. Further, regeneration or recharging of the dehumidification filter 63 is necessary, the heater 62 is turned on while the fan modules 15 and 16 are stopped to evaporate the moisture absorbed by the dehumidification filter 63.

As shown in FIG. 28, the dust collecting filter 47 is mounted on the upper end of the dust collector 44, and the dust collector 44 is detachably coupled to the bottom surface of the dust connector 45. The dust collector 44 may be mounted or removed from the bottom surface of the dust connector 45 in a screw-in method. When removed, the dust collection filter 47 may be separated from the dust connector 45. The dust collecting filter 47 may be provided in the form of a flat circular filter corresponding to the shape of the upper surface of the dust collector 44.

A suction grill 454 is centrally formed on a bottom surface of the dust connector 45, to which the dust collector 44 is mounted. A fan mount housing space 455 is located in an inner center of the dust connector 45 and located above the upper side of the suction grill 454. The suction fan module 46 having a suction fan 461 and a fan motor 462 is accommodated in the fan mount housing space 455. The

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discharge grill 456 is formed in the center of the upper surface of the dust connector 45 above the upper side of the suction fan 461.

When the dust container 41 is raised toward the bottom of the dust connector 45 by the lift 70, the conical dust collector 44 is penetrates through the dust collector through hole 433 centrally formed in the top cover 43. The suction duct 453 rotates the damper 432 while being inserted into the duct insertion hole 431 of the top cover 43. When the dust container 41 is completely coupled to the dust connector 45, the bottom surface of the lifting plate 71 is planar with the bottom surface of the main body 11 to cover the dust container access hole 1208. Further, as the damper 432 rotates, the suction duct 453 and the inside of the dust container 41 communicate with each other.

When the mounting sleeve 451 extending along the edge of the dust inlet 452 and the sleeve 614 extending from the bottom of the lower case 61 come into contact with each other, the dust adhering to the outer surface the pre-filter 65 passes through the suction duct 453 to be collected into the dust container 41.

A process in which the dust collected on the pre-filter 65 is removed to the dust container 41 will be described. When the filter cleaning mode is turned on automatically based on pre-set cleaning cycle or manually by a user inputting a filter cleaning command and based on the conveyor belt configuration of the pre-filter 65, the driven roller 67 is rotated by the belt motor 671 and the pre-filter 65 moves along an outer circumferential surface of the belt motor 67. The suction fan module 46 is turned on to suction the air inside the dust container 41 through is suctioned through the filtering holes 441 formed in the dust collector 44. The suctioned air passes through the dust collection filter 47, and is discharged back into the room through the suction grill 454 and the discharge grill 456.

Because of the suction of air from inside the dust bin 41, a negative pressure is created inside the dust bin 41, and as a result, the air inside the suction duct 453 is sucked into the dust bin 41. At this time, the dust accumulated on the outer surface of the pre-filter 65 are separated by the pressure difference and are collected in the dust container 41.

Among the dust and foreign substances collected inside the dust container, dust and foreign substances which cannot fit through the filtering holes 441 of the dust collector 44 fall to the bottom of the dust container 41, and dust and foreign substances passing through the filtering holes 441 are filtered out by the dust collecting filter 47. Similar to a cyclone dust container applied to a vacuum cleaner, a cyclone phenomenon occurs inside the dust container 41 to separate dust and air based on the operation of the suction fan 461.

The dust cleaning mode may be programmed to be performed for a set time after the dust cleaning command is input, or may be programmed to be performed periodically at a predetermined time interval even if the user does not input the dust cleaning command. In addition, when it is determined that the dust collected in the dust bin 41 has reached a set amount, a message informing that the dust bin is empty may be output through the display unit and/or the speaker.

With reference to FIGS. 29 and 30, an induction unit IU may be used to generate heated dry air. The induction unit IU may be used in addition or as an alternative to the heater 62, shown in FIG. 26, mounted on the lower side of the dehumidifying filter 63 to generate dry air. The induction unit IU may be mounted on a rear surface of the main fan housing 1211, functioning as a heater.

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The induction unit IU may be an induction heater using an induced current as a heat source. When power is applied to the induction unit IU, the main fan housing 1211 is heated by the electromagnetic field to act as a heater. When the main fan 152 rotates in this state, the air sucked into the main fan housing 1211 by the main fan 152 is heated and then discharged to the main discharge duct 1212 through a discharge outlet 1501 of the main fan housing 1211.

Although the above drawing shows that the induction unit IU is mounted only on the rear surface of the main fan housing 1211, the induction unit IU may extend to the rear surface of the suction guide 1214 to discharge heated air to the sub discharge duct 1216. Alternatively, a separate induction unit IU may be mounted on the rear surface of the suction guide 1214.

By allowing at least a portion of the main fan housing 1211 and the suction guide 1214 to operate as a heater, there is an advantage in that it is possible to minimize the possibility of thermal damage to the filters constituting the filtering module 60. Further, additional source of heated dry air assist with drying of clothes or items on the hanging bar 52.

FIG. 31 is a front perspective view of a hanger device 10a according to another embodiment of the present disclosure. FIG. 32 is a bottom perspective view of the hanger device 10a. In this embodiment of the clothes hanger device 10a, a through hole 101 is not formed in the center of the body. The air flow path and the structure thereof may be different compared to the previous embodiment. However, the filtering module provided inside the main body, the fan module, the steam supply device, and the filter cleaning device may be applied in the same or similar manner, and the same reference numerals are applied to the same components. Although the filter cleaning device is not shown or discussed below, the filter cleaning device 40 shown in FIGS. 21 to 28 may be installed in the hanger device 10a in a same or similar way.

The hanger device 10a includes a main body 110 forming an external shape, and a hanging unit 50 connected to the lower side of the main body 110. The hanging unit 50 may include a pair of supports 51 and a hanging bar 52. The main body 110 includes a cabinet 120 having a component mounting space formed therein and a cover 130 covering the opened upper surface of the cabinet. The cabinet 120 may include a lower cabinet 122 and an upper cabinet 123 covering an upper surface of the lower cabinet 122.

The bottom surface of the lower cabinet 122 may be formed to be inclined, and the air discharge part 1223 may be formed by being depressed upwardly on the inclined surface. The inclined surface may start from a predetermined distance from the rear end of the bottom surface of the lower cabinet 122 and extend to the front end of the bottom surface of the lower cabinet 122. Alternatively, the inclined surface may be inclined upwardly from the rear end to the front end of the lower cabinet 122.

With reference to FIGS. 33 to 35, a cover 130 may be slidably and detachably coupled to the upper cabinet 123. The cover 130 may include a cover body 131 and a recessed pull handle 132 formed at the front end of a cover body 131. A suction grill 133 formed in a grid pattern may be provided on the rear section of the cover body 131. The recess pull handle 132 may be provided as a recessed groove of a predetermined depth and width along a front end of the cover body 131.

One or more stepped recesses 1233 may be formed in the upper cabinet 123, and the stepped recesses 1233 are exposed or covered by the cover 130. In this example, two

step recesses **1233** are provided on the left and right sides of the upper cabinet **123**. A partition **1234** divides each of the stepped recesses **1233** into a first space (or front space) and a second space (or rear space). The filtering module **60** is seated into the second space, and the first space may serve as a storage space, for example, for storing replacement filters.

The second space serves a filter seat **1231**, and a filter outlet **1232** in the form of a grill may be formed at the bottom of the filter seat **1231**. As the outlet **1232** is formed in a grid-shaped grill shape, the air passes through the filtering module **60** to the cabinet **120** while stably supporting the filtering module **60** placed on the filter seat **1231**. A steam generator **31** of the steam supply device **30** may be provided in an interior of the lower cabinet **122** and between the left and right filter seats **1231** of the upper cabinet **123**. Although not shown, the filter cleaner **40** may be disposed on the rear side of the steam generator **31** by adjusting the width of the steam generator **31** in the front-rear direction.

A fan module **15a** is placed under the filter seat **1231**. The fan module **15a** includes a fan **153**, a fan motor **154** for driving the fan **153**, and a fan housing **155** for receiving **153**. The fan **153** may be a cross flow fan. As previously discussed, an induction unit IU may be mounted on the rear surface of the fan housing **155** such that the fan housing **155** functions as a heater to heat the suctioned air.

The cabinet **120** includes a middle cabinet **124** disposed between the upper cabinet **123** and the lower cabinet **122**, and a plate **125** serving as a flow path partition disposed between the middle cabinet **124** and the lower cabinet **122**.

The middle cabinet **124** and the flow path partition plate **125** are positioned below the front spaces of the upper cabinet **123**. The middle cabinet **124** is placed downward a first predetermined distance from the upper cabinet **123**, and the flow path partition plate **125** is placed downward a second predetermined distance from the middle cabinet **124**.

A fan mounting space **1221** on which the fan module **15a** is placed is formed inside the lower cabinet **122**, and an air discharge part **1223** is formed in front of the fan mounting part **1221**. A water tank access opening **1222** may be formed at the bottom of the fan mounting space **1221**. In the case of a model in which the filter cleaner **40** is mounted, a dust container access may be formed in the bottom of the lower cabinet **122** spaced apart from the water tank access opening **1222**.

The air discharge part **1223** is a part formed by bending a bottom section of the lower cabinet **122** upward a plurality of times such that the bent sections form a front section, an upper section, and a rear section. The front section and the rear section may be inclined in a direction toward each other and toward the upper section.

A front discharge port **1223a**, an upper discharge port **1223b**, and a rear discharge port **1223c** are formed on the front section, the upper section, and the rear section. Discharge vanes or dampers **19** are mounted on each of the discharge ports **1223a**, **1223b**, **1223c**, and the discharge vanes **19a** are tilted at a predetermined angle by the vane motors **191** to open the discharge ports **1223a**, **1223b**, **1223c** and to control the air flow direction.

A pair of discharge vanes **19** may be mounted on the upper discharge port **1223b** to selectively open and close some or all of the upper discharge outlet **1223b**. For example, one of the pair of discharge vanes **19** is rotatably mounted to the front end of the upper discharge port **1223b**, and the other is to be rotatably mounted to the rear end of the upper discharge port **1223b**.

As in the previous embodiment, a wind guard housing **1242** is formed at the front side of the lower cabinet **122**, and the wind guard **20a** may be inserted into the wind guard housing **1242**. The wind guard **20a** may be provided to be slidably withdrawable downward from the wind guard housing **1242**. The wind guard housing **124** may be a part of the middle cabinet **124** that further extends from the front end of the middle cabinet **124**. Alternatively, a separate structure bent in an n-shape may be mounted on a back surface of the front end of the lower cabinet **122**, and a through hole through which the wind guard **20a** passes is formed in the lower cabinet **122**.

The flow path partition plate **125** is provided between the air discharge part **1223** and the middle cabinet **124** and between the middle cabinet **124** and the lower cabinet **122** such that air flow is divided into an upper flow path PU and a lower flow path PL. The lower flow path PL is defined by a duct formed by the flow path partition plate **125** coupled to the upper section of the air discharge part **1223** and the upper surface of the air discharge part **1223**. The upper flow path PU is defined by a duct from by the flow path partition plate **125** and the middle cabinet **124**. The flow path partition plate **125** may be a separate member coupled to the upper end of the front section for the air discharge part **1223**. Alternatively, the flow path partition plate may be injection-molded as a part of the lower cabinet **122**.

The rear end of the middle cabinet **124** may be positioned adjacent to the fan **153**, and the front end may be connected to the bottom of the lower cabinet **122**. The middle cabinet **124**, similar to the structure of the air discharge part **1223**, includes a front section, an upper surface section, and a rear section. The air discharged from the fan **153** is discharged through the rear discharge port **1223c** and directed through the upper flow path PU to discharge at the front discharge port **1223a**.

The steam supply pipe **32** of the steam supply device **30** extends from the steam generator **31** and extends along the upper surface of the middle cabinet **124**. The main supply pipe **321** of the steam supply pipe **32** extends along the upper edge of the middle cabinet **124**. The discharge nozzles **323** protruding from the bottom surface of the main supply pipe **321** penetrate the middle cabinet **124** and are exposed in a space between the middle cabinet **124** and the flow path partition plate **125**. The sub supply pipe **322** of the steam supply pipe **32** extends along a rear edge and a side end of the middle cabinet **124** to extend into the support **51** of the hanging unit **50**.

The discharge nozzles **323** extending from the bottom surface of the main supply pipe **32** penetrate the top surface of the middle cabinet **124** and are exposed to the upper flow path PU. A portion of the steam supplied to the upper flow path PU is provided into the lower flow path PL to flow. By placing some of the discharge nozzles **323** close to the rear end of the middle cabinet **124**, e.g., close to the discharge outlet **1501** of the fan module **15a**, some steam supplied from the discharge nozzles **323** passes to the lower flow path PL.

With reference to FIGS. **35** and **36**, a space is formed between the upper cabinet **123** and the middle cabinet **124**, and the steam supply pipe **32** is placed on the upper surface of the middle cabinet **124**. When the fan module **15a** operates, indoor air outside the hanger device **10a** passes through the filtering module **60** and flow into the suction port of the fan module **15a**. Further, any air present in the space between the upper cabinet **123** and the lower cabinet **122** flows into the suction port. The suction port of the fan

module **15a** may be understood as a space defined between the rear end of the middle cabinet **124** and the rear end of the fan housing **155**.

Although omitted from the drawings, in order to prevent air flow from occurring in the space between the middle cabinet **124** and the upper cabinet **123** when the fan module **15a** is driven, a rear end of the middle cabinet **124** may be connected to the edge of the filter outlet **1232** formed on the bottom of the upper cabinet **123**. Hence, only the air passing through the filtering module **60** may be sucked into the fan module **15a**. Further, by moving the fan module **15a** toward the rear side of the lower cabinet **122** and reducing the front-rear width of the steam generator **31**, a filter cleaner **40** may be placed between steam generator **31** and the rear surface of the lower cabinet **122**.

An outer partition walls **1224** may be installed spaced apart from the left side of the lower cabinet **122** and spaced apart from the right side of the lower cabinet **122**. The outer partition walls **1224** may be installed parallel to the side surface of the lower cabinet **122** and spaced apart by a predetermined distance. The fan motor **154** and the supports **51** of the hanging unit **50** may be disposed in a space defined by the side surfaces of the lower cabinet **122** and the outer partition walls **1224**. Each of the outer partition walls **1224** also forms a side of the duct defining the upper flow path PU and the lower flow path PL.

Further, a pair of inner partition walls **1225** may be arranged side by side inside the lower cabinet **122** corresponding to both sides of the steam generator **31**. The pair of inner partition walls **1225** form the other side surfaces of the duct defining the upper flow path PU and the lower flow path PL. The steam generator **31** and/or the filter cleaner **40** may be disposed between the pair of inner partition walls **1225**.

As shown in FIGS. **35** and **37**, a part of the air discharged from the discharge outlet **1501** of the fan module **15a** flows along the upper flow path PU formed on the upper side of the flow path partition plate **125**, and toward the front section of the flow path partition plate **125**. The air is discharged toward the clothes or items hung on the hanging bar **52** through the front discharge port **1223a**.

As shown in FIGS. **35** and **38**, a part of the air discharged from the discharge outlet **1501** of the fan module **15a** is discharged toward the clothes or items hung on the hanging bar **52** through the rear outlet **1223c**. Further, a portion of the air discharged from the discharge outlet **1501** of the fan module **15a** flows along the lower flow path PL formed under the flow path partition plate **125** and passes through the upper discharge port **1223b**. It is discharged toward the clothes hung on the hanging bar **52**.

The amount of air discharged through the upper air outlet **1223b** and the rear outlet **1223c** may be controlled differently based on the tilting angle of the discharge vanes **19a**. Hence, the degree of opening through which the air passes through upper discharge ports **1223b** and rear discharge ports **122c** may be independently controlled, and amount of air discharged can vary.

With reference to FIGS. **39** and **40**, the sub supply pipe **322** of the steam supply pipe **32** extends along the rear edge of the middle cabinet **124** and extends into the support **51**. The upper surface of the support **51** may be shielded or covered, and the sub-supply pipe **322** may be introduced into the support **51** through the cover of the support **51**. The upper surface of the cover for the support **51** may be located higher than the upper surface of the flow path partition plate **125**.

The side connecting the upper surface of the support **51** and the flow path partition plate **125** is opened. Specifically, a flow guide **1251** protrudes from the upper surface of the flow path partition plate **125**, and an upper side of the flow guide **1251** is connected to the lower surface of the middle cabinet **124**. A front end of the flow guide **1251** is connected to the open side of the support **51**. An inlet **1244** is formed between the rear end of the flow guide **1251** and the open side of the support **51**. A side passage **1252** is formed between the inlet **1244** and the open side of the support **51** by the flow guide **1251**.

The inlet **1244** is opened and closed by a gate **126**, and the gate **126** moves in the left and right direction by the driving motor M placed on the upper surface of the support **51**. The gate **126** is erected on the rear surface of the support **51** to be slidable in the left and right directions of the cabinet **12**. A pinion may be mounted on a rotation shaft of the driving motor M, and a gear rack engaged with the pinion may be formed on the upper surface of the gate **126**.

When the driving motor M rotates in one direction, the gate **126** moves in a direction closer to the side surface of the cabinet **12**, and the inlet **1244** is opened. When the inlet **1244** is opened, a part of the air discharged from the discharge outlet **1501** of the fan module **15a** is supplied to the side passage **1252** through the inlet **1244**, and introduced into the support **51**. The air passing through the support **51** is discharged through the outlets **522** formed on the bottom surface of the hanging bar **52**, where clothes and items are hung.

The side flow path **1252** is selectively opened or closed by the gate **126** or the degree of opening may adjusted to supply more or less air either the left side or the right side of the hanging unit **50**. For example, when the signals from the pressure sensor PS indicates more clothes or items are hung on the right side than on the left side of the hanging bar **52**, the gate **126** on the right side may be opened more than the gate **126** on the left side such that more air may be supplied to the support **51** connected to the right edge of the hanging bar **52**.

With reference to FIG. **40**, similarly to the hanger device **10** according to the previous embodiment, the wind guard **20a** may be installed in the hanger device **10a** of this embodiment. The wind guard **20a** descends a predetermined length downward from the bottom surface of the front end of the cabinet **12**, and prevents dispersion of dry air, warm air, and/or steam discharged through the air discharge unit **1223**. The dispersion may cause discomfort to users in front of the hanger device **10a**, and reduces drying effect or wrinkle removal effect of drying air, warm air, and/or steam discharged to the clothes or items hung on the hanging bar **52**. In order to minimize such problems, the wind guard **20a** may be controlled to descend during a drying mode or the wrinkle removal mode of the hanger device **10a**. As previously discussed, a mood light or a sterilization module may be also installed on the rear surface of the wind guard **20a** to additionally perform a lighting function or a sterilization function.

Referring to FIG. **42**, the hanging unit **50a** according to another embodiment may include a pair of length-adjustable supports **51a** and a hanging bar **52** connecting the pair of supports **51a**. As can be appreciated, the hanging unit **50a** may be incorporated into the previous embodiments to replace the hanging unit **50** previously discussed. The length of the support **51a** can be adjusted through a multi-stage extension, and the number of extension stages can be changed according to design conditions. In this example, the multi-stage

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extension of the support **51a** may include a stationary arm **53**, an intermediate arm **54**, and a movable arm **55**.

The stationary arm **53** may be fixed to the main body **11**, **110** of the previous embodiments, and communicates with a flow path of suctioned air into the main body **11**. A part of the suctioned air which has passed through the filtering modules **60** and **60a** is introduced into the stationary arm **53** and discharged to the outside through the outlet **522** of the hanging bar **52**. Further, a combination of the sub-supply pipe **322** and the heater **33** of the steam supply device **30** extends into the support **51a**.

With reference to FIGS. **43** and **44**, the stationary arm **53** includes an inner cover **531** having an open side and an outer cover **532** covering the open side of the inner cover **531**. A first accommodating space or a first channel (**533**) is formed inside the stationary arm **53** by the coupling of the inner cover **531** and the outer cover **532**, and the intermediate arm **54** is vertically movable in the channel.

The intermediate arm **54** includes a middle inner cover **541** having one side open, and a middle outer cover **542** covering the open side of the middle inner cover **541**. A second accommodating space or a second channel (**543**) is formed inside the middle arm **54** by the coupling of the middle inner cover **541** and the middle outer cover **542**, and the movable arm **55** is accommodated in the channel to be raised up and down in the vertical direction. Based on design considerations, a plurality of the intermediate arms **54** may be provided, where multiple second channels are by the intermediate arms, and intermediate arms from widest to narrowest width are sequentially provided in corresponding channels having widest to narrowest width.

The movable arm **55** includes a vertical arm **551** ascending and descending inside the second channel of the intermediate arm **54**, and a horizontal arm **552** extending from a lower end of the vertical arm **551**. The vertical arm **551** also includes a third channel **553** for passage of air, hot/warm dry air, or steam, and the horizontal arm **552** includes an opening **554** in communication with the third channel to allow air, hot/warm dry air or steam to be forced into the hanger bar **52**. An end of the hanging bar **52** is coupled to the horizontal arm **552**. The horizontal arm **552** is inserted into the hanging bar **52** so that an end of the hanging bar **52** is connected to the movable arm **55**.

A first lift unit **56** is coupled to the upper end of the movable arm **55**. A gear rack **5411** is formed on the inner surfaces of the middle inner cover **541** and the middle outer cover **542** such that a pair of gear racks face each other. The gear rack **5411** may be recessed to a predetermined depth at the center of the middle inner cover **541** and the middle outer cover **542** and are configured to engage a pair of first pinions (**5622** of FIG. **45**) of the first lift unit **56**. A stopping protrusion **5412** is formed at a predetermined height above a lower end of the intermediate arm **54** and are formed on the inner surfaces of the middle inner cover **541** and the middle outer cover **542**. The stopping protrusion **5412** limits the descent of the movable arm **55** and prevents movable arm **54** from sliding off the intermediate arm **54**.

A second lift unit **57** may be provided at an outer surface of the intermediate arm **54**. The second unit **57** may be provided on each of the side surfaces of the intermediate arm **54** facing each other. The second lifting unit **57** includes a mover **571** coupled to the outer surface at the upper end of the intermediate arm **55** and a stator **572** configured to penetrate the mover **571**. The pair of stators **572** may be coupled to the inner side surface of the stationary arm **53**.

When the intermediate arm **55** moves up and down, the mover **571** moves up and down along the stator **572**. The

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second lift unit **57** including the mover **571** and the stator **572** may comprise a linear actuator. Alternatively, a motorized lead screw actuator, a rotating screw configuration or a rotating nut configuration may be used as a second lift unit **57**. Other means may be used to vertically move the intermediate arm **54** within the fixed arm **53** when power is supplied, including but not limited to, a rack and a pinion and a motor for driving the pinion.

Guide grooves **5311** for guiding vertical movement of the intermediate arm **54** are formed on inner surfaces of the inner cover **531** and the outer cover **532** of the stationary arm **53**. A limiting protrusion **5312** is provided at the lower ends of the inner cover **531** and the outer cover **532**. The limiting protrusion **5312** limit the descent of the intermediate arm **54** and prevent the intermediate arm **54** from sliding out of the stationary arm **53**. The pair of stators **572** for the second lift unit **57** may be vertically erected to face each other inside the fixed arm **53**, and may be placed on the locking protrusion **5312**.

Referring to FIG. **45**, the first lift unit **56** includes a support cover **561** and a pair of first and second lifts **562**, **563** mounted on the support cover **561**. The support cover **561** is configured to have a structure that allows the air that has passed through the filtering module **60** to flow into the support **51a** while being coupled to the upper end of the movable arm **55**. The first lift **562** includes a first motor **5621**, a first pinion **5622**, and a first pinion shaft **5623** rotated by the first motor and connected to the first pinion **5622**. The second lift **563** includes a second motor **5631**, a second pinion **5632**, and a second pinion shaft **5633** rotated by the second motor **5631** and connected to the second pinion **5632**. The first and second pinion shafts **5623**, **5633** extends from one end to the other end of the support cover **561** and are parallel to face each other.

The support cover **561** includes a pair of motor mounts **5611**, a steam supply pipe support **5613**, and a pair of bridges **5612** coupling the pair of motor mounts **5611** to the steam supply pipe support **5613**. The pair of motor mounts **5611** comprises a first motor mount and a second motor mount placed in parallel with each other. The first motor **5621** may be accommodated on the first motor mount, and the other end of the first pinion shaft **5623** may be connected to, e.g., a slave motor or a roller bearing mounted to the second motor mount. The second motor **5631** may be accommodated on the second motor mount, and the other end of the second pinion shaft **5633** may be connected to, e.g., a slave motor or a roller bearing mounted to the first motor mount.

Each of the first pinion **5622** and the second pinion **5632** are respectively engaged with the gear rack **5411** extending lengthwise on the inner surfaces of the middle inner cover **541** and the middle outer cover **542**. When the first motor **5621** and the second motor **5631** rotate in any one direction, the support cover **561** and the movable arm **55** rise or descends as a single body. Other means for raising or lowering the movable arm **55** are possible other than the combination of the motor and rack and pinion structure including but not limited to linear motor actuator described above, a motorized lead screw, a rotating screw configuration or a rotating nut.

A guide hole **5614** through which the sub-supply pipe **322** passes is formed in the steam supply pipe support **5613**. The steam supply pipe support **5613** has a cylindrical shape of a prescribed diameter **D** and extend a predetermined length in a vertical direction to stably support the sub supply pipe **322**. Since an air flow path is formed between a pair of motor mounts **5611** facing each other, even when the support cover

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561 is coupled to the upper end of the movable arm 55, the air that has passed through the filtering module 60 may be supplied into the moving arm 55.

As shown in FIG. 46, an extension pipe 3221 may be slidable within the sub-supply pipe 322. A lower opened end of the extension pipe 3221 penetrates through a coupler 3222 and may be fixed to the coupler 3222. The coupler 3222 may have the same or similar shape as the support cover 561 such that the air flowing into the movable arm 55 passes through the coupler 3222 and flows to the hanging bar 52.

Referring to FIG. 47, in a state in which the length of the support 51 or 51a is at a minimum, i.e., in a state where the intermediate arm 54 and the movable arm 55 are fully inserted into the stationary arm 53, the support cover 561 is located on the upper end of the support 51 or 51a. Further, the coupler 3222 is positioned at the inner lower end of the movable arm 55.

An upper step or stop 5511 may be formed on the inner upper side of the movable arm 55. The upper step 5511 is a portion on which the upper end of the coupler 3222 is caught or stopped. Accordingly, from the moment when the upper end of the coupler 3222 is caught on the upper stepped 5511 while the moving arm 55 descends, the moving arm 55 and the extension pipe 3221 descend as one body.

A lower step or stop 5512 may be formed at an inner lower side of the movable arm 55. The lower step 5512 is a portion on which the lower end of the coupler 3222 is caught or stopped. Accordingly, from the moment when the lower end of the coupler 322 is caught on the lower stepped 5512 while the moving arm 55 rises, the moving arm 55 and the extension pipe 3221 rise as one body.

Referring to FIG. 48, a portion of the air that has passed through the filtering module 60 passes through the support 51 or 51a to the hanging bar 52 irrespective of the extended or contracted state. In a state in which the movable arm 55 is maximally descended, the upper end of the extension pipe 3221 is maintained in a state inserted into the sub-supply pipe 322. The intermediate arm 54 may descend until the stator 571 is caught by the limiting protrusion 5312 of the stationary arm 53. The support cover 561 of the movable arm 55 descends in the intermediate arm 54 until the support cover 561 is caught by the stopping protrusion 5412 of the intermediate arm 54.

In a first method to extend the length of the support 51 or 51a, the intermediate arm 54 may be withdrawn first from the stationary arm 53, and thereafter, the movable arm 55 may be withdrawn from the intermediate arm 54. The intermediate or movable arms may or may not be withdrawn to a maximum. The order in which the support 51 or 51a is contracted or raised may be performed in the reverse order of the order in which it is extended, but may not be limited thereto.

In a second method to extend the length of the support 51 or 51a, movable arm may be withdrawn first from the intermediate arm 54, and thereafter, the intermediate arm 54 is withdrawn from the fixed arm. The intermediate or movable arms may or may not be withdrawn to a maximum. The order in which the support 51a is contracted or raised may be performed in the reverse order of the order in which it is extended, but may not be limited thereto.

Referring to FIG. 49-53, the filtering module 60 according to another embodiment may have substantially the same configuration as the filtering module 60 of the previous embodiments, but there are some differences. The filtering module 60 may further include a shield frame 68, a dehumidifying filter 63 and/or deodorizing filter provided in the

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shield frame 68, and a shield plate 69 configured to cover at least one of the dehumidifying filter 63 or the deodorizing filter 63a. A sliding motor SM mounted on a motor support 605 may be powered to slide the shielding plate 69 in a direction between the front and rear of the shield frame 68.

The shield frame 68 is provided between the HEPA filter 64 and the heater 62, and the shield plate 69 is provided between the HEPA filter 64 and the dehumidifying and/or deodorizing filter 63, 63a. the deodorizing filter 63a may be disposed in front of the dehumidifying filter. The dehumidifying filter 63 and the deodorizing filter 63a may have the same left-right length but may have same or different front-rear width. If either of the filters 63, 63a is larger than the other, the shielding plate 69 may have a size corresponding to the larger filter 63 or 63a. In this described embodiment, each of the dehumidifying filter 63 and the deodorizing filter 63a may be sized to halve the shield frame 68 in the front and rear directions. The shielding plate 69 may have a size corresponding to the size of the dehumidifying filter 63 or the deodorizing filter 63a.

The front-rear width of the shield frame 68 may have a size corresponding to the front-rear width of the lower case 61. The left-right length of the shield frame 68 may have the same length of the HEPA filter 64. Accordingly, all or most of the air passing through the HEPA filter 64 is guided to an inner space of the shield frame 68 and passes through one or both of the filters 63, 63a. Slits 681 are formed on the left and right sides of the shield frame 68, and are configured to receive the left and right edges of the shielding plate 69.

The shielding plate 69 includes a main plate 691 and a pair of extension ends 692 (first and second extension ends) extending from left and right ends of the main plate 691. The main plate 691 covers and shields at least one of the dehumidification filter 63 or the deodorization filter 63a. Each of the extension ends 692 is stepped downward from the top surface of the main plate 691 to have a thickness thinner than that of the main plate 691 such that after inserting the extension ends into the slits 681, the upper surface of the main plate 691 and the upper surface of the shield frame 68 are same or substantially planar. The first extension end and the second extension end may have the same or different left-right lengths. A gear rack 693 is formed on a bottom surface of one of the first and second extension ends 692.

A moving member for moving the shielding plate 69 in the front-rear direction is configured to be coupled with the gear rack 693. The moving member includes a pinion 601 configured to engage with the gear rack 693, and a sliding motor SM for rotating the pinion 601. A motor support 605 may be provided on a side surface of the shield frame 68. The motor support 605 has a shape of rectangular pole having a predetermined left-right width with an opening on the upper and lower surfaces and has the same front and rear width of the shield frame 68. The motor support 605 is provided with a slit 605a facing the slit 681 of the shield frame 68 and contacts the shield frame 68. When the first extended end 692 of the shield plate 69 having the gear rack 693 is inserted into the slits 681, 651a, the pinion 601 engages with gear rack 693. As can be appreciated, another type of structure for stably supporting the sliding motor SM may be used, or a structure in which the sliding motor SM is fixed to the edge of the upper surface of the heater 62 may be used.

When the pinion 601 rotates in a first direction to move the shielding plate 69 and to contact with the front end of the shield frame 68, the shielding plate 69 completely covers the deodorizing filter 63a. In this instance, all of the air sucked

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into the filtering module **60** passes through the dehumidifying filter **63** such that only the dehumidifying function is performed.

When the pinion **601** rotates in a second direction opposite the first direction to move the shielding plate **69** and to contact with the rear end of the shield frame **68**, the shielding plate **69** completely shields the dehumidifying filter **63**. In this instance, all of the air sucked into the filtering module **60** passes through the deodorization filter **63a** such that only the deodorization function is performed.

When the shield plate **69** is positioned at the center of the shield frame **68**, a part of the air sucked into the filtering module **60** passes through the deodorization filter **63a** and the remaining air passes through the dehumidifying filter **63** such that the dehumidifying and deodorizing functions are simultaneously performed. As can be appreciated, the area covering the dehumidification filter **63** and the area covering the deodorization filter **63a** may be set differently according to the position of the shielding plate **69** in the front-rear direction.

As can be appreciated, both the dehumidifying filter and the deodorizing filter **63a** need not be placed inside the shield frame **68**. For example, when the dehumidifying filter **63** is placed in an inner half of the shield frame **68** and the other half may be left as an empty space, the deodorizing function will not be performed, and only the dehumidifying function will be selectively performed. Similarly, when only the deodorizing filter **63a** is provided, the dehumidifying function may not be performed and only the deodorizing function may be selectively performed. In an alternative embodiment, the inner half of the shield frame **68** may be filled by the dehumidifying filter **63**, and the other half is left as an empty space, and the deodorizing filter **63a** is placed on the upper side of the shielding plate **69**. In this alternative, the deodorization filter **63a** may be provided having the same size as the size of the shield frame **68** such that the deodorization function is always performed, and only the dehumidification function may be selectively performed by the shield plate **69**.

With reference to FIGS. **54** and **55**, the hanger device **10** according to another embodiment may further include a moving device for horizontally moving the main body **11** in the front-rear direction from the installation wall surface. The moving device may include a support plate (SP) and a pair of support rails (**1101**) extending from the front surface of the support plate (SP). The pair of support rails **1101** extend into the main body **11** through the rear surface of the main body **11**. The support plate SP is fixed to the installation wall and may have a rectangular plate shape. The support rail **1101** may extend from a front left edge and a front right edge of the support plate SP.

A sensing device for detecting the width of clothes hung on a hanger may be mounted on the front surface of the support plate SP. The sensing device may be provided in various forms, for example, an infrared sensor **1300**. The infrared sensor **1300** includes a light emitter **1301** mounted on one side of the front lower end of the support plate SP, and a light receiver **1302** mounted on the other side of the front lower end of the support plate SP and facing the light emitter **1301**. When the light irradiated from the light emitter **1301** is sensed by the light receiver **1302**, the control unit of the hanger device **10** determines that the clothes or items hung on the hanger do not touch the wall. On the other hand, if the light irradiated from the light emitter **1301** is not detected by the light receiver **1302**, the control unit determines that the clothes or items on the hanger are in contact

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with the wall, and the main body **11** horizontally moves away from the installation wall in the front-rear direction.

Referring to FIG. **56** (for explanation purposes only, the position of the upper and lower cabinets is reversed in FIG. **56**), the moving device further includes a guide rail **1102** and a power generating member coupled to the support rail **1101**. The guide rail **1102** is coupled to the inner edge of the main body **11**, specifically, the cabinet **12**, and moves as one body with the main body **11**. The support rail **1101** is coupled to the guide rail **1102** to pass through the guide rail **1102**. A gear rack **1101a** is formed on the bottom surface of the support rail **1101**.

The power generating member includes a motor **1203** and a pinion **1201** coupled to a rotation shaft **1202** of the motor **1203**. The motor **1203** may be fixed to an upper surface of the discharge guide **1215** accommodated in the body **11**. A motor seat **1215a** on which the motor **1203** is seated may be formed on the upper surface of the discharge guide **1215**, and the motor seat **1215a** is in the form of a depression accommodating a portion of the motor **1203**. The pinion **1201** is engaged with the gear rack **1101a** of the support rail **1101**.

When power is applied to the motor **1203** and the pinion **1201** rotates in one direction based on the sensor detection, the guide rail **1102** moves forward along the support rail **1101**. Since the guide rail **1102** is formed as one body with the main body **11** and as the guide rail **1102** moves forward along the support rail **1101**, the main body **11** moves away from installation wall surface. Conversely, when the pinion **1201** rotates in the other direction, the guide rail **1102** and the main body **11** moves toward a direction closer to the installation wall surface.

As can be appreciated, a moving device for moving the main body **11** in the front-rear direction, a rail structure is presented, but is not limited thereto, and various forms including a motorized lead screw, a linear motor, or an electric cylinder actuator capable of multi-stage withdrawal may be used. In such a case, a moving unit such as the lead screw or a linear motor connects the support plate SP and the main body **11** so that the main body **11** moves away from the support plate SP or is close to the support plate SP.

Referring to FIG. **58**, the eccentricity applied to the hanging bar **52** may be extracted using pressure values transmitted from the pressure sensors PS attached to the hanger groove **521** as described for the previous embodiments. See, e.g., FIG. **1**, FIG. **31**, FIG. **42**, or FIG. **54** embodiments. In addition to or separate from the pressure sensors PS, a gyro sensor **523** may be installed on the hanger bar **52**. The gyro sensor **523** may be mounted on the center of the top surface or the center of the bottom surface of the hanging bar **52**. Alternatively, an accelerometer may be used.

When clothes or items are hung on either one of the left and right sides of the hanging bar **52**, an inclination of the hanging bar **52** may be detected by the gyro sensor **523**. According to the eccentricity value sensed by the gyro sensor **523**, more air, warm dry air, or steam may be supplied to the side of the hanger having larger incline detected by the gyro sensor **523**. For example, in the FIG. **1** embodiment, the rotational speed of the main fan module **15** and the sub-fan module **16** corresponding to the side on which more clothes are items are hung is adjusted or increased. In another example for FIG. **31** embodiment, the opening of inlet **1244** may be adjusted by the gate **126** described in FIGS. **39** and **40**.

The pair of supports **51** may include an elastic material in order for the gyro sensor **523** to detect the inclination of the

hanging bar **52**. For example, a portion or a part of each of the pair of supports **51** may be formed to include a corrugated pipe such that the length of the support **51** may be changed by the force applied to the hanging bar **52**. In addition to this method, the support **51** and the hanging bar **52** may be connected by an elastic body such that when different magnitudes of force are applied to both ends of the hanging bar **52**, the hanging bar **52** is inclined.

When more clothes or items are hung on one side of the hanging bar **52**, a notification message is output through a display unit provided on the front of the main body **10** or a speaker provided on the main body **10**. For example, when clothes are hung on the left side of the bar **52**, a notification message requesting to hang clothes by evenly distributing the clothes to the other side can be output in the form of text, image, video, or voice.

FIG. **59** shows a schematic of the electronic components or module for the various described embodiments. The electronic components may include a communication unit **310**, an input unit **320**, a learning processor **330**, a sensing unit **340**, an output unit **350**, a memory **370**, and a processor **380** (also referred to as a control unit).

The communication unit **310** may transmit/receive data to and from external devices such as another hanger device or an external server using wired/wireless communication technology. For example, the communication unit **310** may transmit and receive sensor information, a user input, a learning model, a control signal, and the like, from external devices. Presently, the communication technology used by the communication unit **310** may include GSM (Global System for Mobile communication), CDMA (Code Division Multi Access), LTE (Long Term Evolution), 5G, WLAN (Wireless LAN), Wi-Fi (Wireless-Fidelity), Bluetooth™, RFID (Radio Frequency Identification), Infrared Data Association (IrDA), ZigBee, NFC (Near Field Communication), etc.

The input unit **320** may acquire or receive various types of data. The input unit **320** may include a camera **3210** (FIG. **60**) for inputting an image signal, a microphone for receiving an audio signal, a user buttons or touch panel for receiving information from a user. The various data or signal provided by the input unit may be referred to as sensing data or sensor information.

The learning processor **330** may train a model composed of an artificial neural network by using training data. The learned artificial neural network may be referred to as a learning model. The learning model may be used to infer a result value with respect to new input data other than the training data, and the inferred value may be used as a basis for a decision to perform a certain operation. The learning processor **330** may include a memory integrated or implemented in the hanger device. Alternatively, the learning processor **330** may use the memory **370**, an external memory directly coupled to the hanger, or a memory maintained in an external device.

The sensing unit **340** may acquire at least one of internal information of the clothes hanger devices **10** and **10a**, surrounding environment information of the clothes hanger devices **10** and **10a**, and user information by using various sensors. The sensors may include a proximity sensor, an illuminance sensor, an acceleration sensor, a magnetic sensor, a gyro sensor, an inertial sensor, an RGB sensor, an IR sensor, a fingerprint recognition sensor, an ultrasonic sensor, a light sensor, a microphone, Lidar, Radar, etc.

The output unit **350** may generate an output related to sight, hearing, or touch. The output unit **350** may include a display unit that outputs visual information, a speaker that

outputs auditory information, a haptic module that outputs tactile information, and the like.

A memory **370** may store data to support various functions of the hanger device. The memory **370** may store input data obtained from the input unit **320**, learning data, a learning model, a learning history, and the like. The memory may comprise one or more of RAM, SRAM, ROM, solid state hard drive, etc.

The processor **380** may determine at least one executable operation of the hanger device based on information determined or generated using a data analysis algorithm or a machine learning algorithm. In addition, the processor **380** may control the components of the hanger devices to perform various operation. The processor **380** may request, search, receive, or utilize the data of the learning processor **330** or the memory **370**, and perform a predicted operation or an operation determined to be desirable among the at least one executable operation.

The processor **380** may also control various component to operate a prescribe function of the hanger device based on data received through the communication unit **310**. When the connection of the external device is required to perform an operation, the processor **380** may generate a control signal for controlling the corresponding external device and transmit the generated control signal to the corresponding external device.

The processor **380** may obtain information with respect to a user input, and determine a user's requirement based on the obtained information. The processor **380** may use at least one of a speech to text (STT) engine for converting a voice input into a character string or a natural language processing (NLP) engine for obtaining intention information of a natural language. At least one of the STT engine or the NLP engine may be configured as an artificial neural network, where at least a part of which is learned according to a machine learning algorithm. At least one of the STT engine and the NLP engine may be learned by the learning processor **330**.

The processor **380** may collect history information including user feedback on the operation contents or operation of the hanger device, and store the information in the memory **370** or the learning processor **330**, or transmit the information to an external device. The collected historical information may be used to update the learning model. The processor **380** may also control at least some of the components of the clothes hanger device in order to drive an application program stored in the memory **370**. The processor **380** may operate by combining two or more of the components included in the hanger devices to drive the application program.

As shown in FIG. **60**, a camera **3210** may be disposed under the body **11** in a direction toward the hanging unit **50** of the FIG. **1** embodiment. As shown in FIG. **61**, the camera **3210** may be disposed on the lower surface of the lower cabinet **122** in a direction toward the hanging unit **50**.

With reference to FIG. **62-64**, the camera **3210** may photograph the hanger area including the hanging unit **50** (step **S4601**). The camera **3210** may generate an image by photographing a hanger area including the hanging unit **50**. The processor **380** may acquire the image captured by the camera **3210** (step **S4602**). FIG. **63** is an exemplary view of the image **4701**, the image being a photographed image of the hanger area including the hanging unit **50**.

Based on the image **4701**, the processor **380** may acquire clothes information regarding at least one clothes hung on the hanging unit **50** (step **S4603**). The clothes information may include at least one of information on the number of

clothes hung on the hanging unit **50**, or area information of each of at least one clothes and location information. The location information may correspond to coordinate values of clothes recognized in the image **4701**.

The processor **380** may input the hanger image **4701** to the image recognition model **4801** to obtain clothes information **4802** output from the image recognition model **4802**. The image recognition model **4801** may be an artificial neural network (ANN) trained to recognize clothes included in an image file corresponding to the input data, and output clothes information including the total number of recognized clothes, and/or the area and location information of each recognized clothes. An artificial neural network (ANN) is a model used in machine learning, and may refer to an overall model having problem-solving ability, which is composed of artificial neurons (nodes) that form a network by combining synapses. The image recognition model may be a model stored in the memory **370**.

As shown in FIG. **64**, the processor **380** may input the hanger image **4701** for the image recognition model **4801**. The processor **380** may obtain the clothes information **4802** output from the image recognition model **4801**. The output clothes information **4802** may include a total number of recognized clothes, an area information of each recognized clothes, and a location information. For example, the output clothes information **4802** may include a total number '3' for the clothes **4705**, **4706**, and **47007** recognized in the hanger image **4701**, an area information of each of the recognized clothes **4705**, **4706** and **47007**, and a location information. The area information may refer to information about at least one area of the hanger area in which clothes are recognized. The location information may correspond to a coordinate value in an image of an area where clothes are recognized.

The processor **380** may determine whether clothes are hung on the hanging unit **50** based on the clothes information (step **S4604**). For example, the processor **380** may determine whether clothes are hung on the hanging unit **50** based on information on the total number of recognized clothes included in the clothes information.

When clothes are hung on the hanging unit **50**, the processor **380** may change the illuminance or color of the mood light (step **S4604**). For example, when it is determined that clothes are hung on the hanging unit **50**, the processor **380** may turn on the mood light **17** and allow light to be gently emitted through the through hole **101** to notify the user that clothes are hung on the hanger devices, e.g., **10** and **10a**. Further, power is applied to the hanger devices, e.g., **10** and **10a**, in preparation for hot/dry air supply mode (or drying mode) or steam supply mode (the humidification mode or the wrinkle removal mode).

The processor **380** may determine an area with more clothes from among a plurality of divided areas of the hanger area based on area information and location information of each piece of clothing included in the clothes information (step **4605**). For example, referring to FIG. **63**, the clothes hanger image **4701** may be divided into a first area **4703** and a second area **4704** on the left and right based on a central dividing line **4708** in the entire area. The processor **380** may determine which of the first and second areas the recognized clothes **4705**, **4706**, and **4707** belong to based on location information of the recognized clothes information. For example, the processor **380** may determine that the first clothes **4705** belong to the first area **4703** and the second clothes **4706** and the third clothes **4707** belong to the second area **4704** based on the location information of the clothes.

Based on the result of step **S4605**, the processor **380** may supply more air, warm air, or steam to an area where more

clothes are hung (step **S4606**). For example, in the case of the hanger device **10** according to the FIG. **1** embodiment, the processor **380** may change the rotation speed of the main fan module **15** and the sub fan module **16** corresponding to the area or region where more clothes are hung. The rotation speed of the main fan module **15** and the sub-fan module **16** of the region can be controlled to be higher than that of the other region with less clothes. In the case of the hanger device **10a** of FIG. **311** embodiment, the processor **380** adjusts the amount of movement of the gate **126** described in FIGS. **39** and **40** such that the inlet **1244** is more open on the side of the lower cabinet **122** where more clothes are hung.

Referring to FIG. **65**, the hanger device **10** of the FIG. **1** embodiment may include at least one camera, which may comprise a front camera **3210a** on the front surface of the main body **11**, a left camera **3210bL** a left side surface of the main body **11**, and/or a right camera **3210bR** on a right side surface of the main body **11**. The cameras may be arranged in a direction in which the surrounding area of the hanger device **10** may be photographed. As shown in FIG. **66**, the cameras **3210a**, **3210bL** and/or **3210bR** are similarly arranged for the hanger device **10a** of the FIG. **31** embodiment.

With reference to FIGS. **67** to **70**, at least one of a front, left or right camera **3210a**, **3210bL**, **3210bR** captures at least one image of the peripheral/surrounding region (step **S4901**). The processor **380** acquire an image of the peripheral/surrounding region captured by at least one camera (step **S4602**). The peripheral region image may include at least one of a front area image captured by the front camera **3210a**, a left area image captured by the left camera **3210bL**, and a right area image captured by the right camera **3210bR**.

For example, the processor **380** may acquire a right region image including the first object **4708R** from the right side by the right camera **3210bR**. The processor **380** may acquire a left area image including the second object **4708L** approached from the left by the left camera **3210bL**. See FIG. **68**. The processor **380** may acquire a front area image including a third object **4709** approached from the front by the front camera **3210a**.

The processor **380** may acquire proximity information regarding an object approaching or present around the hanger devices **10** and **10a** based on at least one image of the peripheral region (**S4603**). The proximity information may include at least one of object information about an object approaching, present around or retreating from the hanger devices **10** and **10a**, proximate area information, and proximate distance information about a distance from the hanger devices **10** and **10a**. The object information may include information on whether an object is a human, an animal, or an object. The proximate area information may be information regarding whether to approach from the front, left, or right area.

As shown in FIG. **70**, the processor **380** may obtain proximity information **4903** output from the image recognition model **4801a** by inputting at least one image **4901** of the peripheral region to the image recognition model. The image recognition model may be an artificial neural network (ANN) trained to output proximity information including object information and approach distance information by recognizing an object included in an image file with respect to an image file that is input data. An artificial neural network (ANN) is a model used in machine learning, and may refer to an overall model having problem-solving ability, which is composed of artificial neurons (nodes) that

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form a network by combining synapses. The image recognition model may be a model stored in the memory 370.

The processor 380 may control the operation of the hanger device based on the obtained proximity information (step S4604). The processor 380 may determine whether a person is approaching the hanger devices 10 and 10a based on the object information and the proximate distance information included in the proximity information.

For example, when the power of the hanger devices 10 and 10a is not turned on or when it is determined that a person is approaching the hanger device based on the proximity information in the standby mode, the processor 380 supplies power to the hanger devices 10 and 10a or the hot air mode (or drying mode) or steam supply mode (humidification mode or wrinkle removal mode) is turned on to prepare the hanger devices 10 and 10a for operation. Further, the mood light 17 may be turned on and illuminance or color of light may be adjusted for emission through the through hole 101. The mood light may inform the user that the hanger devices 10 and 10a have recognized the approach of a person.

The processor 380 may also determine a region or an area to which the object is present based on the proximity information. For example, the processor 380 may determine whether the object is approaching from the front, left, or right area/region based on the proximate area information included in the proximity information. Based on the proximity information, the processor 380 may reduce or change the supply of air, warm/hot air, or steam in an area of the hanger device corresponding to the proximate area information when the hanger device (10, 10a) is operating in the hot air supply mode (or drying mode) or steam supply mode (humidification mode or wrinkle removal mode).

For the hanger device 10 of FIG. 1 embodiment, the processor 380 may determine the rotational speed of the main fan module 15 and the sub-fan module 16 corresponding to the area to which the object is approaching based on the proximity information. For example, the processor 380 may reduce the rotational speeds of the left main fan module 15 and the left sub-fan module 16 corresponding to the left area/region to which the object is approaching. When the object approaches from the front, the processor 380 may change the rotational speed of both left and right main fan modules 15 and the sub-fan modules 16 to be lower than the existing rotation speed.

In the case of the hanger device 10a according to FIG. 31 embodiment, the processor 380 may adjust the movement of the gate 126 described in FIGS. 39 and 40 to decrease or increase the opening of the inlet 1244. For example, the processor 380 may control the movement of the gate 126 on the left side to reduce the supply of air, warm/hot air, or steam if the object is approaching from the left side of the hanger device 10a.

A hanger device of present disclosure includes a main body (11) including a at least one cabinet (12) and a cover (3) to cover the at least one cabinet, the cover having at least one opening and the cabinet having at least one discharge port (1207). A fan module (15, 16) is provided in the cabinet and is configured to suction air through the opening of the cover and force the air to the discharge port. A filtering module (60) is disposed over the fan module to filter air as the air is suctioned through the opening of the cover. A discharge tube (14) extends through upper and lower surfaces of the main body and has at least one discharge duct (141, 142) which communicate with the at least one discharge port of the cabinet. A hanging unit (50) extends from the bottom surface of the main body.

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A hanger device of the present disclosure may include a body (11, 110) having at least opening (openings of cover 13, 1232) to allow air to be suctioned into the body and at least one discharge port (1207, 1223a, 1223b, or 1223c) to allow suctioned air to be forced out of the body. A fan module (15, 15a) is provided inside the body and is configured to suction air through the opening and force the suction air to the at least one discharge port. A filtering module (60) removes foreign substances contained in the suctioned air. A hanging unit (50a) is coupled to the main body, wherein the hanging unit includes: a pair of supports (51a) extending from a bottom of the main body and having at least one lift unit (56, 57) to change a length of the supports in a vertical direction; a hanging bar (52) connecting the ends of the pair of supports; and an air flow path (L2) provided in the pair of supports to allow portion of the suctioned air to pass through the pair of supports.

A hanger device according to the present disclosure may comprise a body (11, 110) having at least opening (openings of cover 13, 1232) to allow air to be suctioned into the body and at least one discharge port (1207, 1223a, 1223b, or 1223c) to allow suctioned air to be forced out of the body. A fan module (15, 15a) is provided inside the body and is configured to suction air through the opening and force the suction air to the at least one discharge port. A filtering module (60) removes foreign substances contained in the suctioned air. A steam supply device (30) is provided inside the body to generate steam and supply the generated steam to the discharge port. A hanging unit (50, 50a) is configured to extend from a bottom surface of the body.

A hanger device according to the present disclosure may comprise a body (11, 110) having at least opening (openings of cover 13, 1232) to allow air to be suctioned into the body and at least one discharge port (1207, 1223a, 1223b, or 1223c) to allow suctioned air to be forced out of the body. A fan module (15, 15a) may be provided inside the body and may be configured to suction air through the opening and force the suction air to the at least one discharge port. A dust removal filter (64, 65) is configured to remove dust contained in the air suctioned in by the fan module. A dehumidifying filter (63) may absorb moisture contained in the air that has passed through the dust removing filter. A heater (62, IU) may be configured to heat the air that has passed through the dehumidifying filter. A hanging unit (50, 50a) is configured to extend from the bottom surface of the body and configured to allow passage of a portion of suctioned air being forced out of the body.

A hanger device according to the present disclosure may comprise a body (11, 110) having at least one opening (openings of cover 13, 1232) to allow air to be suctioned into the body and at least one discharge port (1207, 1223b, or 1223c) to allow suctioned air to be forced out of the body. A fan module (15, 15a) may be provided inside the body. A filter (60) may be configured to remove foreign substances contained in the suctioned air. A hanging unit (50, 50a) extends from the bottom surface of the main body. At least one wind guard (20a) may be provided in front of the hanging unit, and configured to move in a vertical direction, and a driving unit (18a) may slide the wind guard in a vertical direction.

A hanger device of the present disclosure may comprise a body (11, 110) having at least one opening (openings of cover 13, 1232) to allow air to be pulled into the body and at least one discharge port (1207, 1223b, or 1223c) to allow pulled air to be pushed out of the body. A fan module (15, 15a) provided inside the body, and a filtering module (60) is disposed over the fan module. A hanging unit (50, 50a)

extends from a bottom surface of the body. The filtering module may include a dehumidifying filter (63) for absorbing moisture contained in the air drawn in by the fan module; and a shielding plate (69) configured to move in a horizontal direction over an upper surface of the dehumidifying filter to shield at least part of the dehumidifying filter.

A hanger device according to the present disclosure may include a body (11, 110) having at least one opening (openings of cover 13, 1232) to allow air to be pulled into the body and at least one discharge port (1207, 1223b, or 1223c) to allow pulled air to be pushed out of the body. A fan module (15, 15a) is configured to pull air into the body and push air toward the discharge port, and a filter (63, 63a, 64 or 65) is provided over the fan module to filter air drawn in by the fan module. A hanging unit (50, 50a) extends from a bottom surface of the body. A filter cleaner (40) is provided inside the body to clean to the filter, and the filter cleaner may be positioned between the filter and the hanging unit.

A hanger device according to the present disclosure may include a body (11, 110) having at least one opening (openings of cover 13, 1232) to allow air to be drawn into the body and at least one discharge port (1207, 1223b, or 1223c) to allow drawn air to be pushed out of the body. A fan module (15, 15a) is provided inside the body, and a filter (60) is configured to remove foreign substances contained in the air being drawn in by the fan module. A hanging unit (50, 50a) may extend from the bottom surface of the body. At least one wind guard (20a) may be provided in front of the hanging unit and configured to move in a vertical direction. A driving unit (18a) may slide the wind guard in the vertical direction, and a sterilization module (100) may be provided on a rear surface of the wind guard.

A hanger device according to the present disclosure may include a body (11, 110) having at least one opening (openings of cover 13, 1232) to allow air to be pulled into the body and at least one discharge port (1207, 1223b, or 1223c) to allow pulled air to be pushed out of the body. The body is configured to move away or toward a wall surface. A fan module (15, 15a) is provided inside the body, and a filtering module (60) disposed over the fan module. A hanging unit (50, 50a) extends from a bottom surface of the body.

A hanger device according to the present disclosure may include a body (11, 110) having at least one opening (openings of cover 13, 1232) to allow air to be pulled into the body and at least one discharge port (1207, 1223b, or 1223c) to allow pulled air to be pushed out of the body. A fan module (15, 15a) is provided inside the body, and a filtering module (60) is disposed over the fan module. At least one air passage (MAP1, MAP2, SAP1, SAP2 or PL, PU) is formed between an outlet (1501) of the fan module and the discharge port. A hanging unit (50, 50a) includes a hanging bar (52) extending horizontally in the width direction of the body and is provided under the discharge port, and a pair of supports (51) extends from a lower surface of the main body to support both ends of the hanging bar. A camera disposed under the body to face toward the hanging unit to photograph a hanger region including the hanging unit.

A hanger device according to the present disclosure may include a body (11, 110) having at least one opening (openings of cover 13, 1232) to allow air to be pulled into the body and at least one discharge port (1207, 1223b, or 1223c) to allow pulled air to be pushed out of the body. A fan module (15, 15a) is provided inside the body, and a filtering module (60) is disposed over the fan module. At least one air passage (MAP1, MAP2, SAP1, SAP2 or PL,

PU) is formed between an outlet (1501) of the fan module and the discharge port. A hanging unit (50, 50a) includes a hanging bar (52) extending horizontally in the width direction of the body and provided under the discharge port, and a pair of supports (51) extends from a lower surface of the body to support both ends of the hanging bar. At least one camera is disposed on the body in a direction to photograph the surrounding region of the body.

It will be understood that when an element or layer is referred to as being “on” another element or layer, the element or layer can be directly on another element or layer or intervening elements or layers. In contrast, when an element is referred to as being “directly on” another element or layer, there are no intervening elements or layers present. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

It will be understood that, although the terms first, second, third, etc., may be used herein to describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms are only used to distinguish one element, component, region, layer or section from another region, layer or section. Thus, a first element, component, region, layer or section could be termed a second element, component, region, layer or section without departing from the teachings of the present disclosure.

Spatially relative terms, such as “lower”, “upper” and the like, may be used herein for ease of description to describe the relationship of one element or feature to another element(s) or feature(s) as illustrated in the figures. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation, in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “lower” relative to other elements or features would then be oriented “upper” relative to the other elements or features. Thus, the exemplary term “lower” can encompass both an orientation of above and below. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly.

The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the disclosure. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

Embodiments of the disclosure are described herein with reference to cross-section illustrations that are schematic illustrations of idealized embodiments (and intermediate structures) of the disclosure. As such, variations from the shapes of the illustrations as a result, for example, of manufacturing techniques and/or tolerances, are to be expected. Thus, embodiments of the disclosure should not be construed as limited to the particular shapes of regions illustrated herein but are to include deviations in shapes that result, for example, from manufacturing.

Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this disclosure belongs. It will be further understood that terms, such as those defined in commonly used diction-

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aries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

Any reference in this specification to “one embodiment,” “an embodiment,” “example embodiment,” etc., means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the disclosure. The appearances of such phrases in various places in the specification are not necessarily all referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with any embodiment, it is submitted that it is within the purview of one skilled in the art to effect such feature, structure, or characteristic in connection with other ones of the embodiments.

Although embodiments have been described with reference to a number of illustrative embodiments thereof, it should be understood that numerous other modifications and embodiments can be devised by those skilled in the art that will fall within the spirit and scope of the principles of this disclosure. More particularly, various variations and modifications are possible in the component parts and/or arrangements of the subject combination arrangement within the scope of the disclosure, the drawings and the appended claims. In addition to variations and modifications in the component parts and/or arrangements, alternative uses will also be apparent to those skilled in the art.

The invention claimed is:

1. A hanger device comprising:

a body having at least opening to allow air to be suctioned into the body and at least one discharge port to allow suctioned air to be forced out of the body;

a fan module provided inside the body and configured to suction air through the opening and force the suction air to the at least one discharge port;

a filtering module for removing foreign substances contained in the suctioned air;

a hanging unit coupled to the main body, wherein the hanging unit includes:

a pair of supports extending from a bottom of the main body and having at least one lift unit to change a length of the supports in a vertical direction;

a hanging bar connecting the ends of the pair of supports; and

an air flow path provided in the pair of supports to allow portion of the suctioned air to pass through the pair of supports,

wherein each of the pair of supports comprises:

a stationary arm fixed to the body;

an intermediate arm movable in a vertical direction while coupled to the stationary arm; and

at least movable arm movable in the vertical direction while coupled to the intermediate arm,

wherein the at least one lift unit includes:

a first lift unit provided on the movable arm and configured to move the movable arm relative to the intermediate arm; and

a second lift unit provided on the intermediate arm and configured to move the intermediate arm relative to the stationary arm,

wherein at least one of the first lift unit or the second lift unit includes one of rack and pinion with a motor, a linear actuator or a motorized lead screw.

2. The hanger device of claim 1, wherein the air flow path extends into the hanging bar, which is hollow.

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3. The hanger device of claim 1, further comprising a steam supply device provided in the body, the steam supply device including:

a steam generator configured to convert liquid to steam;

a main supply pipe extending from the steam generator extending inside the main body to provide steam from the steam generator to the at least one discharge port;

a sub-supply pipe extending from the steam generator and extending into the pair of supports; and

an extension pipe configured to be movable relative to the sub-supply pipe such that the extension pipe moves vertically with movement of at least one of the intermediate arm or the movable arm.

4. The hanger device of claim 3, wherein a coupler is mounted at the lower end of the extension tube, and the coupler is accommodated in the movable arm.

5. The hanger device of claim 1, the filtering module includes:

a dust removing filter for filtering dust contained in the suctioned;

a dehumidifying filter for absorbing moisture contained in the suctioned air that has passed through the dust removing filter; and

a deodorizing filter for removing odor from the suctioned air that has passed through the dust removal filter.

6. The hanger device of claim 5, wherein a heater is provided in the filtering module for at least one of heating the suctioned air or recharging the dehumidifying filter.

7. The hanger device of claim 1, further comprising an induction heater provided in the body and adjacent to the fan module to heat the suctioned air.

8. The hanger device of claim 1, wherein the stationary arm comprises a first inner cover and a first outer cover coupled to each other and forming a first channel to allow vertical movement of the intermediate arm within the stationary arm, and a limiting protrusion formed at the bottom of the first channel to limit the downward movement of the intermediate arm.

9. The hanger device of claim 8, wherein the intermediate arm comprises a second inner cover and a second outer cover coupled to each other and configured to slide along the first channel of the stationary arm, the first and second cover forming a second channel to allow vertical movement of the movable arm within the intermediate arm, and a stopping protrusion formed at the bottom of the second channel to limit the downward movement of the movable arm.

10. The hanger device of claim 9, wherein the movable arm comprises a vertical arm and a horizontal arm provided at a bottom of the vertical arm, the vertical arm having a third channel, and an opening is provided in the horizontal arm for communication with the third channel.

11. The hanger device of claim 10, wherein the hanger bar is coupled to the horizontal arm.

12. The hanger device of claim 10, wherein the first, second and third channels form the air flow path.

13. The hanger device of claim 10, wherein the first lift unit is provided on top of the vertical arm, the first lift unit including a support cover and at least one lift provided on the support cover, the at least one lift includes a pinion, a motor and a shaft coupled to the pinion and the motor, and a gear rack is formed vertically in the second channel of the intermediate arm.

14. The hanger device of claim 13, wherein the support cover comprises:

a motor mount to support the motor;

a pipe support configured to allow passage of a sub-supply pipe; and

a bridge coupling the motor mount and the pipe support and having a narrower width than the second channel and the third channel such that the second and third channels communicate with each other.

15. The hanger device of claim **9**, wherein the second lift unit includes a mover is provided on a top outer side surface of the intermediate arm, and a stator is support on the limiting protrusion of the stationary arm, the mover and stator forming a linear actuator.

16. The hanger device of claim **8**, wherein at least one of the first inner cover or the first outer cover includes a guide groove to guide the movement of the intermediate arm along the first channel.

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